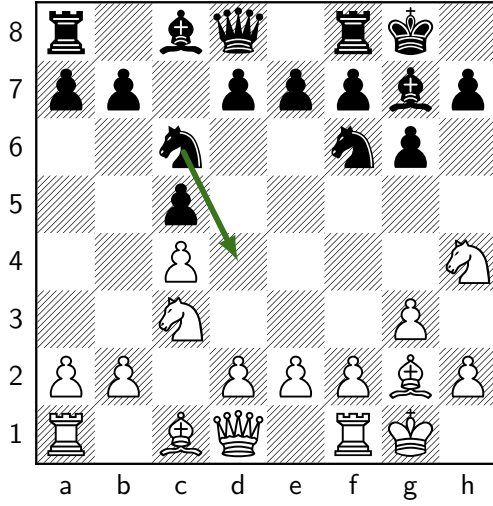


Melanie-BOTS (CCRL engines) — English, Symmetrical

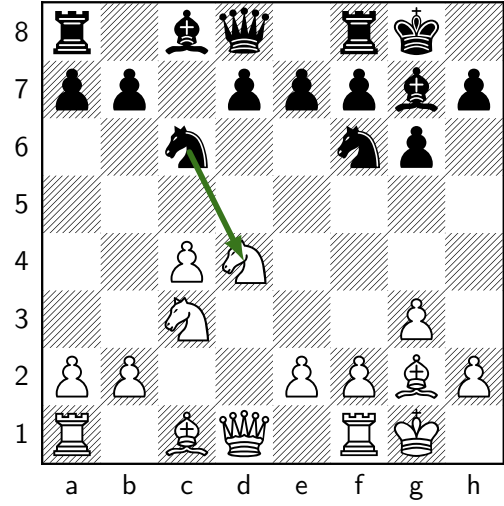
Melanie (Deep_Vision)

Abstract

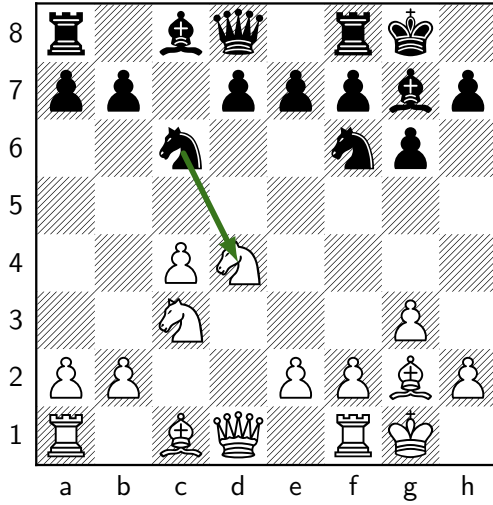
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



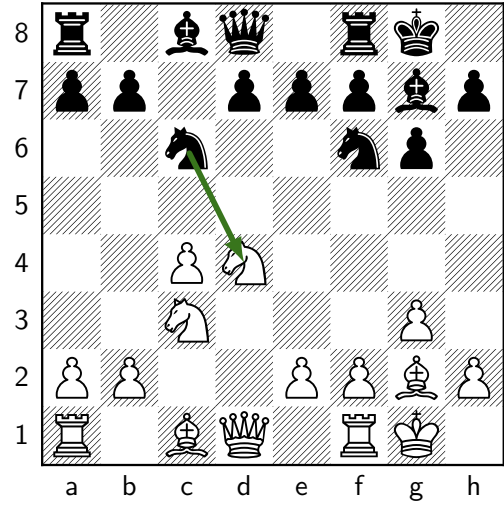
(a)



(b)

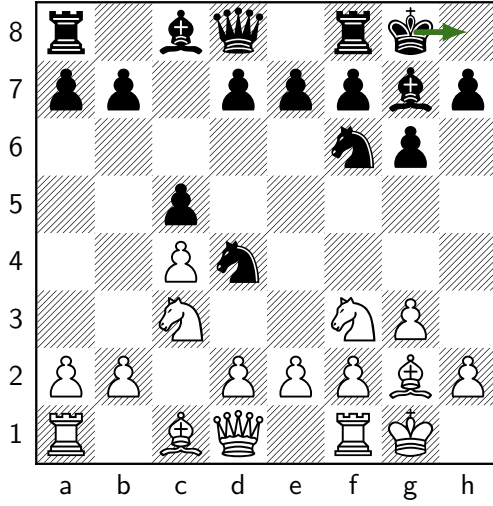


(c)

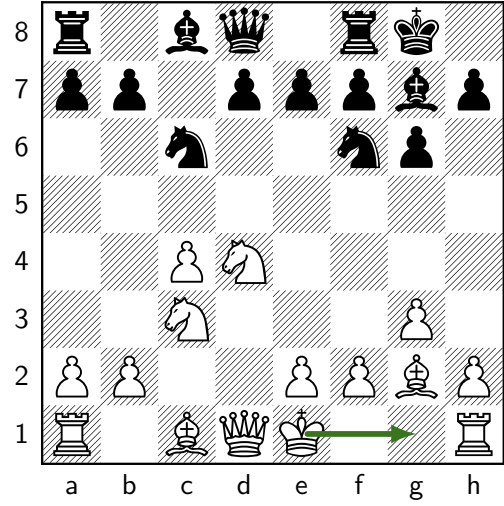


(d)

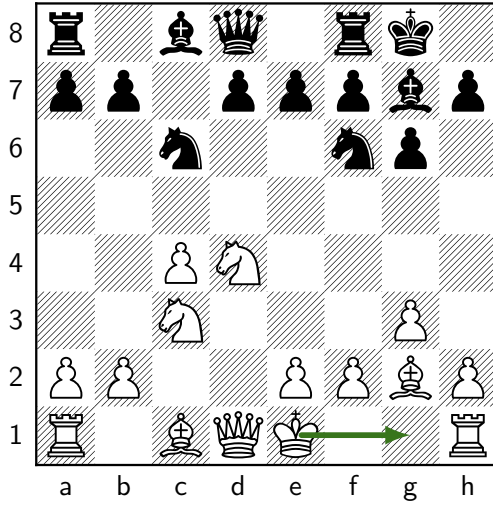
Figure 1: **Move 7, Black: Nd4**. (a) Our position; Melanie plays Nd4 (c6d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#249` (ply 15) [structural distance $d=28$, QAP obj=28]: the master played Nxd4; under μ our Nc3 plays the role of that game’s Nc3; cross-type roles: our pb2 as their Qd8, our Qd8 as their Pa7; transferred move c6d4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#250` (ply 15) [structural distance $d=28$, QAP obj=28]: the master played Nxd4; under μ our Nc3 plays the role of that game’s Nc3; cross-type roles: our pb2 as their Qd8, our Qd8 as their Pa7; transferred move c6d4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1920` (ply 15) [structural distance $d=28$, QAP obj=28]: the master played Nxd4; under μ our Nc3 plays the role of that game’s Nc3; cross-type roles: our pb2 as their Qd8, our Qd8 as their Pa7; transferred move c6d4 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Nd4 (summed QAP objective 112, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



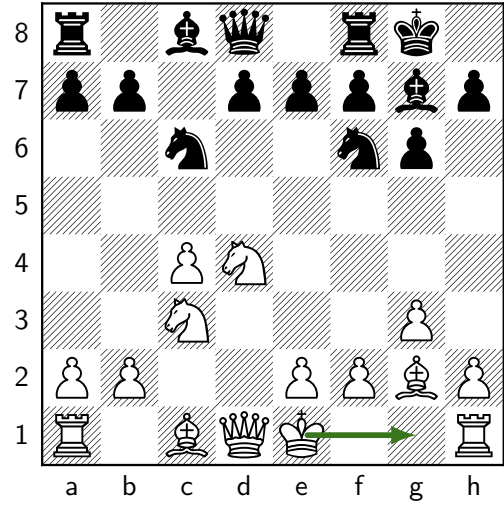
(a)



(b)

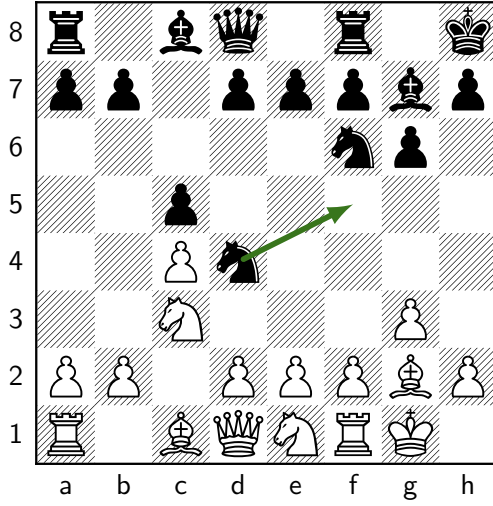


(c)

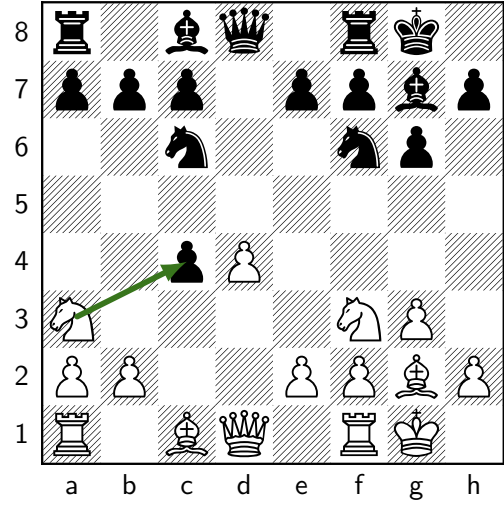


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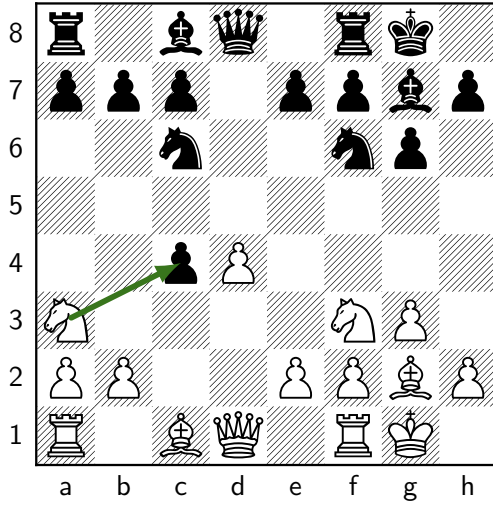
Figure 2: **Move 8, Black: Kh8.** (a) Our position; Melanie plays Kh8 (g8h8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#372 (ply 14) [structural distance $d=52$, QAP obj=52]: the master played 0-0; under μ our Kg1 plays the role of that game’s Ke1; cross-type roles: our Bg7 as their Ra1, our pg3 as their Bc1; transferred move g8h8 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#373 (ply 14) [structural distance $d=52$, QAP obj=52]: the master played 0-0; under μ our Kg1 plays the role of that game’s Ke1; cross-type roles: our Bg7 as their Ra1, our pg3 as their Bc1; transferred move g8h8 (**endorses this move**). (d) Retrieved from ccr1_top_p0001#3221 (ply 14) [structural distance $d=52$, QAP obj=52]: the master played 0-0; under μ our Kg1 plays the role of that game’s Ke1; cross-type roles: our Bg7 as their Ra1, our pg3 as their Bc1; transferred move g8h8 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Kh8 (summed QAP objective 415, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



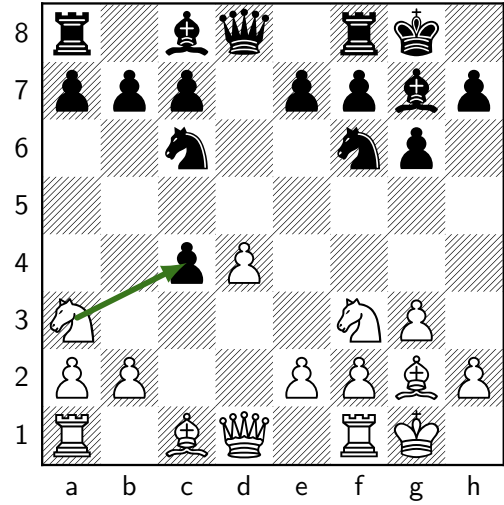
(a)



(b)

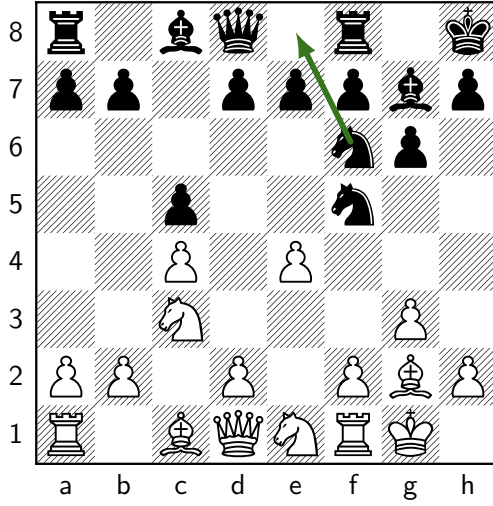


(c)

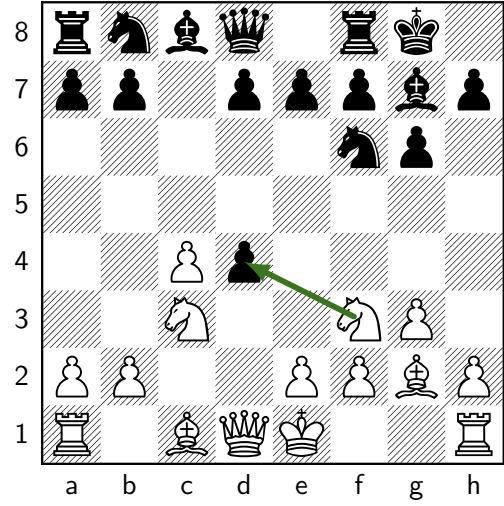


(d)

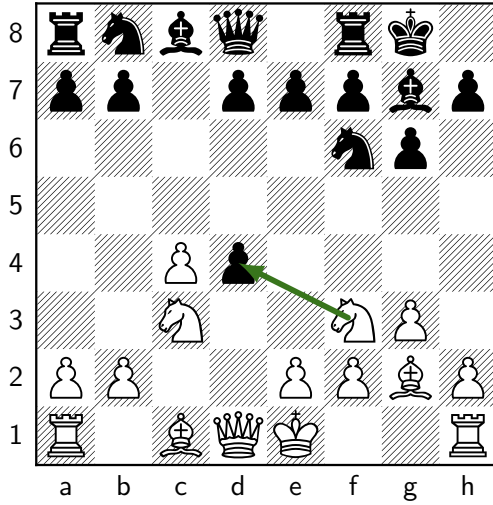
Figure 3: **Move 9, Black: Nf5**. (a) Our position; Melanie plays Nf5 (d4f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#512` (ply 14) [structural distance $d=72$, QAP obj=53]: the master played Nxc4; under μ our Nd5 plays the role of that game’s Na3; cross-type roles: our Pd7 as their Ra1, our qd1 as their Bc1; transferred move d4f5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#513` (ply 14) [structural distance $d=72$, QAP obj=53]: the master played Nxc4; under μ our Nd5 plays the role of that game’s Na3; cross-type roles: our Pd7 as their Ra1, our qd1 as their Bc1; transferred move d4f5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1412` (ply 14) [structural distance $d=72$, QAP obj=53]: the master played Nxc4; under μ our Nd5 plays the role of that game’s Na3; cross-type roles: our Pd7 as their Ra1, our qd1 as their Bc1; transferred move d4f5 (**endorses this move**). Synthesis: 14 retrieved analogs transferred to Nf5 (summed QAP objective 739, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



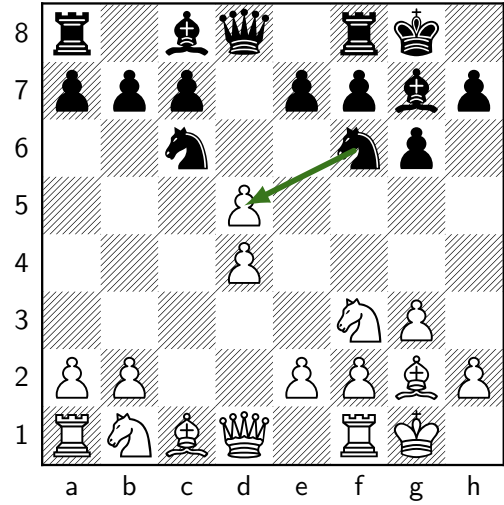
(a)



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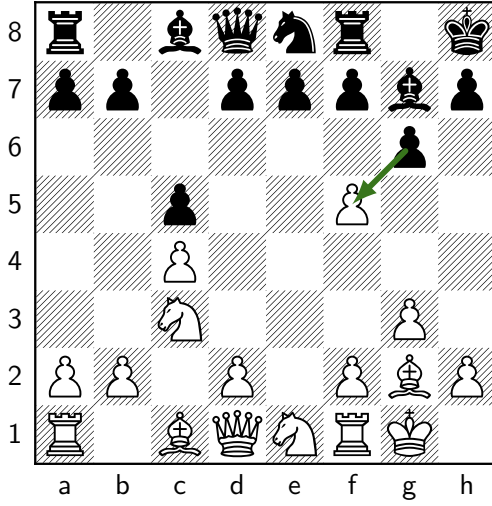


(c)

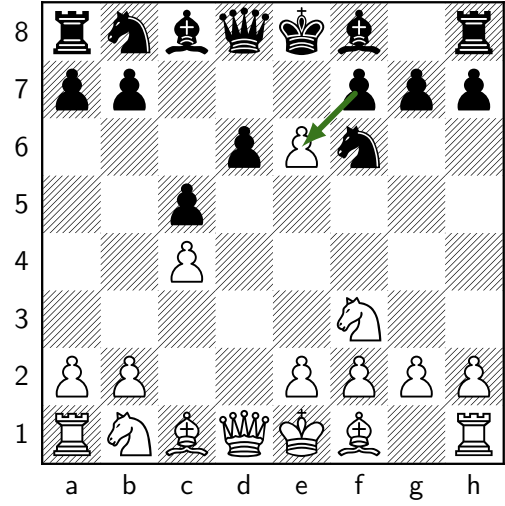


(d)

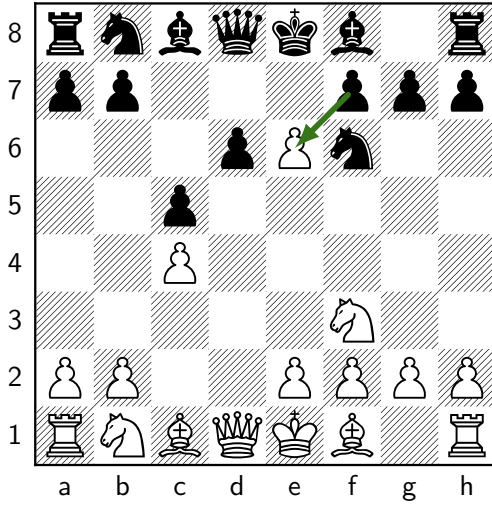
Figure 4: **Move 10, Black: Ne8 (f6e8).** (a) Our position; Melanie plays Ne8 (f6e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `cctl_top_p0001#3615` (ply 12) [structural distance $d=71$, QAP obj=36]: the master played Nxd4; under μ our Nf3 plays the role of that game’s Nf3; cross-type roles: our Pa7 as their Bc1, our Nf6 as their Qd1; transferred move f6e8 (**endorses this move**). (c) Retrieved from `cctl_top_p0001#3616` (ply 12) [structural distance $d=71$, QAP obj=36]: the master played Nxd4; under μ our Nf3 plays the role of that game’s Nf3; cross-type roles: our Pa7 as their Bc1, our Nf6 as their Qd1; transferred move f6e8 (**endorses this move**). (d) Retrieved from `cctl_top_p0000#2171` (ply 13) [structural distance $d=70$, QAP obj=38]: the master played Nxd5; under μ our Nf3 plays the role of that game’s Nf3; cross-type roles: our pd2 as their Ra8, our Pc5 as their Bc8; transferred move f6h5. Synthesis: 2 retrieved analogs transferred to Ne8 (summed QAP objective 72, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



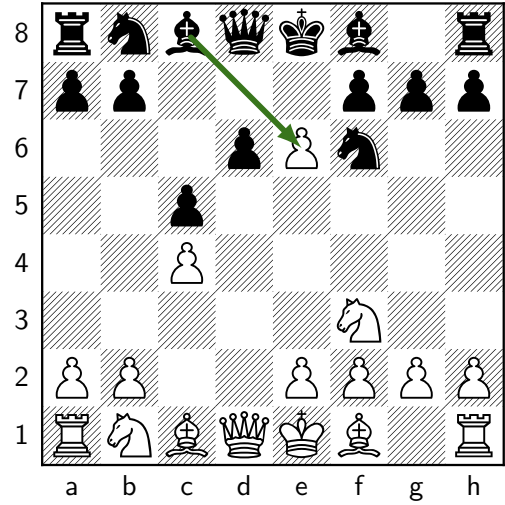
(a)



(b)

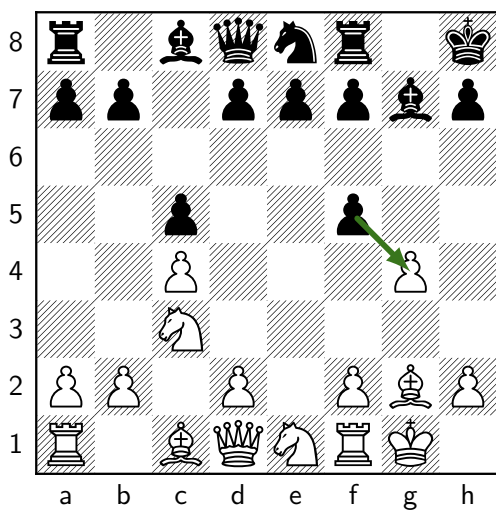


(c)

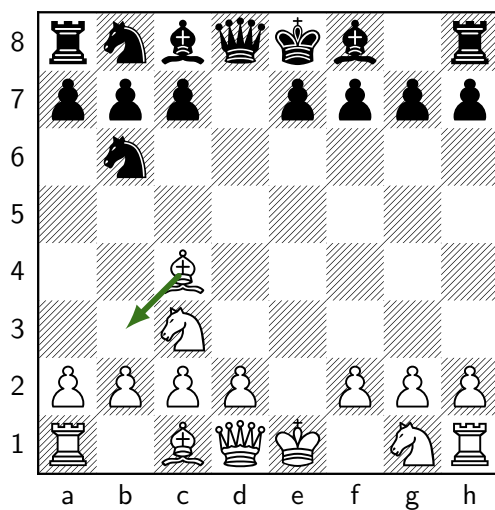


(d)

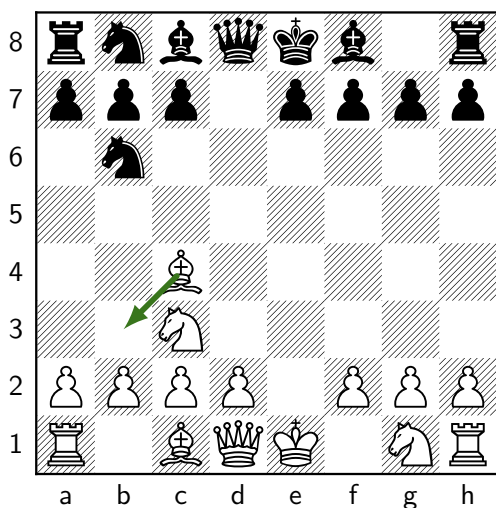
Figure 5: **Move 11, Black: gxf5.** (a) Our position; Melanie plays **gxf5** (**g6f5**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1218` (ply 9) [structural distance $d=62$, QAP obj=66]: the master played **fxe6**; under μ our $Pg3$ plays the role of that game’s $Pf2$; cross-type roles: our $Bc8$ as their $Ra8$, our $ra1$ as their $Nb8$; transferred move **g6f5** (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#1219` (ply 9) [structural distance $d=62$, QAP obj=66]: the master played **fxe6**; under μ our $Pg3$ plays the role of that game’s $Pf2$; cross-type roles: our $Bc8$ as their $Ra8$, our $ra1$ as their $Nb8$; transferred move **g6f5** (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#413` (ply 9) [structural distance $d=62$, QAP obj=66]: the master played **Bxe6**; under μ our $Pg3$ plays the role of that game’s $Bc1$ (a **cross-type** correspondence); cross-type roles: our $Bc8$ as their $Ra8$, our $ra1$ as their $Nb8$; transferred move **g6f5** (**endorses this move**). Synthesis: 9 retrieved analogs transferred to **gxf5** (summed QAP objective 525, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



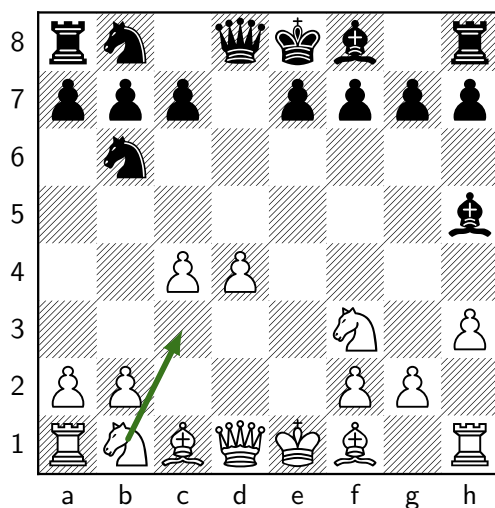
(a)



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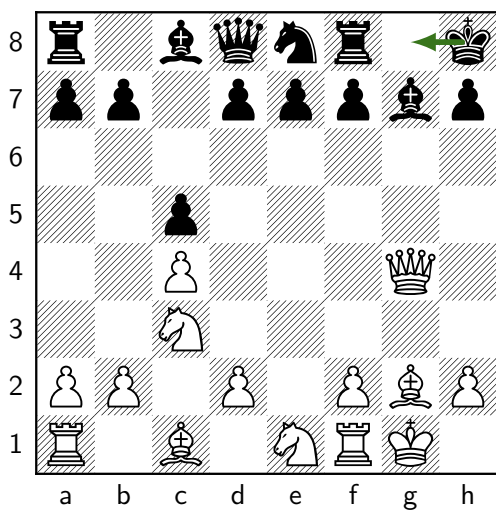


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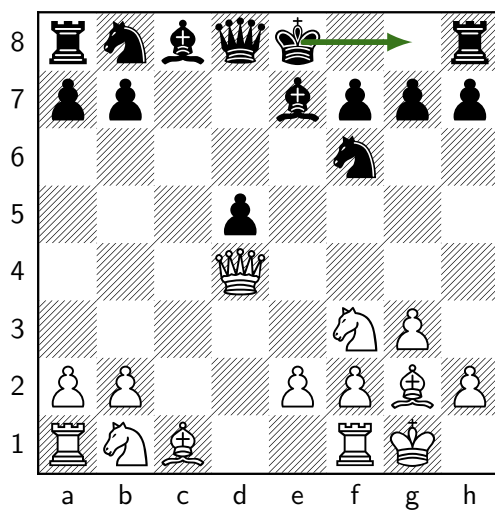


(d)

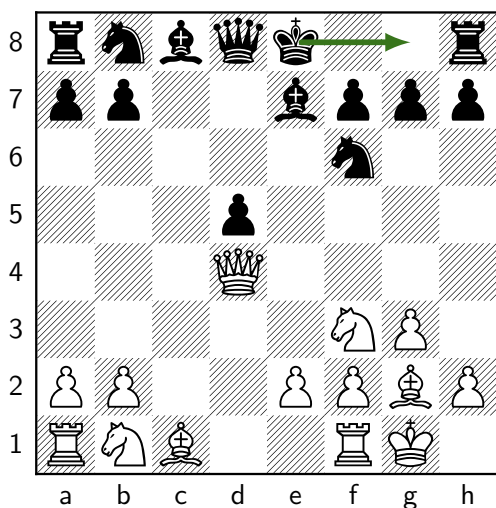
Figure 6: **Move 12, Black: f5g4.** (a) Our position; Melanie plays f5g4 (f5g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#2798 (ply 8) [structural distance $d=75$, QAP obj=35]: the master played Bb3; under μ our Pf4 plays the role of that game’s Bc4 (a **cross-type** correspondence); cross-type roles: our nc3 as their Ra1, our Rf8 as their Bc1; transferred move f5g4 (**endorses this move**). (c) Retrieved from ccr1_top_p0003#2799 (ply 8) [structural distance $d=75$, QAP obj=35]: the master played Bb3; under μ our Pf4 plays the role of that game’s Bc4 (a **cross-type** correspondence); cross-type roles: our nc3 as their Ra1, our Rf8 as their Bc1; transferred move f5g4 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#2684 (ply 12) [structural distance $d=68$, QAP obj=22]: the master played Nc3; under μ our Qd1 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our bg2 as their Ra1, our Pf7 as their Bc1; transferred move d8c7. Synthesis: 2 retrieved analogs transferred to f5g4 (summed QAP objective 70, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



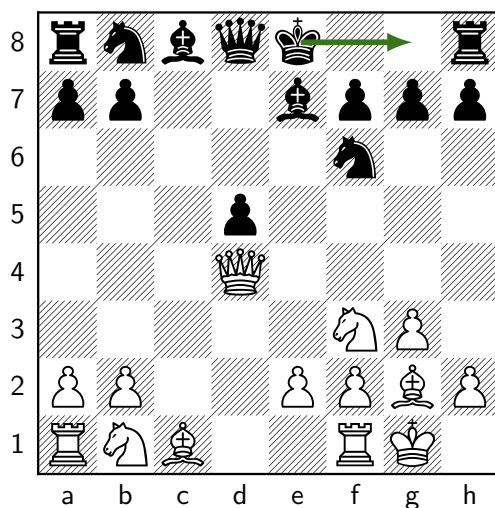
(a)



(b)

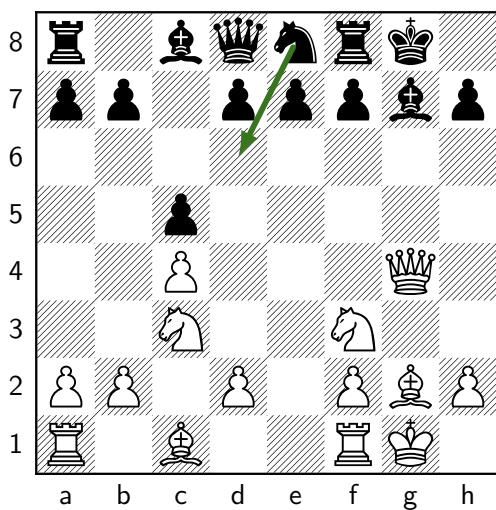


(c)

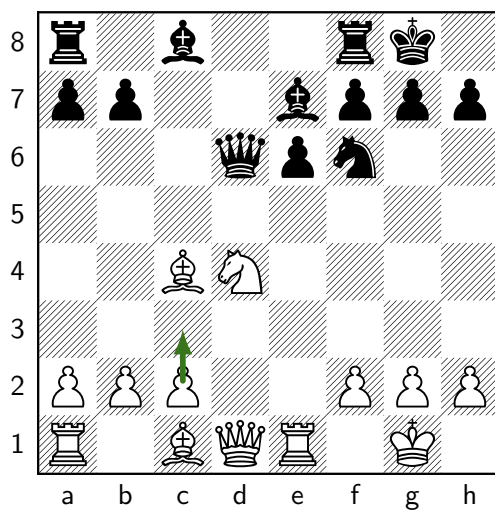


(d)

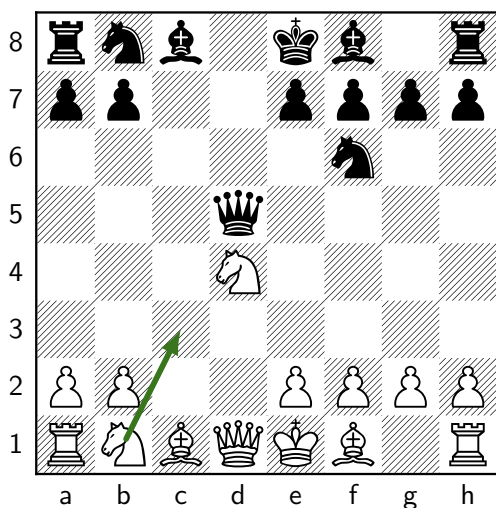
Figure 7: **Move 13, Black: Kg8.** (a) Our position; Melanie plays Kg8 (h8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#232` (ply 15) [structural distance $d=81$, QAP obj=50]: the master played 0-0; under μ our *Kh1* plays the role of that game’s *Ke1*; cross-type roles: our *Bc8* as their *Ra8*, our *Qd8* as their *Nb8*; transferred move h8g8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#233` (ply 15) [structural distance $d=81$, QAP obj=50]: the master played 0-0; under μ our *Kh1* plays the role of that game’s *Ke1*; cross-type roles: our *Bc8* as their *Ra8*, our *Qd8* as their *Nb8*; transferred move h8g8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1989` (ply 15) [structural distance $d=81$, QAP obj=50]: the master played 0-0; under μ our *Kh1* plays the role of that game’s *Ke1*; cross-type roles: our *Bc8* as their *Ra8*, our *Qd8* as their *Nb8*; transferred move h8g8 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Kg8 (summed QAP objective 388, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



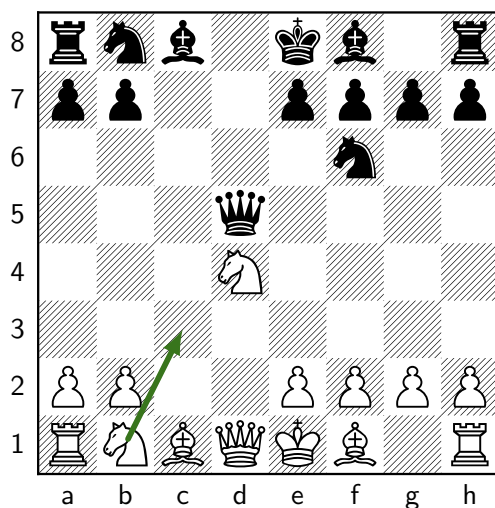
(a)



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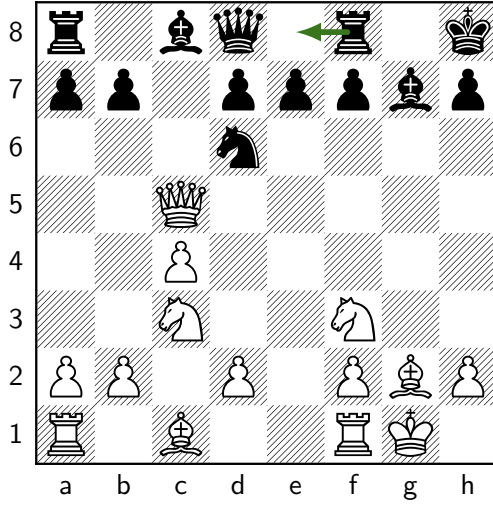


(c)

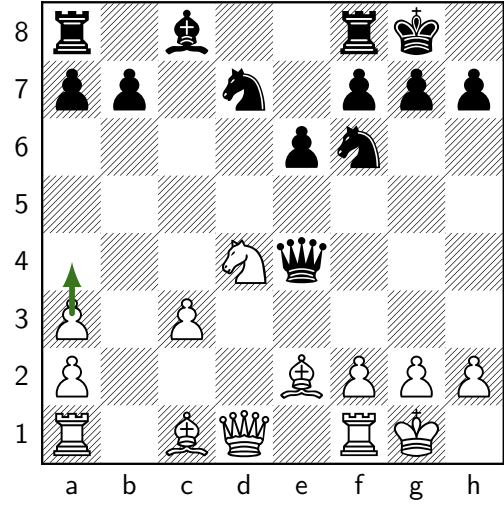


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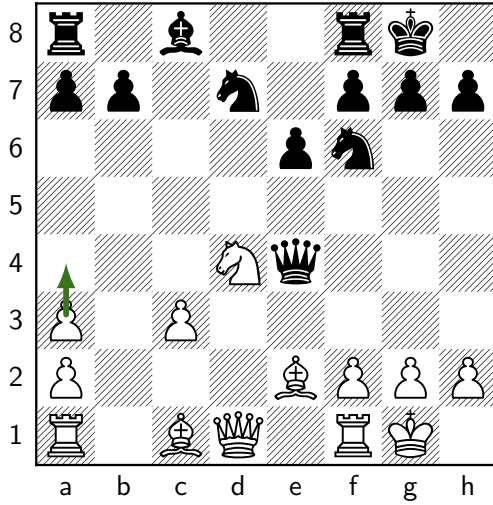
Figure 8: **Move 14, Black: Nd6**. (a) Our position; Melanie plays Nd6 (e8d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2954` (ply 22) [structural distance $d=88$, QAP obj=74]: the master played c3; under μ our *Ne1* plays the role of that game’s *Pc2* (a **cross-type** correspondence); cross-type roles: our *Bc8* as their *Ra1*, our *Pf7* as their *Bc1*; transferred move e8d6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2602` (ply 10) [structural distance $d=67$, QAP obj=36]: the master played Nc3; under μ our *Ne1* plays the role of that game’s *Nb1*; cross-type roles: our *Bc8* as their *Ra1*, our *ph2* as their *Bc1*; transferred move e8d6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2603` (ply 10) [structural distance $d=67$, QAP obj=36]: the master played Nc3; under μ our *Ne1* plays the role of that game’s *Nb1*; cross-type roles: our *Bc8* as their *Ra1*, our *ph2* as their *Bc1*; transferred move e8d6 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Nd6 (summed QAP objective 147, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



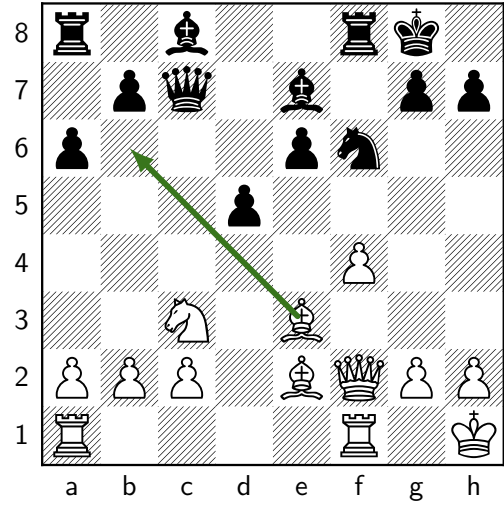
(a)



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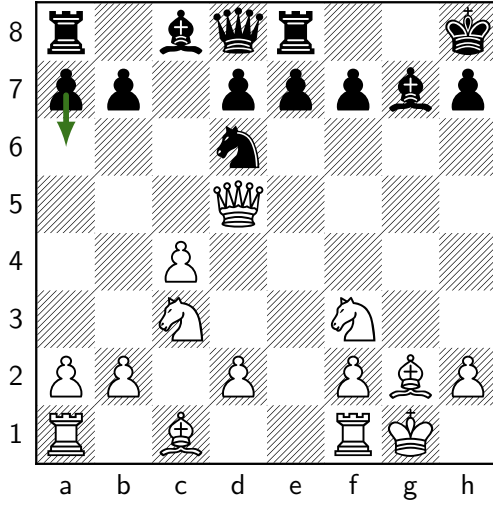


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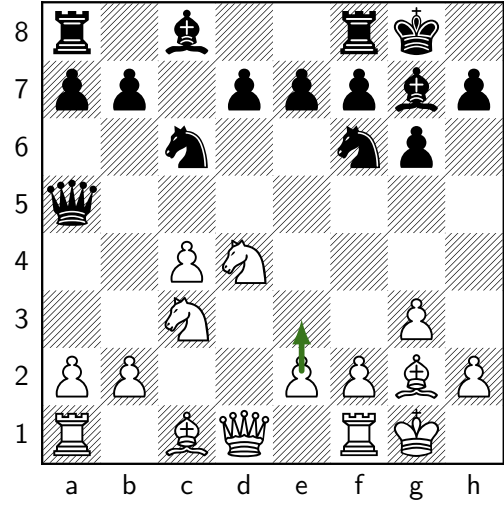


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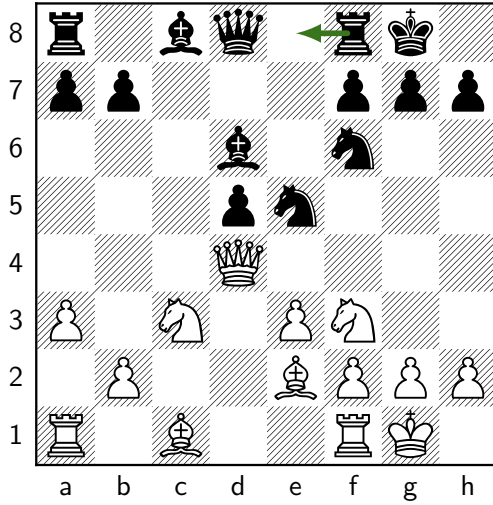
Figure 9: **Move 16, Black: Re8**. (a) Our position; Melanie plays **Re8** (**f8e8**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2235` (ply 22) [structural distance $d=78$, QAP obj=61]: the master played **a4**; under μ our *Rf1* plays the role of that game’s *Pa3* (a **cross-type** correspondence); cross-type roles: our *pf2* as their *Ra1*, our *nc3* as their *Bc1*; transferred move **f8e8** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2236` (ply 22) [structural distance $d=78$, QAP obj=61]: the master played **a4**; under μ our *Rf1* plays the role of that game’s *Pa3* (a **cross-type** correspondence); cross-type roles: our *pf2* as their *Ra1*, our *nc3* as their *Bc1*; transferred move **f8e8** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2331` (ply 28) [structural distance $d=77$, QAP obj=22]: the master played **Bb6**; under μ our *Rf1* plays the role of that game’s *Be3* (a **cross-type** correspondence); cross-type roles: our *pb2* as their *Ra1*, our *Bg7* as their *Rf1*; transferred move **f8e8** (**endorses this move**). Synthesis: 4 retrieved analogs transferred to **Re8** (summed QAP objective 160, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



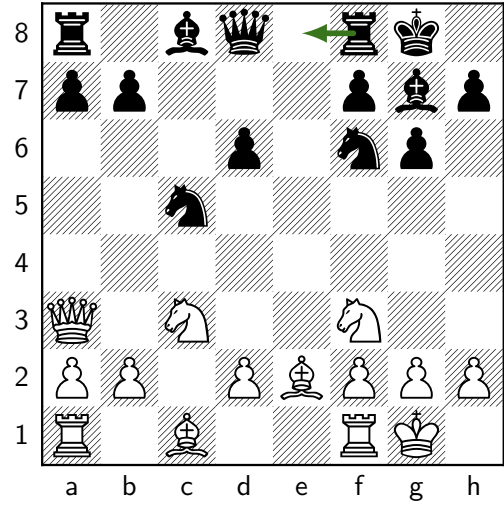
(a)



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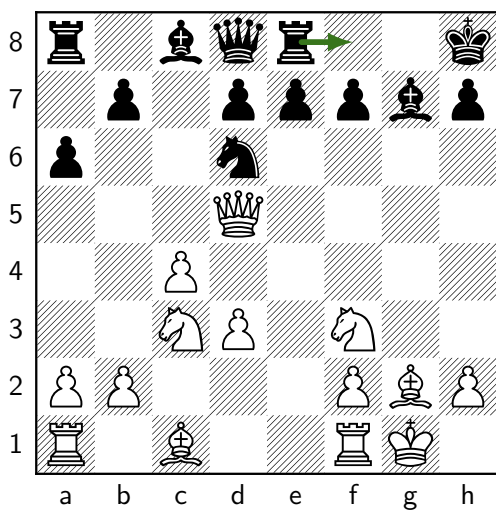


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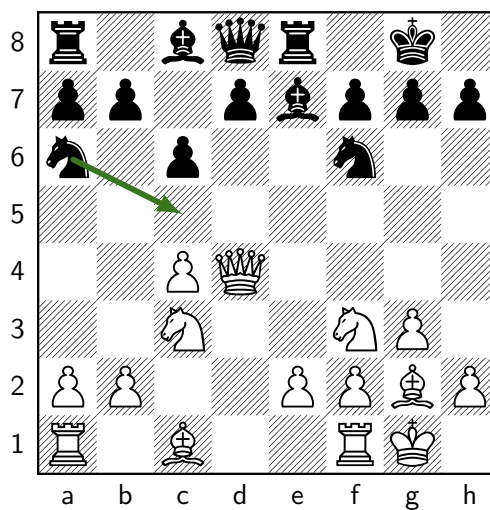


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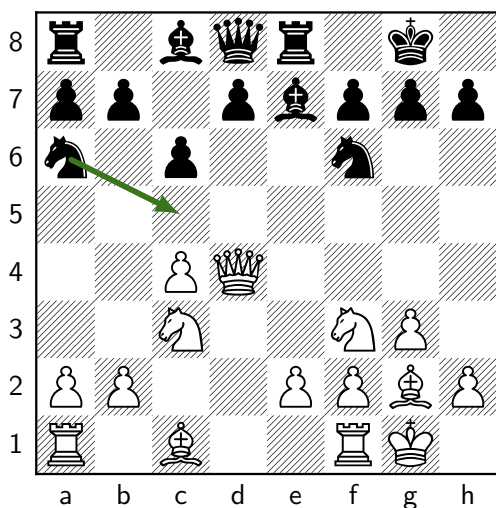
Figure 10: **Move 17, Black: a6.** (a) Our position; Melanie plays a6 (a7a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#372` (ply 16) [structural distance $d=90$, QAP obj=58]: the master played e3; under μ our *Pa2* plays the role of that game’s *Pe2*; cross-type roles: our *Bc8* as their *Ra1*, our *pa2* as their *Bc1*; transferred move a7a6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#50` (ply 23) [structural distance $d=64$, QAP obj=57]: the master played Re8; under μ our *Bg2* plays the role of that game’s *Rf1* (a **cross-type** correspondence); cross-type roles: our *Ph7* as their *Ra8*, our *nf3* as their *Pf7*; transferred move g7h6. (d) Retrieved from `ccr1_top_p0001#2420` (ply 21) [structural distance $d=66$, QAP obj=46]: the master played Re8; under μ our *Bg2* plays the role of that game’s *Rf1* (a **cross-type** correspondence); cross-type roles: our *Bc8* as their *Ra8*, our *Re8* as their *Bc8*; transferred move g7f8. Synthesis: 1 retrieved analog transferred to a6 (summed QAP objective 58, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



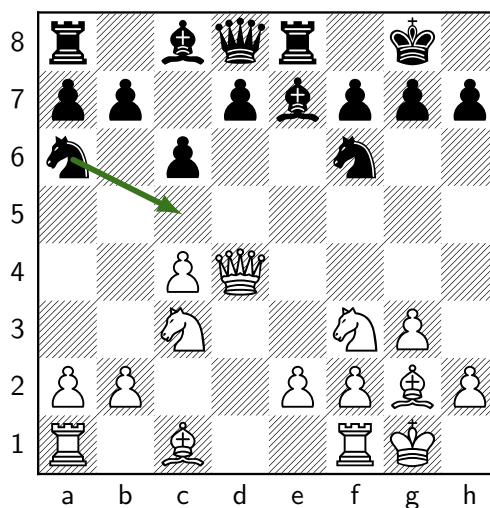
(a)



(b)

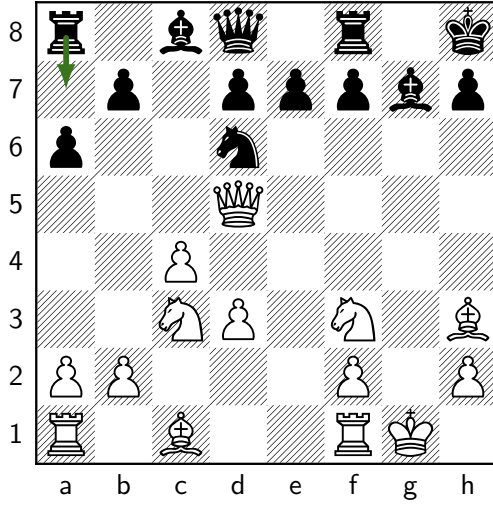


(c)

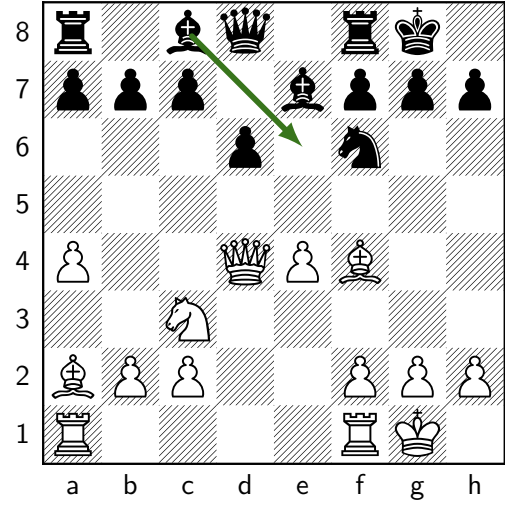


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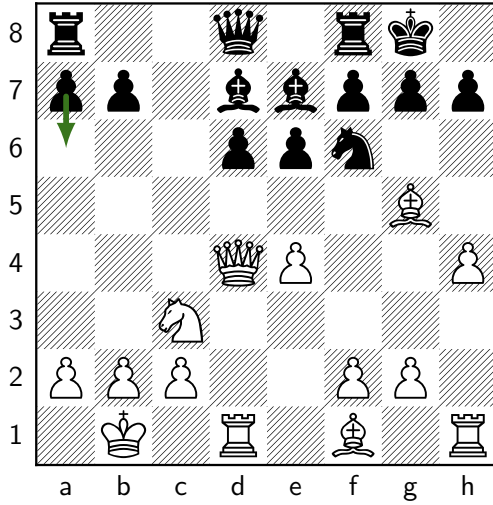
Figure 11: **Move 18, Black: Rf8 (e8f8).** (a) Our position; Melanie plays Rf8 (e8f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2619` (ply 19) [structural distance $d=57$, QAP obj=51]: the master played Nc5; under μ our *Re1* plays the role of that game's *Na3* (a **cross-type** correspondence); cross-type roles: our *Pd7* as their *Qd8*, our *Bg7* as their *Re8*; transferred move e8f8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2620` (ply 19) [structural distance $d=57$, QAP obj=51]: the master played Nc5; under μ our *Re1* plays the role of that game's *Na3* (a **cross-type** correspondence); cross-type roles: our *Pd7* as their *Qd8*, our *Bg7* as their *Re8*; transferred move e8f8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2125` (ply 19) [structural distance $d=57$, QAP obj=51]: the master played Nc5; under μ our *Re1* plays the role of that game's *Na3* (a **cross-type** correspondence); cross-type roles: our *Pd7* as their *Qd8*, our *Bg7* as their *Re8*; transferred move e8f8 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Rf8 (summed QAP objective 239, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



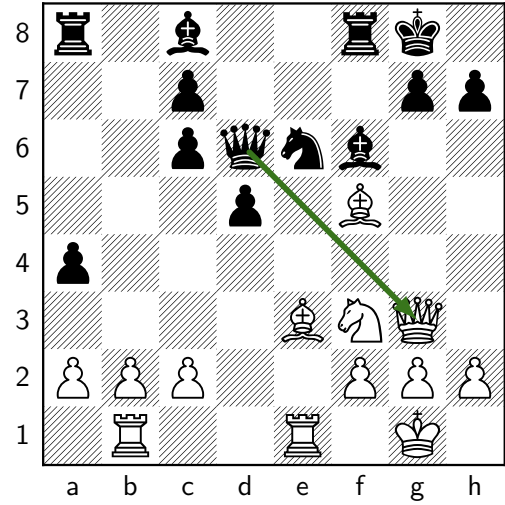
(a)



(b)

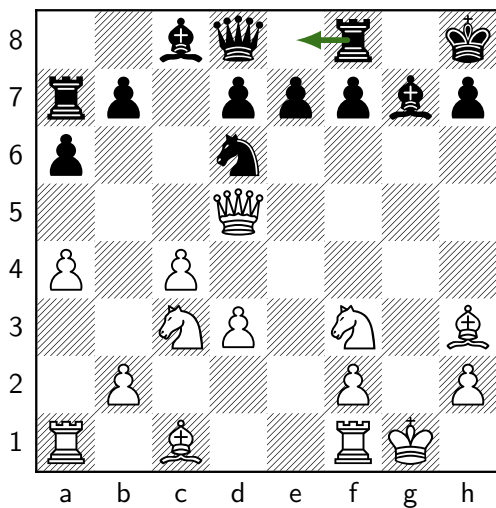


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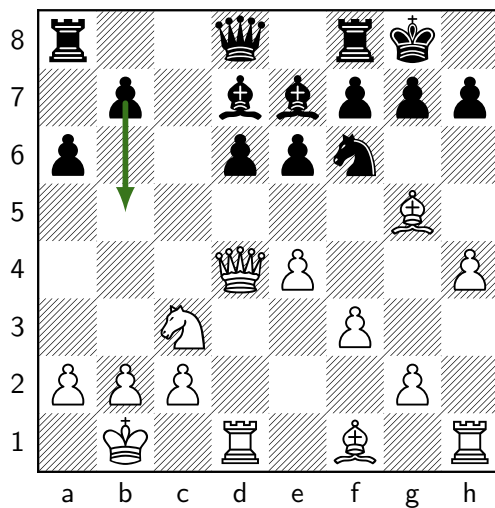


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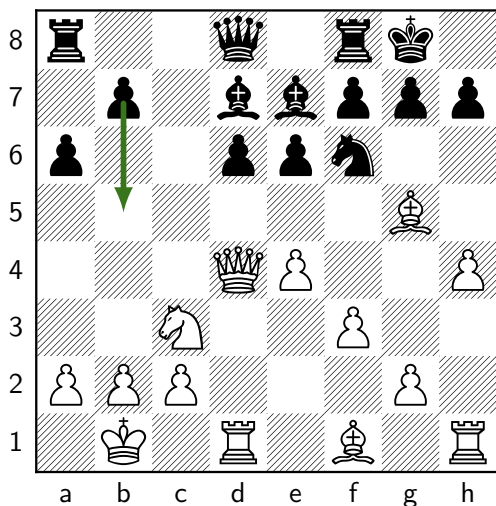
Figure 12: **Move 19, Black: Ra7.** (a) Our position; Melanie plays Ra7 (a8a7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#800` (ply 21) [structural distance $d=58$, QAP obj=84]: the master played Be6; under μ our Ra1 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Ra8, our pb2 as their Qd8; transferred move a8a7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1549` (ply 21) [structural distance $d=58$, QAP obj=97]: the master played a6; under μ our Nd3 plays the role of that game’s Pa2 (a **cross-type** correspondence); cross-type roles: our Pa6 as their Qd8, our Bc8 as their Pb7; transferred move d6f5. (d) Retrieved from `ccr1_top_p0001#2527` (ply 33) [structural distance $d=70$, QAP obj=56]: the master played Qxg3; under μ our Nd3 plays the role of that game’s Qd3 (a **cross-type** correspondence); cross-type roles: our Pd7 as their Bc8, our Bg7 as their Rf8; transferred move d6c4. Synthesis: 1 retrieved analog transferred to Ra7 (summed QAP objective 84, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



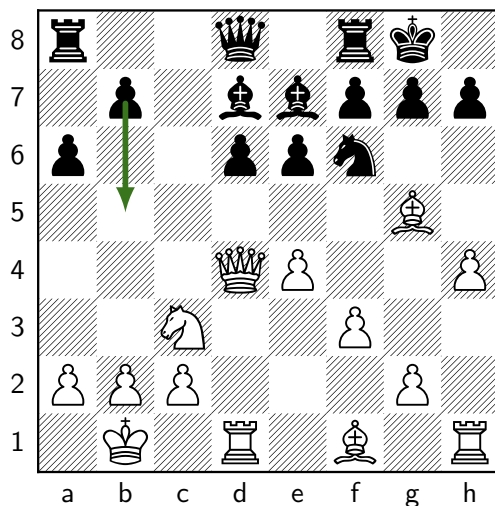
(a)



(b)

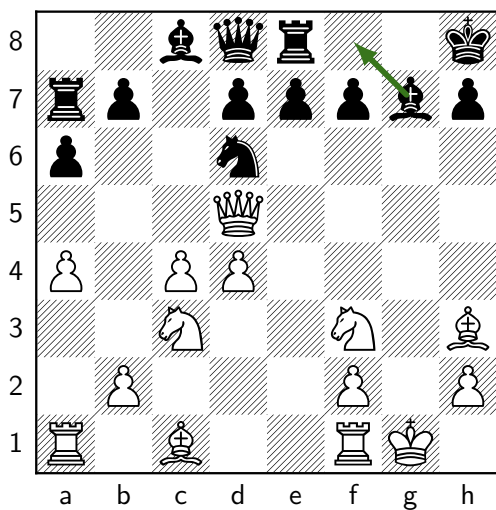


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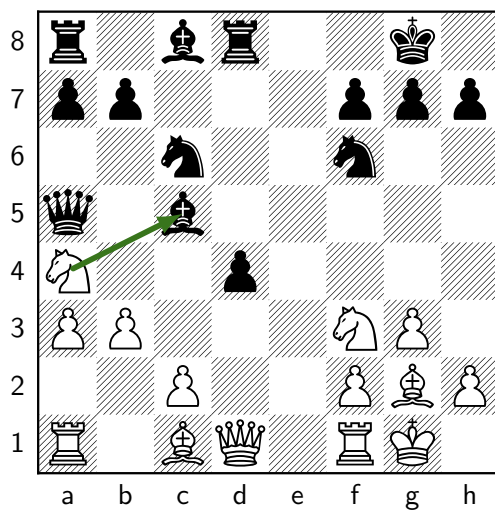


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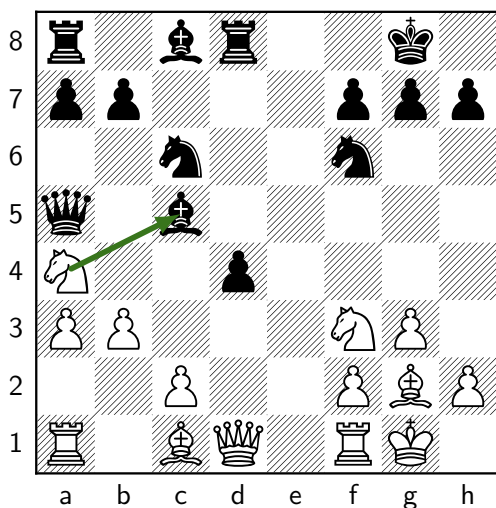
Figure 13: **Move 20, Black: Re8**. (a) Our position; Melanie plays Re8 (f8e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2708` (ply 23) [structural distance $d=62$, QAP obj=60]: the master played b5; under μ our *Rf1* plays the role of that game's *Pb2* (a **cross-type** correspondence); cross-type roles: our *Bc8* as their *Ra8*, our *pa4* as their *Qd8*; transferred move f8e8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1159` (ply 23) [structural distance $d=62$, QAP obj=60]: the master played b5; under μ our *Rf1* plays the role of that game's *Pb2* (a **cross-type** correspondence); cross-type roles: our *Bc8* as their *Ra8*, our *pa4* as their *Qd8*; transferred move f8e8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#930` (ply 23) [structural distance $d=62$, QAP obj=60]: the master played b5; under μ our *Rf1* plays the role of that game's *Pb2* (a **cross-type** correspondence); cross-type roles: our *Bc8* as their *Ra8*, our *pa4* as their *Qd8*; transferred move f8e8 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Re8 (summed QAP objective 299, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



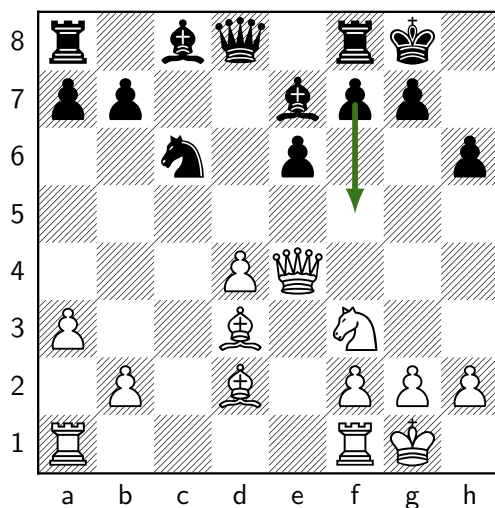
(a)



(b)

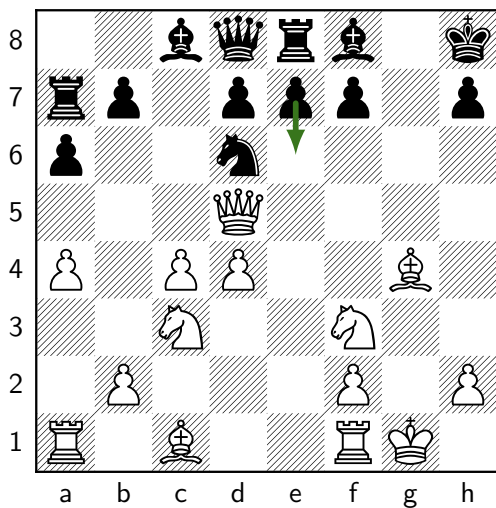


(c)

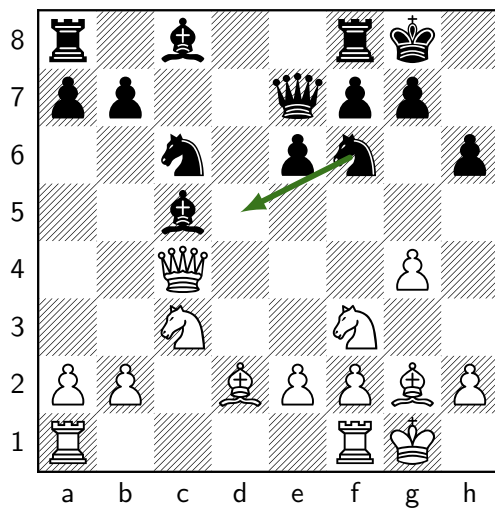


(d)

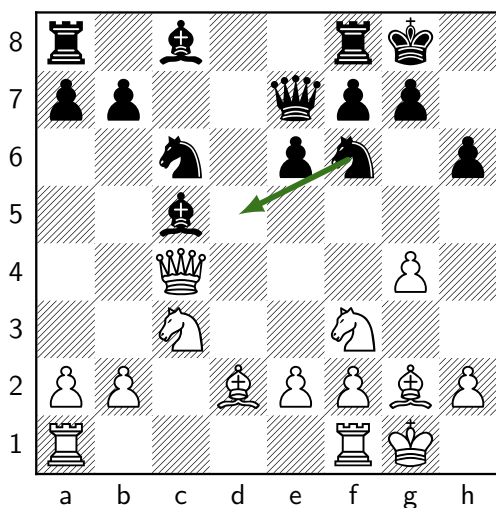
Figure 14: **Move 21, Black: Bf8.** (a) Our position; Melanie plays Bf8 (g7f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2180` (ply 24) [structural distance $d=91$, QAP obj=55]: the master played Nxc5; under μ our Nd3 plays the role of that game’s Na4; cross-type roles: our Bc8 as their Ra1, our Bg7 as their Qd1; transferred move d6c4. (c) Retrieved from `ccr1_top_p0003#2634` (ply 24) [structural distance $d=91$, QAP obj=55]: the master played Nxc5; under μ our Nd3 plays the role of that game’s Na4; cross-type roles: our Bc8 as their Ra1, our Bg7 as their Qd1; transferred move d6c4. (d) Retrieved from `ccr1_top_p0000#1273` (ply 27) [structural distance $d=82$, QAP obj=43]: the master played f5; under μ our Pe2 plays the role of that game’s Pf2; cross-type roles: our Bg7 as their Qd8, our Pa6 as their Be7; transferred move e7e6. Synthesis: no retrieved analog transferred to Bf8 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



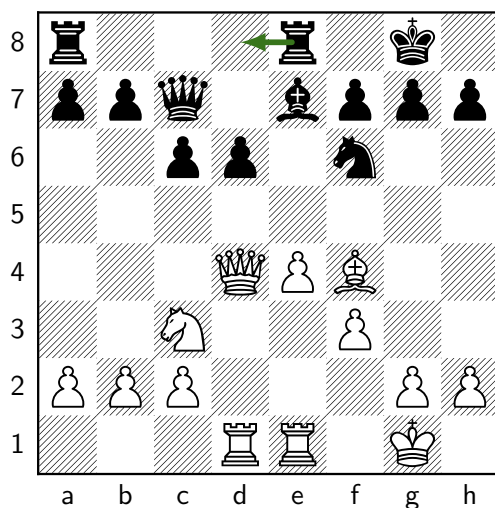
(a)



(b)

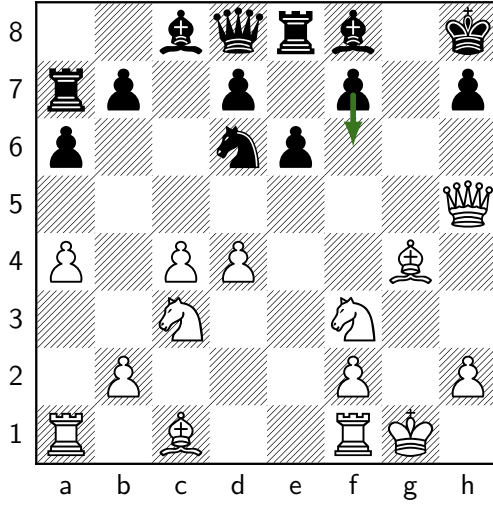


(c)

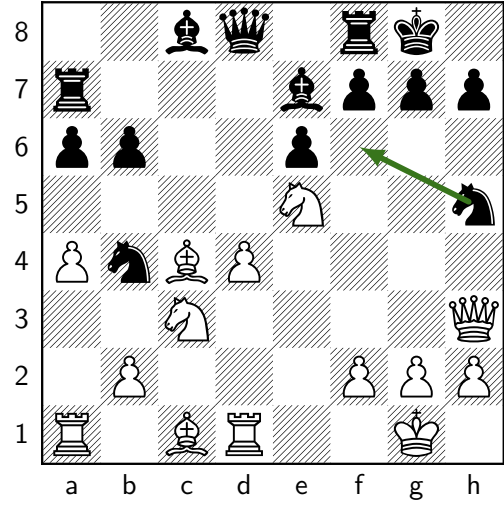


(d)

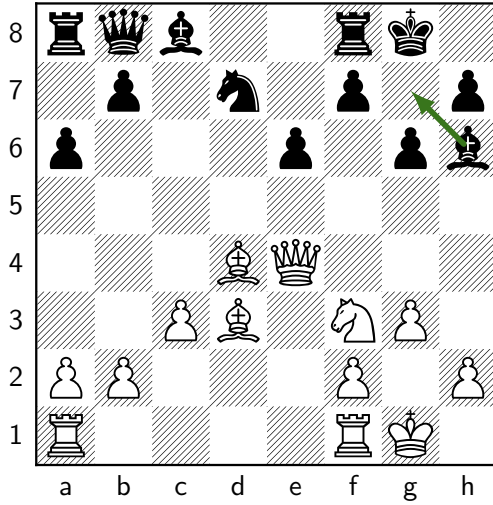
Figure 15: **Move 22, Black: e6**. (a) Our position; Melanie plays e6 (e7e6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3725` (ply 27) [structural distance $d=110$, QAP obj=33]: the master played Nd5; under μ our *Pe2* plays the role of that game’s *Nf3* (a **cross-type** correspondence); cross-type roles: our *Qd8* as their *Ra8*, our *Pa6* as their *Bc8*; transferred move e7e6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1726` (ply 27) [structural distance $d=110$, QAP obj=33]: the master played Nd5; under μ our *Pe2* plays the role of that game’s *Nf3* (a **cross-type** correspondence); cross-type roles: our *Qd8* as their *Ra8*, our *Pa6* as their *Bc8*; transferred move e7e6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#2364` (ply 27) [structural distance $d=102$, QAP obj=49]: the master played Red8; under μ our *Bf1* plays the role of that game’s *Re1* (a **cross-type** correspondence); cross-type roles: our *Bc8* as their *Ra8*, our *Pe7* as their *Be7*; transferred move f8g7. Synthesis: 2 retrieved analogs transferred to e6 (summed QAP objective 65, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



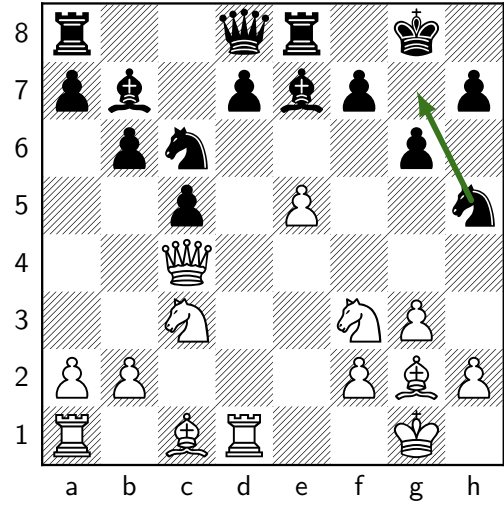
(a)



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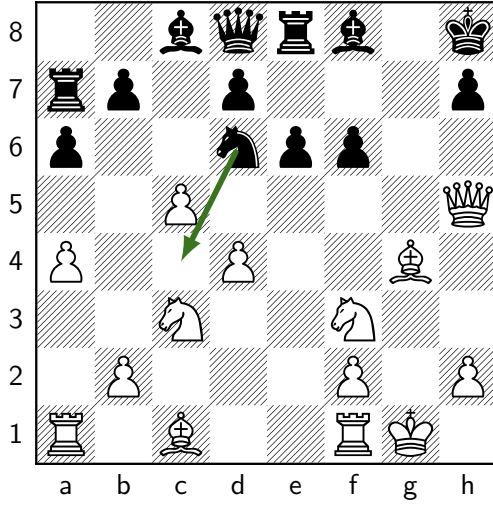


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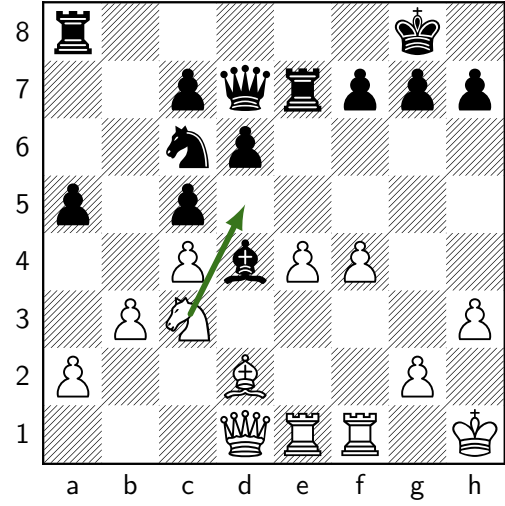


(d)

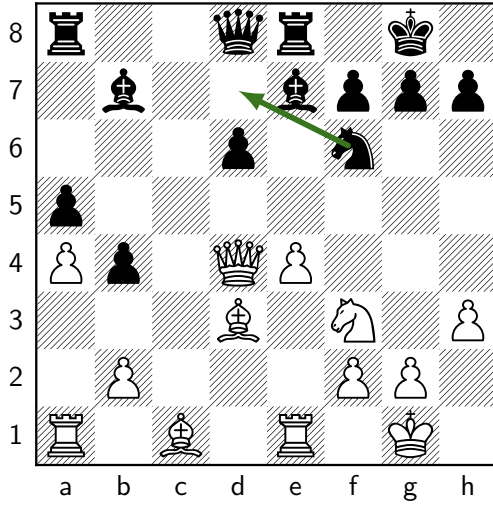
Figure 16: **Move 23, Black: f6**. (a) Our position; Melanie plays f6 (f7f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#6` (ply 31) [structural distance $d=109$, QAP obj=50]: the master played Nf6; under μ our Pf2 plays the role of that game’s Nh4 (a **cross-type** correspondence); cross-type roles: our Nd6 as their Bc8, our pf2 as their Ra7; transferred move f7f6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1171` (ply 37) [structural distance $d=111$, QAP obj=120]: the master played Bg7; under μ our Pb2 plays the role of that game’s Bh3 (a **cross-type** correspondence); cross-type roles: our Bf8 as their Ra8, our Nd6 as their Qb8; transferred move b7b6. (d) Retrieved from `ccr1_top_p0002#3509` (ply 31) [structural distance $d=97$, QAP obj=79]: the master played Ng7; under μ our Nd3 plays the role of that game’s Nh4; cross-type roles: our pd4 as their Ra8, our Bc8 as their Qd8; transferred move d6c4. Synthesis: 1 retrieved analog transferred to f6 (summed QAP objective 50, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



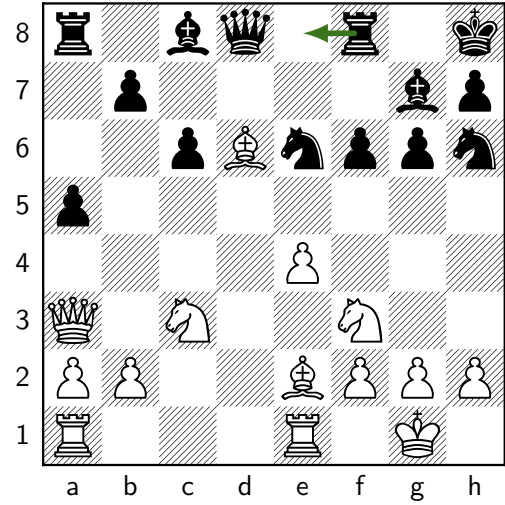
(a)



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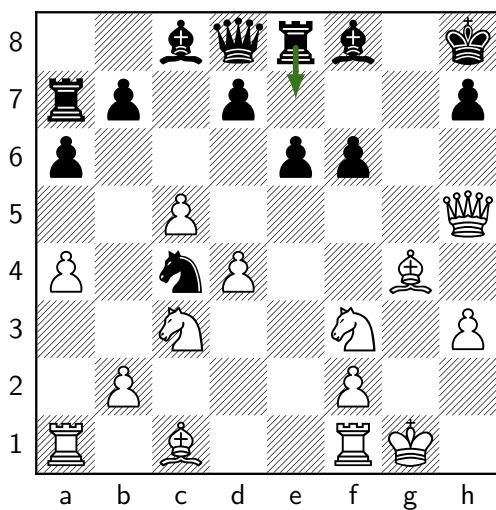


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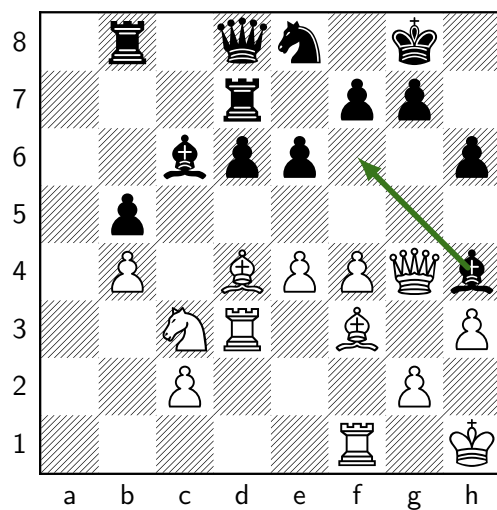


(d)

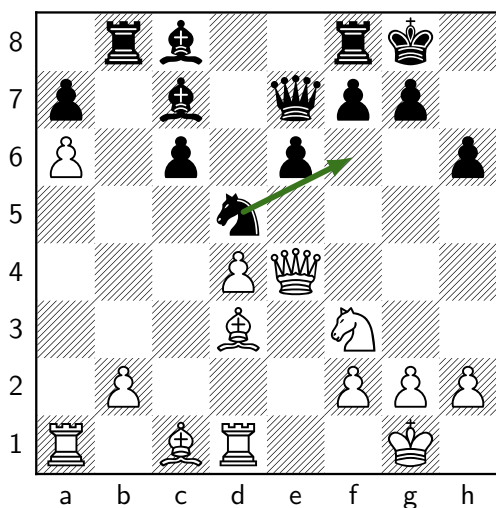
Figure 17: **Move 24, Black: Nc4**. (a) Our position; Melanie plays Nc4 (d6c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1703` (ply 38) [structural distance $d=88$, QAP obj=34]: the master played Nd5; under μ our Nd3 plays the role of that game’s Nc3; cross-type roles: our bg4 as their Qd1, our pd4 as their Re1; transferred move d6c4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1773` (ply 37) [structural distance $d=92$, QAP obj=57]: the master played Nd7; under μ our Re1 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our bg4 as their Ra8, our Nd6 as their Qd8; transferred move e8e7. (d) Retrieved from `ccr1_top_p0003#1259` (ply 33) [structural distance $d=97$, QAP obj=42]: the master played Re8; under μ our Nd3 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our Pb7 as their Bg7, our Ra7 as their Ph7; transferred move d6f7. Synthesis: 1 retrieved analog transferred to Nc4 (summed QAP objective 34, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



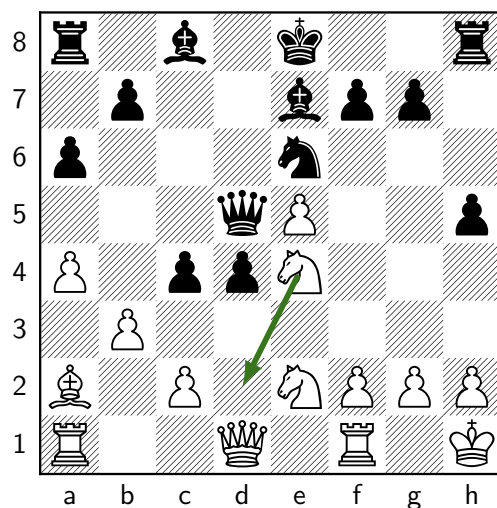
(a)



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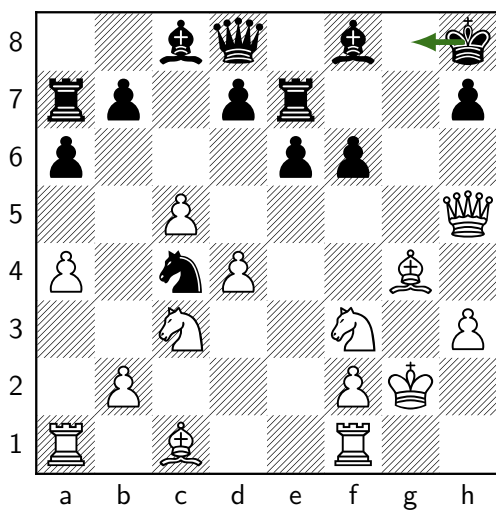


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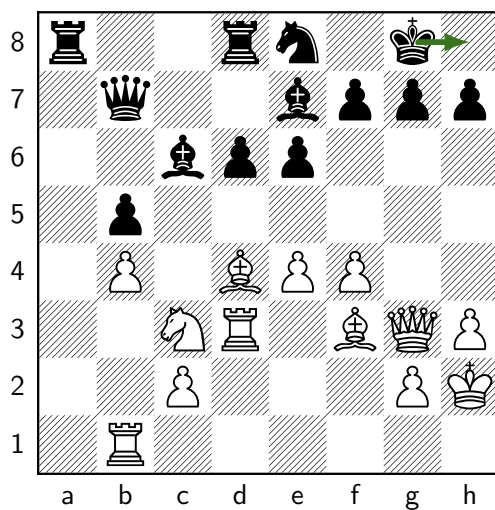


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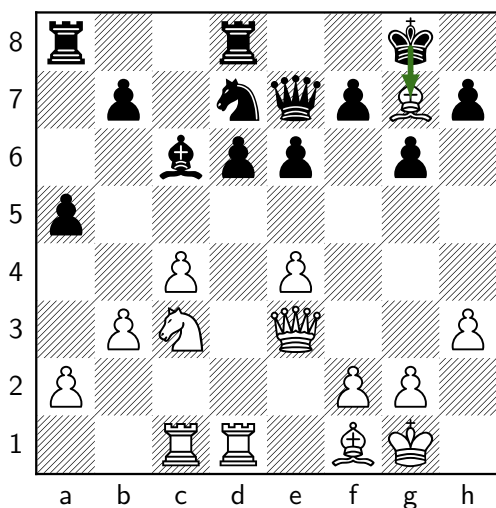
Figure 18: **Move 25, Black: Re7 (e8e7)**. (a) Our position; Melanie plays Re7 (e8e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#3960` (ply 49) [structural distance $d=97$, QAP obj=84]: the master played Bf6; under μ our Re1 plays the role of that game’s Bh5 (a **cross-type** correspondence); cross-type roles: our bc1 as their Rb8, our Bf8 as their Ne8; transferred move e8e7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2412` (ply 39) [structural distance $d=93$, QAP obj=55]: the master played Nf6; under μ our Re1 plays the role of that game’s Nd4 (a **cross-type** correspondence); cross-type roles: our nc3 as their Rb8, our Pf6 as their Rf8; transferred move e8e7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#2433` (ply 34) [structural distance $d=88$, QAP obj=26]: the master played Nd2; under μ our Re1 plays the role of that game’s Ne4 (a **cross-type** correspondence); cross-type roles: our bg4 as their Ra1, our ph3 as their Rf1; transferred move e8e7 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Re7 (summed QAP objective 165, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



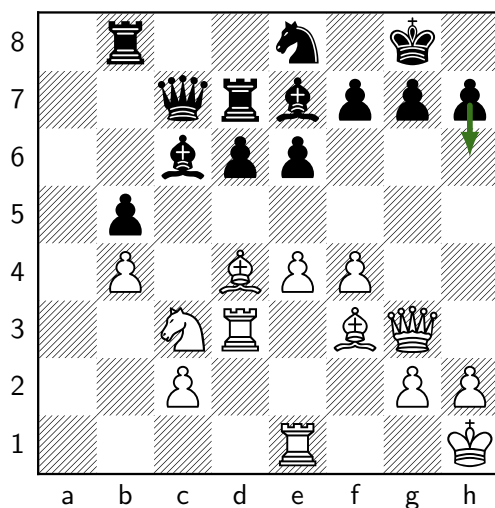
(a)



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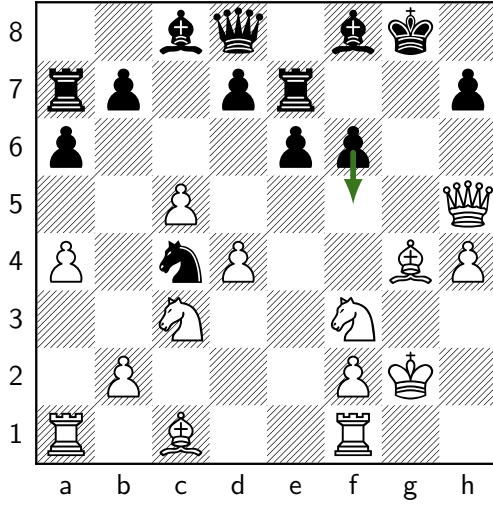


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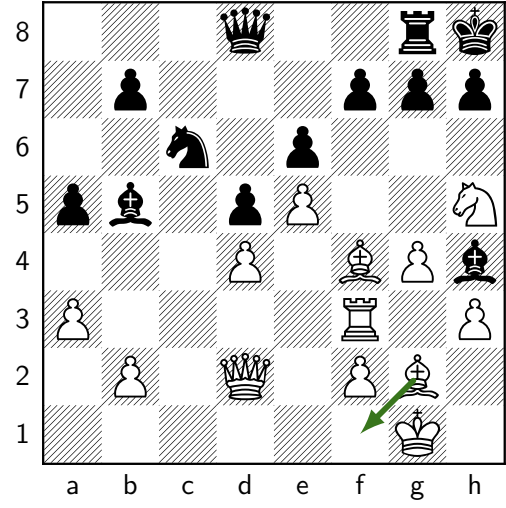


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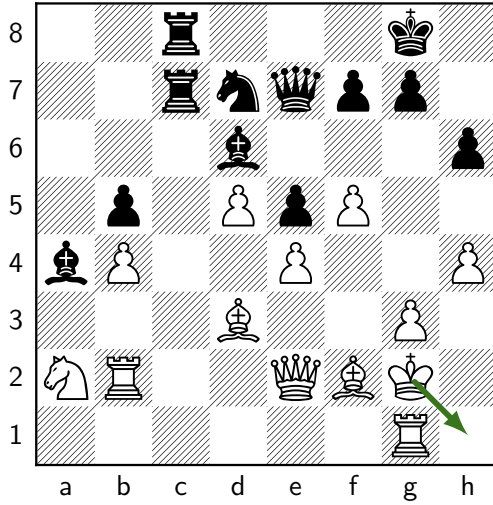
Figure 19: **Move 26, Black: Kg8.** (a) Our position; Melanie plays Kg8 (h8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3884` (ply 53) [structural distance $d=104$, QAP obj=87]: the master played Kh8; under μ our Kh1 plays the role of that game’s Kg1; cross-type roles: our nc3 as their Rd8, our Re7 as their Ne8; transferred move h8g8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1094` (ply 43) [structural distance $d=117$, QAP obj=14]: the master played Kxg7; under μ our Kh1 plays the role of that game’s Kg1; cross-type roles: our Bc8 as their Ra8, our Pe6 as their Rd8; transferred move h8g8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3960` (ply 43) [structural distance $d=107$, QAP obj=76]: the master played h6; under μ our Re2 plays the role of that game’s Ph2 (a **cross-type** correspondence); cross-type roles: our bc1 as their Rb8, our pd4 as their Ne8; transferred move e7e8. Synthesis: 3 retrieved analogs transferred to Kg8 (summed QAP objective 178, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



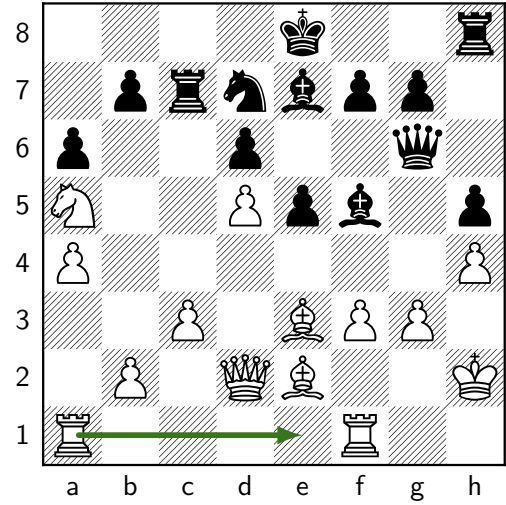
(a)



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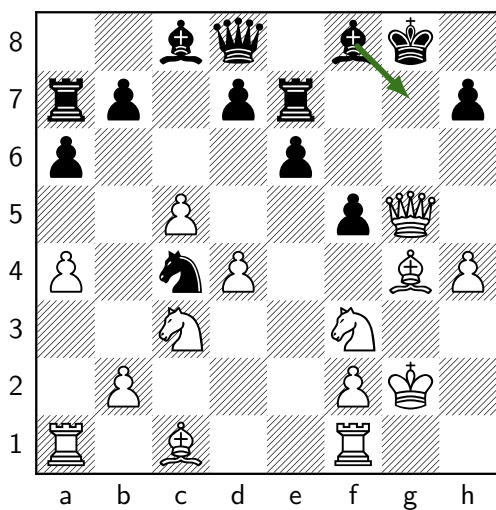


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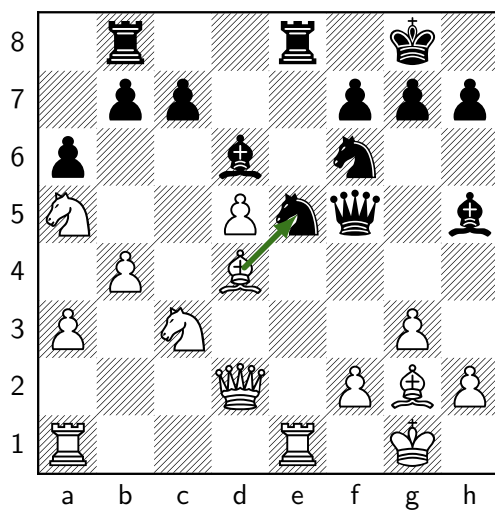


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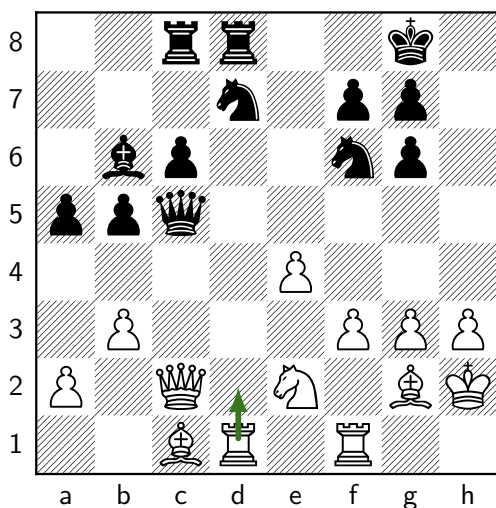
Figure 20: **Move 27, Black: f5**. (a) Our position; Melanie plays f5 (f6f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2560` (ply 54) [structural distance $d=105$, QAP obj=78]: the master played Bf1; under μ our Pf3 plays the role of that game’s Bg2 (a **cross-type** correspondence); cross-type roles: our Nc4 as their Qd2, our Bf8 as their Rf3; transferred move f6f5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2420` (ply 60) [structural distance $d=111$, QAP obj=59]: the master played Kh1; under μ our Kg1 plays the role of that game’s Kg2; cross-type roles: our Qd8 as their Rg1, our Re7 as their Na2; transferred move g8g7. (d) Retrieved from `ccr1_top_p0000#2587` (ply 38) [structural distance $d=100$, QAP obj=47]: the master played Rae1; under μ our Pd2 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our pa4 as their Rf1, our ra1 as their Qd2; transferred move d7d6. Synthesis: 1 retrieved analog transferred to f5 (summed QAP objective 78, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



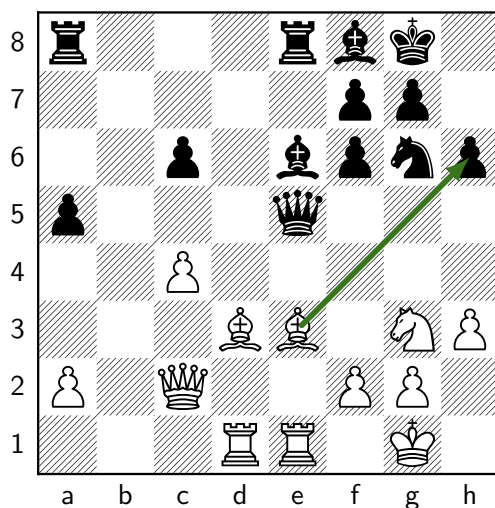
(a)



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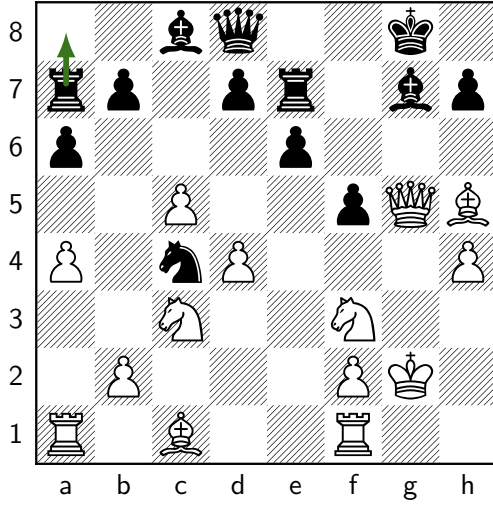


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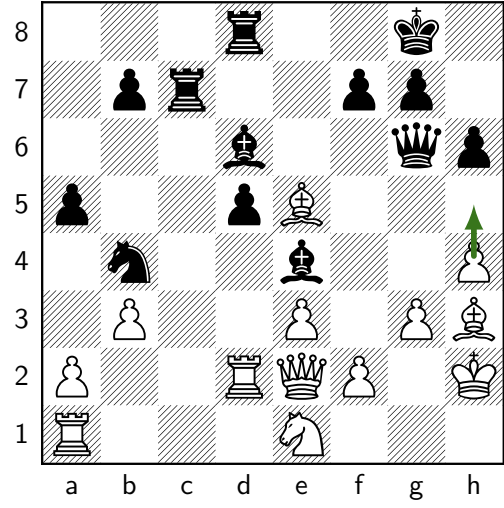


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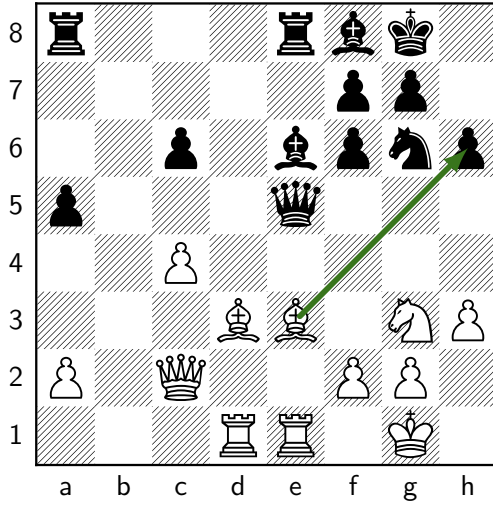
Figure 21: **Move 28, Black: Bg7.** (a) Our position; Melanie plays Bg7 (f8g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1463` (ply 46) [structural distance $d=119$, QAP obj=11]: the master played Bxe5; under μ our Bf1 plays the role of that game’s Bd4; cross-type roles: our bc1 as their Ra1, our qg5 as their Re1; transferred move f8g7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1483` (ply 52) [structural distance $d=121$, QAP obj=5]: the master played Rd2; under μ our Bf1 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our Qd8 as their Bc1, our Bc8 as their Rf1; transferred move f8g7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#133` (ply 44) [structural distance $d=120$, QAP obj=23]: the master played Bxh6; under μ our Re2 plays the role of that game’s Be3 (a **cross-type** correspondence); cross-type roles: our Bf8 as their Rd1, our Qd8 as their Re1; transferred move e7g7. Synthesis: 2 retrieved analogs transferred to Bg7 (summed QAP objective 15, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



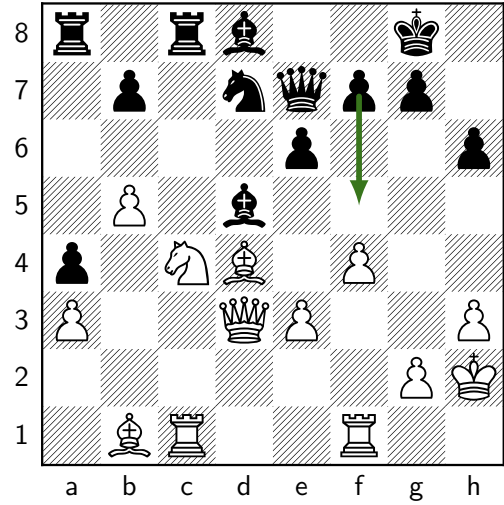
(a)



(b)

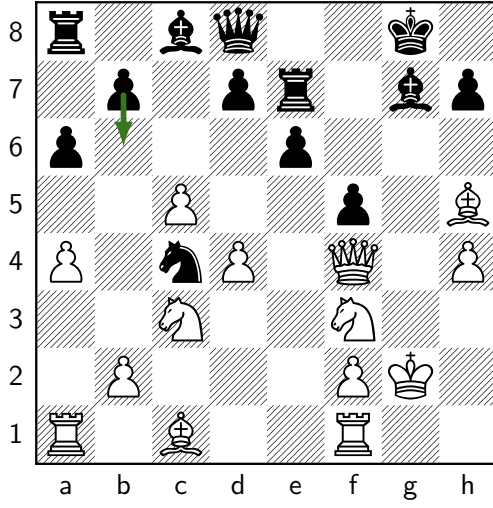


(c)

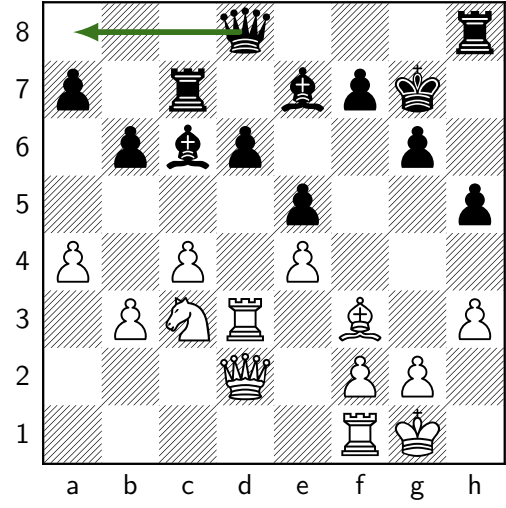


(d)

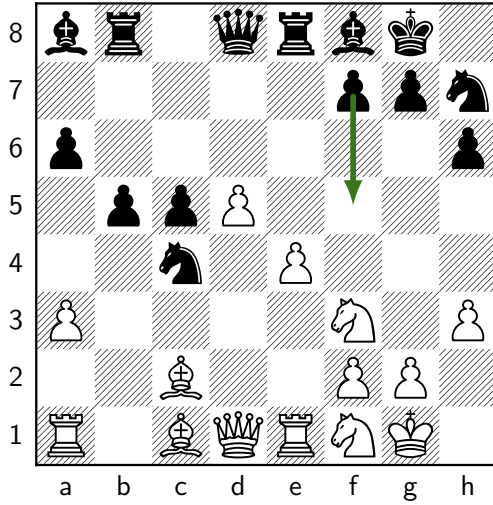
Figure 22: **Move 29, Black: Ra8.** (a) Our position; Melanie plays Ra8 (a7a8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1.top_p0001#1989` (ply 46) [structural distance $d=114$, QAP obj=38]: the master played h5; under μ our *Ra2* plays the role of that game’s *Ph4* (a **cross-type** correspondence); cross-type roles: our *bc1* as their *Ra1*, our *pb2* as their *Ne1*; transferred move a7a8 (**endorses this move**). (c) Retrieved from `ccr1.top_p0000#133` (ply 44) [structural distance $d=115$, QAP obj=90]: the master played Bxh6; under μ our *Re2* plays the role of that game’s *Be3* (a **cross-type** correspondence); cross-type roles: our *Bc8* as their *Rd1*, our *Qd8* as their *Re1*; transferred move e7e8. (d) Retrieved from `ccr1.top_p0001#2360` (ply 51) [structural distance $d=119$, QAP obj=31]: the master played f5; under μ our *Ph2* plays the role of that game’s *Pf2*; cross-type roles: our *bc1* as their *Ra8*, our *qg5* as their *Rc8*; transferred move h7h6. Synthesis: 1 retrieved analog transferred to Ra8 (summed QAP objective 38, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



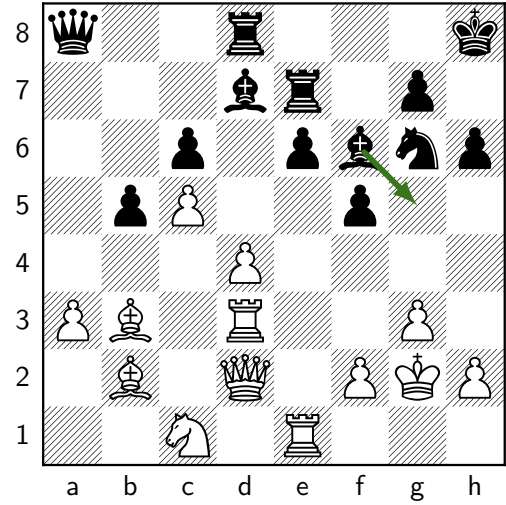
(a)



(b)

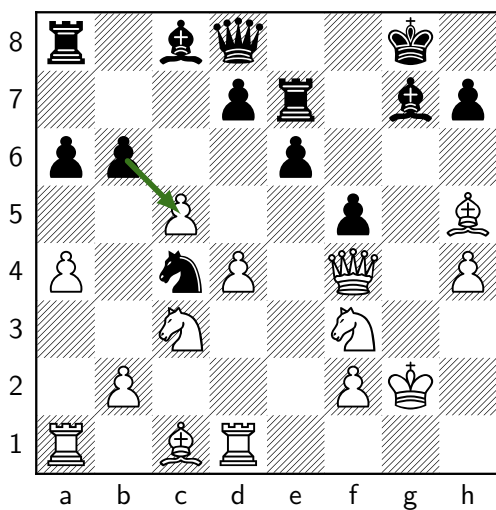


(c)

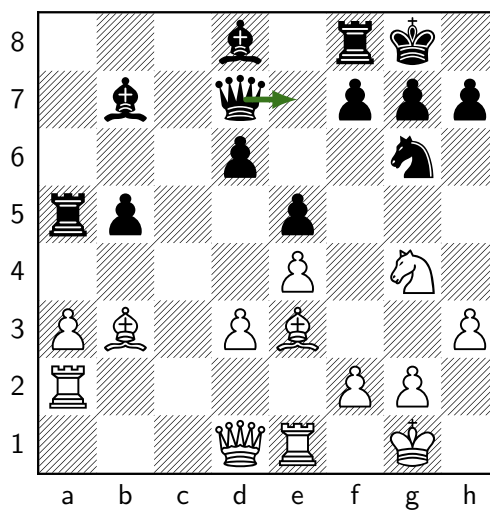


(d)

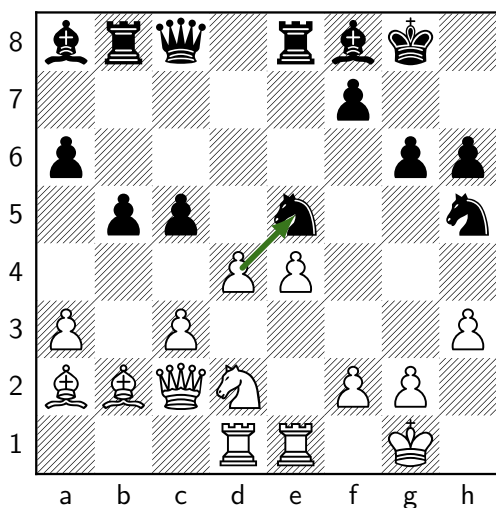
Figure 23: **Move 30, Black: b6.** (a) Our position; Melanie plays b6 (b7b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3490` (ply 75) [structural distance $d=113$, QAP obj=51]: the master played Qa8; under μ our *Pb2* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *ph4* as their *Rh8*, our *bc1* as their *Rc7*; transferred move b7b6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#201` (ply 45) [structural distance $d=113$, QAP obj=51]: the master played f5; under μ our *Qd1* plays the role of that game’s *Pf2* (a **cross-type** correspondence); cross-type roles: our *Bc8* as their *Rb8*, our *Bg7* as their *Qd8*; transferred move d8c7. (d) Retrieved from `ccr1_top_p0002#3841` (ply 75) [structural distance $d=115$, QAP obj=37]: the master played Bg5; under μ our *Bg2* plays the role of that game’s *Bf3*; cross-type roles: our *Ra8* as their *Qa8*, our *Bc8* as their *Rd8*; transferred move g7h6. Synthesis: 1 retrieved analog transferred to b6 (summed QAP objective 51, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



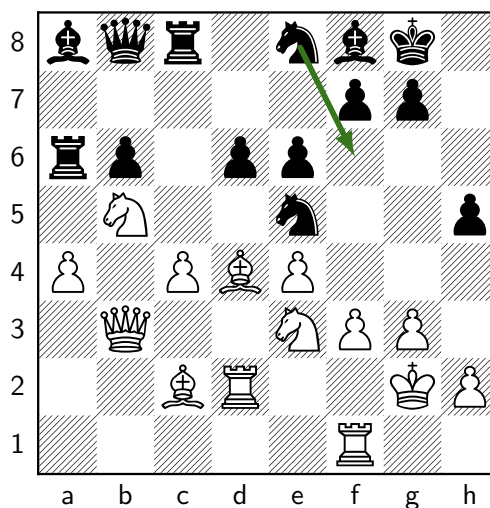
(a)



(b)

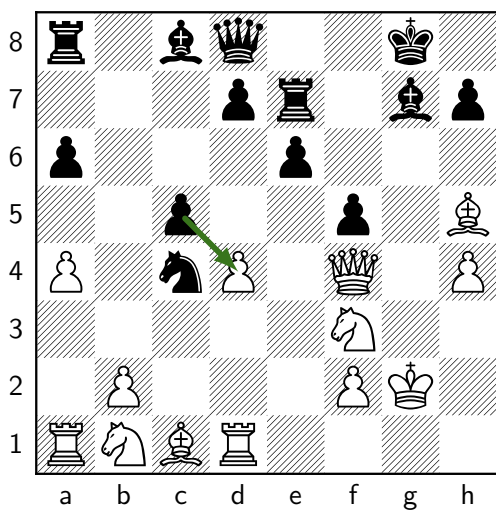


(c)

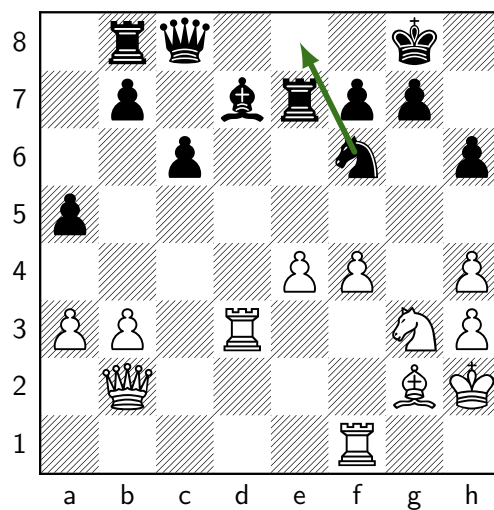


(d)

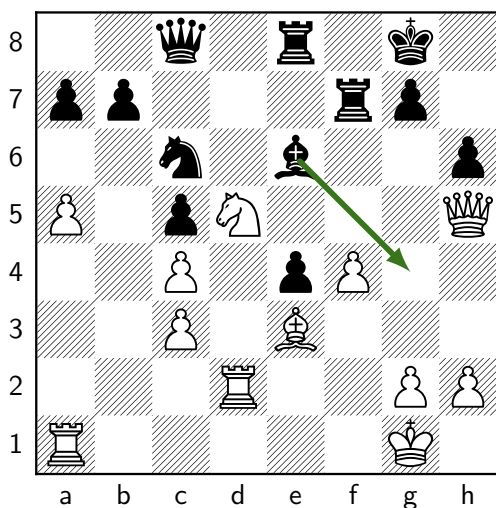
Figure 24: **Move 31, Black: bxc5**. (a) Our position; Melanie plays bxc5 (b6c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#181` (ply 45) [structural distance $d=113$, QAP obj=58]: the master played Qe7; under μ our *Pb3* plays the role of that game’s *Qd2* (a **cross-type** correspondence); cross-type roles: our *Re7* as their *Bd8*, our *Qd8* as their *Rf8*; transferred move b6c5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#971` (ply 46) [structural distance $d=100$, QAP obj=29]: the master played dxe5; under μ our *Pb3* plays the role of that game’s *Pd4*; cross-type roles: our *ph4* as their *Re1*, our *Pf5* as their *Bb2*; transferred move b6c5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3306` (ply 71) [structural distance $d=119$, QAP obj=25]: the master played Nf6; under μ our *Pb3* plays the role of that game’s *Ne1* (a **cross-type** correspondence); cross-type roles: our *Nc4* as their *Ba8*, our *Re7* as their *Qb8*; transferred move b6c5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to bxc5 (summed QAP objective 112, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



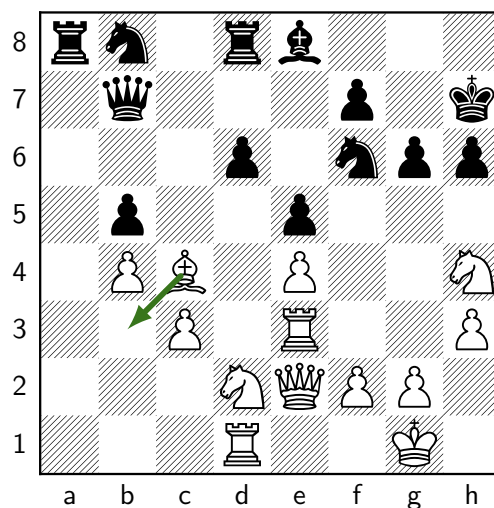
(a)



(b)

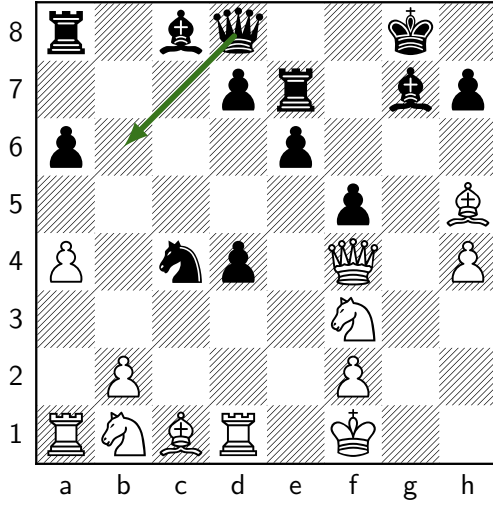


(c)

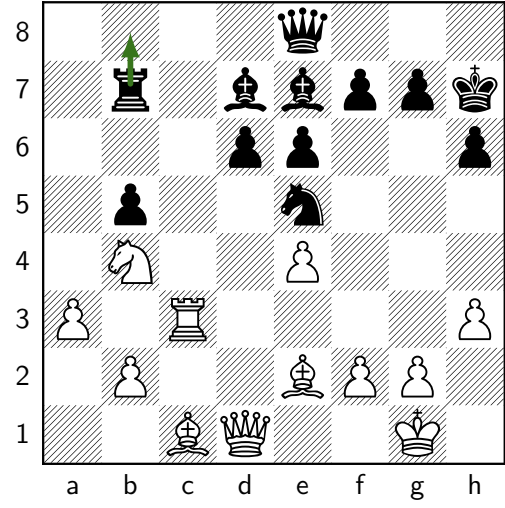


(d)

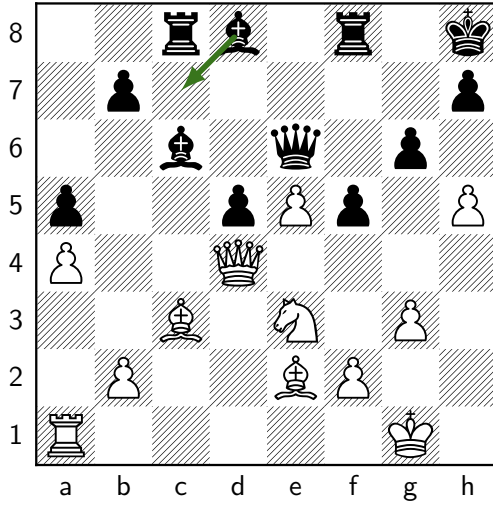
Figure 25: **Move 32, Black: cxd4**. (a) Our position; Melanie plays cxd4 (c5d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#1323 (ply 53) [structural distance $d=111$, QAP obj=43]: the master played Ne8; under μ our Pc4 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Rb8, our Pf5 as their Bd7; transferred move c5d4 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#2562 (ply 53) [structural distance $d=123$, QAP obj=20]: the master played Bg4; under μ our Pc4 plays the role of that game’s Be3 (a **cross-type** correspondence); cross-type roles: our Pd7 as their Qc8, our Qd8 as their Re8; transferred move c5d4 (**endorses this move**). (d) Retrieved from ccr1_top_p0001#915 (ply 62) [structural distance $d=121$, QAP obj=16]: the master played Bb3; under μ our Pc4 plays the role of that game’s Bc4 (a **cross-type** correspondence); cross-type roles: our pa4 as their Rd1, our Bg7 as their Nd2; transferred move c5d4 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to cxd4 (summed QAP objective 99, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



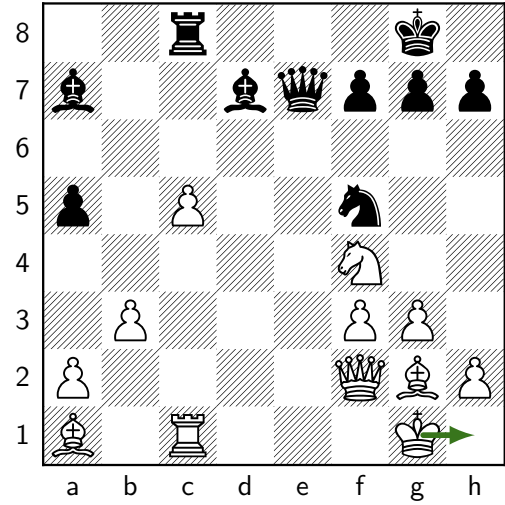
(a)



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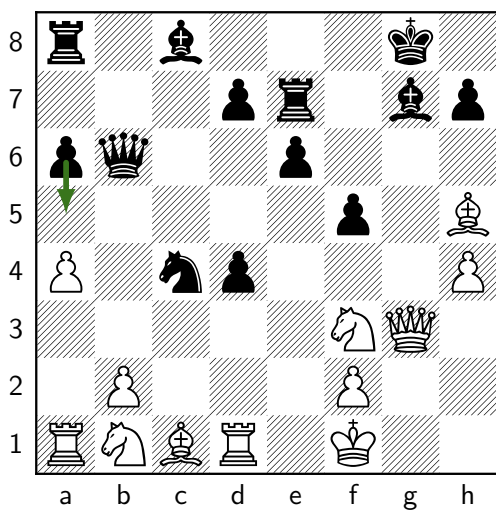


(c)

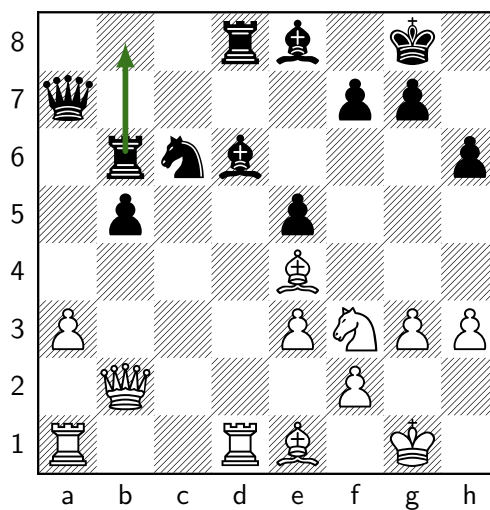


(d)

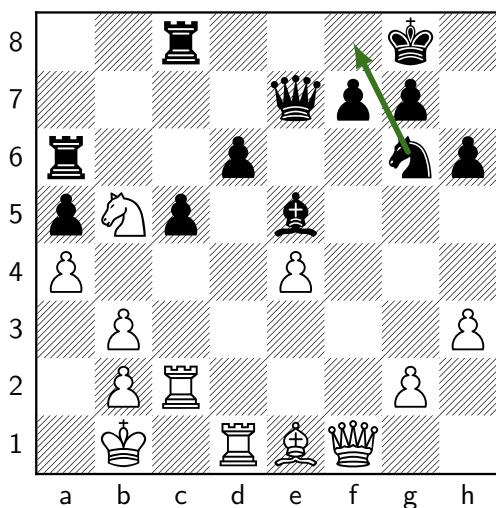
Figure 26: **Move 33, Black: Qb6.** (a) Our position; Melanie plays Qb6 (d8b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2540` (ply 59) [structural distance $d=127$, QAP obj=20]: the master played Rb8; under μ our Qd1 plays the role of that game’s Rb2 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Qe8, our pa4 as their Bd7; transferred move d8b6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2731` (ply 59) [structural distance $d=121$, QAP obj=100]: the master played Bc7; under μ our Pe3 plays the role of that game’s Bd1 (a **cross-type** correspondence); cross-type roles: our ph4 as their Rc8, our Qd8 as their Rf8; transferred move e6e5. (d) Retrieved from `ccr1_top_p0000#2577` (ply 54) [structural distance $d=129$, QAP obj=85]: the master played Kh1; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our Pa6 as their Ba1, our Qd8 as their Rc1; transferred move g8h8. Synthesis: 1 retrieved analog transferred to Qb6 (summed QAP objective 20, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



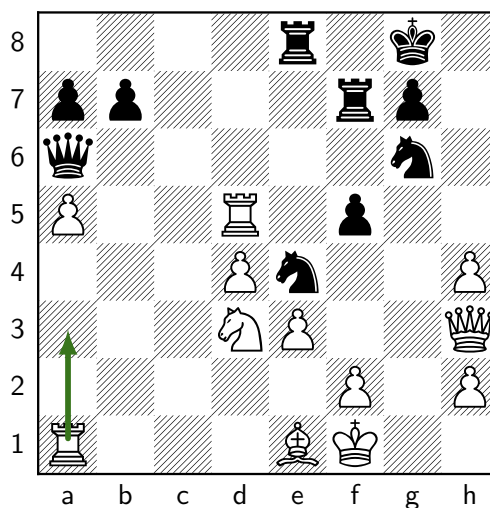
(a)



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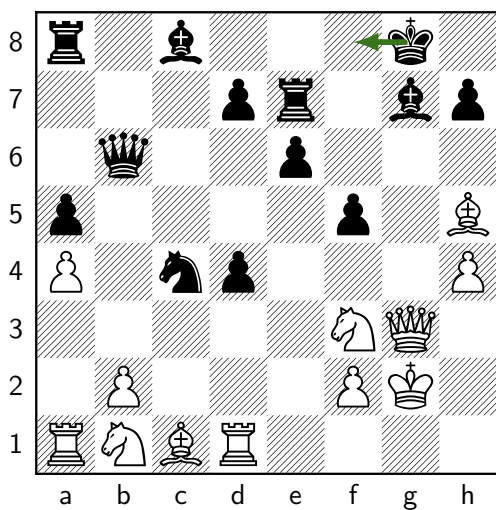


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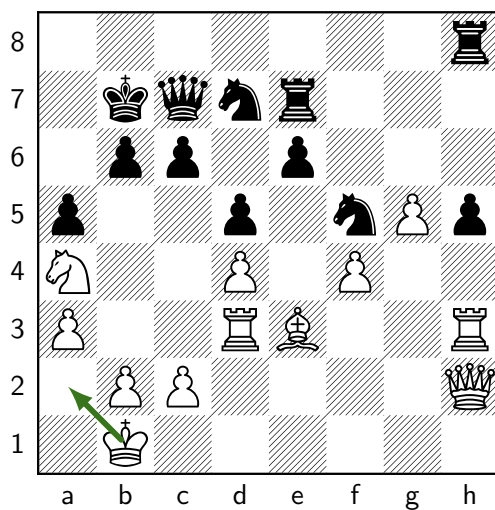


(d)

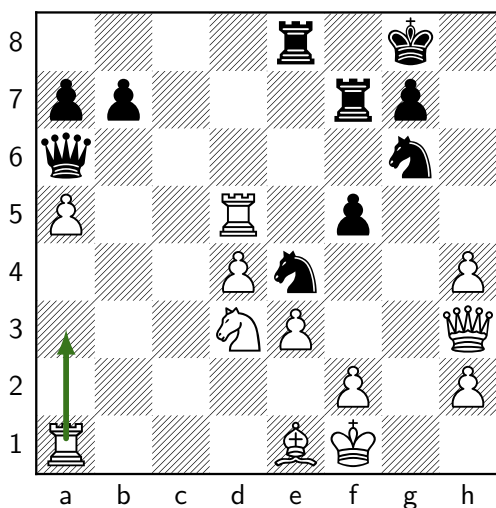
Figure 27: **Move 34, Black: a5.** (a) Our position; Melanie plays a5 (a6a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#2831` (ply 73) [structural distance $d=117$, QAP obj=51]: the master played Rbb8; under μ our *Pa3* plays the role of that game’s *Rb3* (a **cross-type** correspondence); cross-type roles: our *Ph7* as their *Rd8*, our *pa4* as their *Be8*; transferred move a6a5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#3408` (ply 67) [structural distance $d=109$, QAP obj=50]: the master played Nf8; under μ our *Pa3* plays the role of that game’s *Ng3* (a **cross-type** correspondence); cross-type roles: our *ra1* as their *Pg7*, our *bh5* as their *Ra6*; transferred move a6a5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#2750` (ply 54) [structural distance $d=112$, QAP obj=95]: the master played Ra3; under μ our *Re2* plays the role of that game’s *Ra1*; cross-type roles: our *Pe6* as their *Be1*, our *Bg7* as their *Nd3*; transferred move e7e8. Synthesis: 2 retrieved analogs transferred to a5 (summed QAP objective 100, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



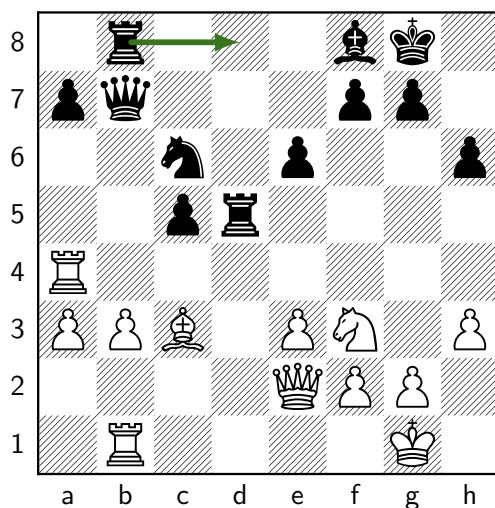
(a)



(b)

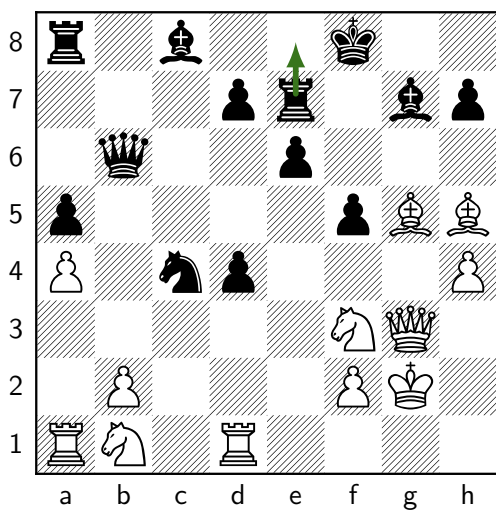


(c)

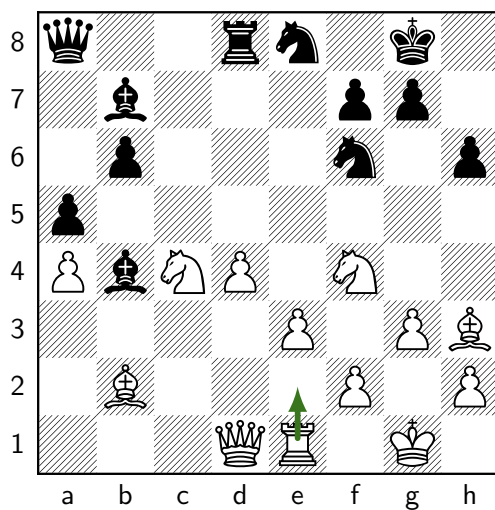


(d)

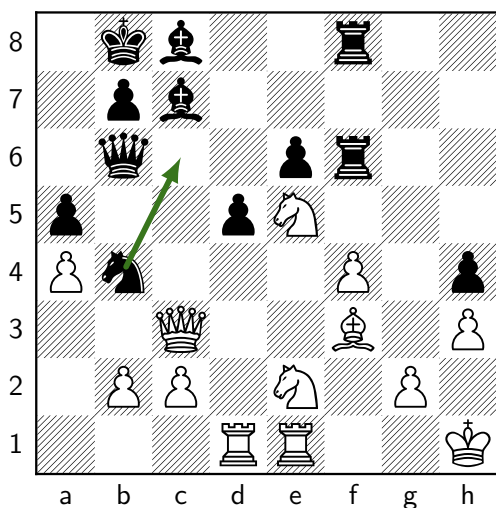
Figure 28: **Move 35, Black: Kf8**. (a) Our position; Melanie plays Kf8 (g8f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3315` (ply 58) [structural distance $d=132$, QAP obj=43]: the master played Ka2; under μ our Kg1 plays the role of that game's Kb1; cross-type roles: our bc1 as their Qh2, our bh5 as their Pa3; transferred move g8f8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2750` (ply 54) [structural distance $d=121$, QAP obj=92]: the master played Ra3; under μ our Re2 plays the role of that game's Ra1; cross-type roles: our Ph7 as their Be1, our Bg7 as their Pf2; transferred move e7e8. (d) Retrieved from `ccr1_top_p0001#461` (ply 83) [structural distance $d=128$, QAP obj=34]: the master played Rbd8; under μ our Bc1 plays the role of that game's Rb1 (a **cross-type** correspondence); cross-type roles: our Ph7 as their Bf8, our Bg7 as their Pg7; transferred move c8a6. Synthesis: 1 retrieved analog transferred to Kf8 (summed QAP objective 43, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



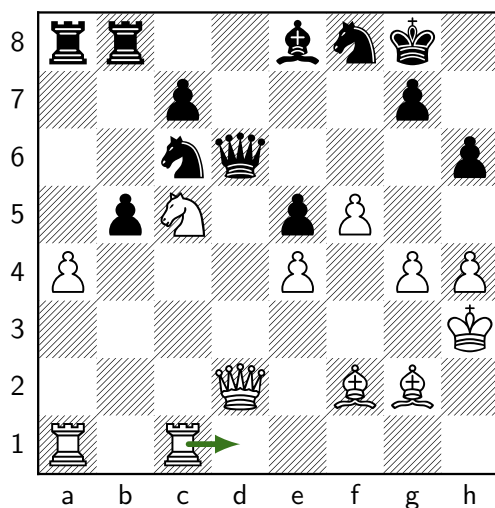
(a)



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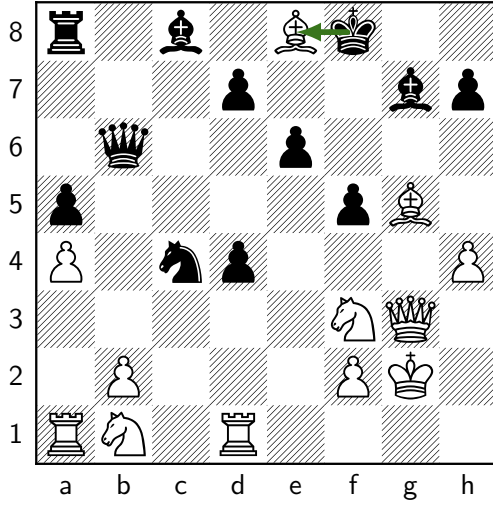


(c)

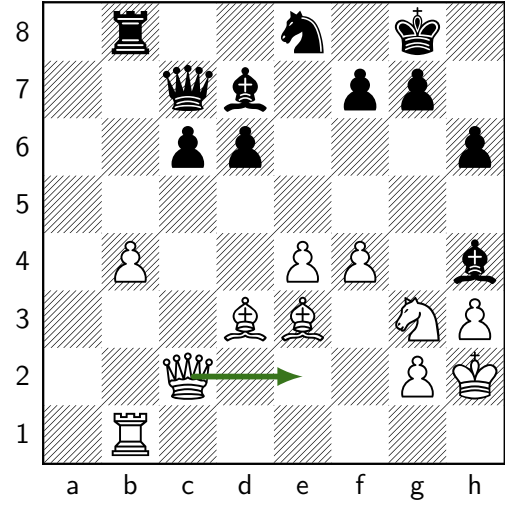


(d)

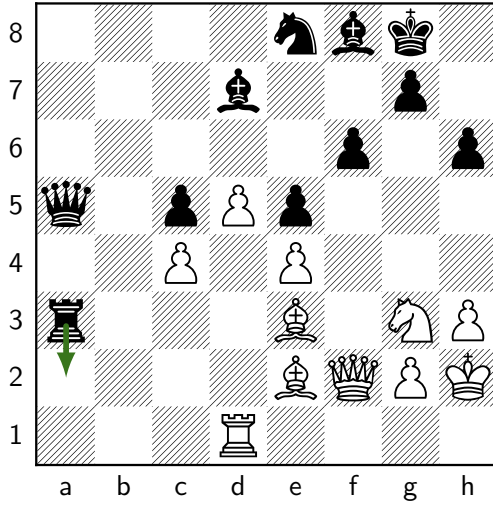
Figure 29: **Move 36, Black: Re8.** (a) Our position; Melanie plays Re8 (e7e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3669` (ply 54) [structural distance $d=118$, QAP obj=37]: the master played Re2; under μ our Re2 plays the role of that game’s Re1; cross-type roles: our Ra8 as their Ph2, our Nc4 as their Pe3; transferred move e7e8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#598` (ply 79) [structural distance $d=128$, QAP obj=34]: the master played Nc6; under μ our Re2 plays the role of that game’s Nb5 (a **cross-type** correspondence); cross-type roles: our Pa5 as their Bc8, our Pe6 as their Rf8; transferred move e7e8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#579` (ply 68) [structural distance $d=130$, QAP obj=53]: the master played Rd1; under μ our Pe3 plays the role of that game’s Rc1 (a **cross-type** correspondence); cross-type roles: our Re7 as their Qd2, our Pd7 as their Bf2; transferred move e6e5. Synthesis: 2 retrieved analogs transferred to Re8 (summed QAP objective 70, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



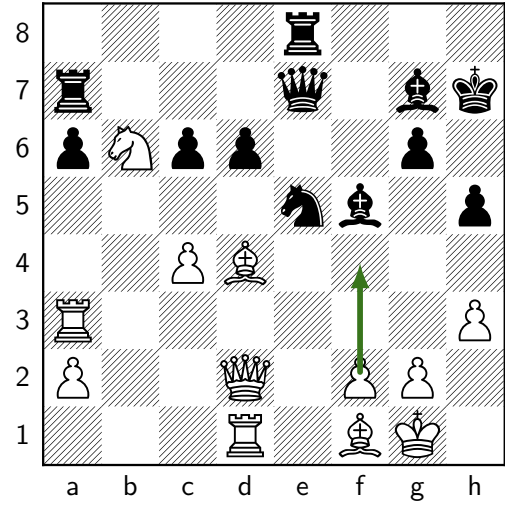
(a)



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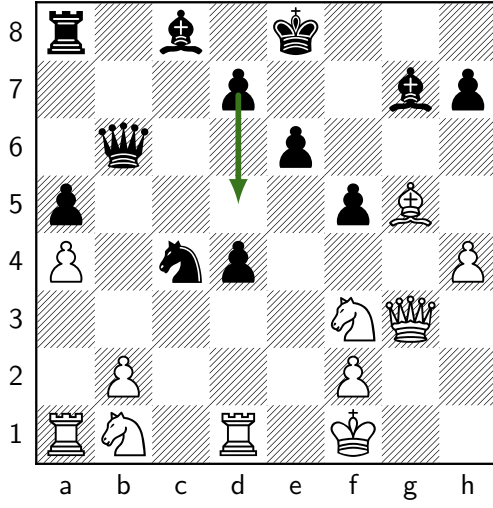


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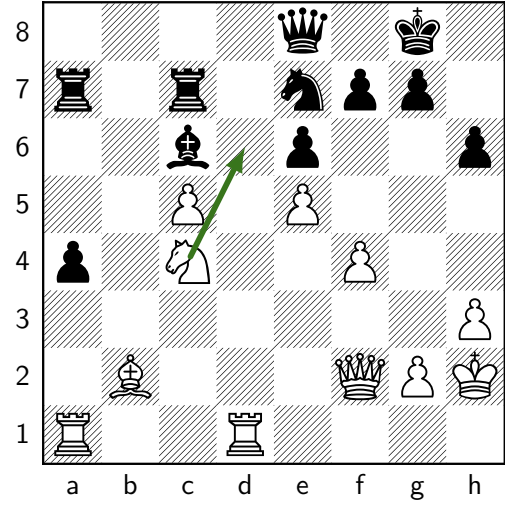


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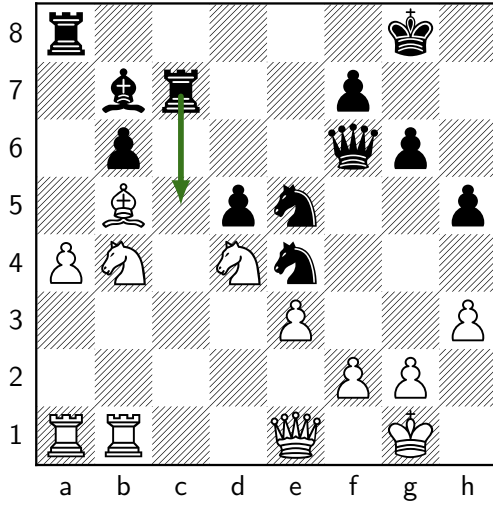
Figure 30: **Move 37, Black: Kxe8.** (a) Our position; Melanie plays Kxe8 (f8e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#169` (ply 76) [structural distance $d=108$, QAP obj=44]: the master played Qe2; under μ our Qb3 plays the role of that game’s Qc2; cross-type roles: our Bc8 as their Pg2, our Pd4 as their Bd3; transferred move b6a6. (c) Retrieved from `ccrl_top_p0001#2201` (ply 89) [structural distance $d=108$, QAP obj=44]: the master played Ra2; under μ our Kf1 plays the role of that game’s Ra6 (a **cross-type** correspondence); cross-type roles: our rd1 as their Ne8, our ra1 as their Pg7; transferred move f8g8. (d) Retrieved from `ccrl_top_p0003#1298` (ply 60) [structural distance $d=111$, QAP obj=42]: the master played f4; under μ our Ph2 plays the role of that game’s Pf2; cross-type roles: our ph4 as their Rd1, our Nc4 as their Qd2; transferred move h7h6. Synthesis: no retrieved analog transferred to Kxe8 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



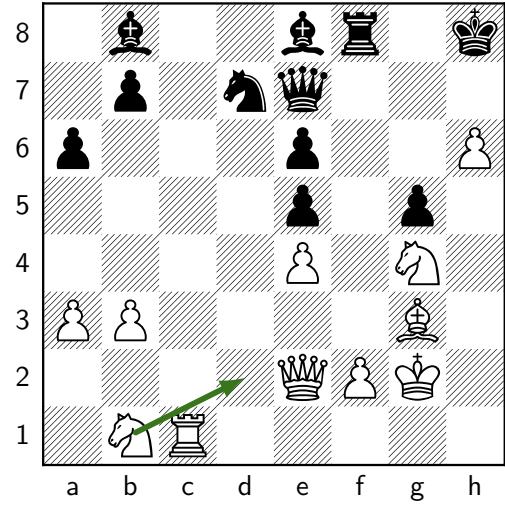
(a)



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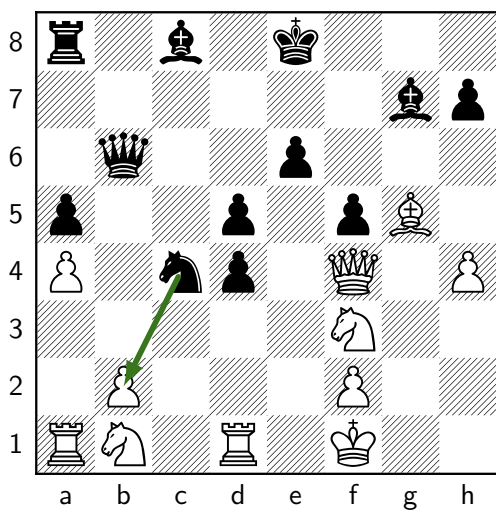


(c)

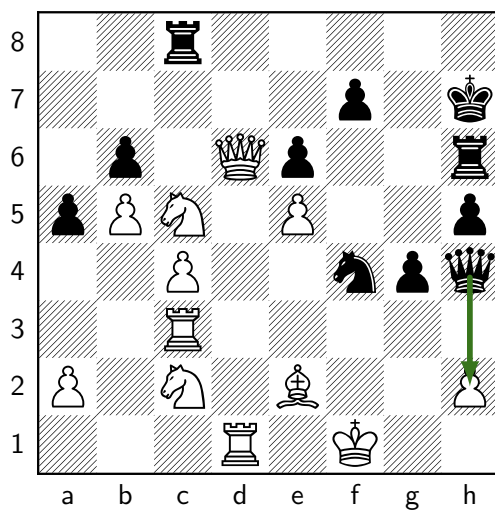


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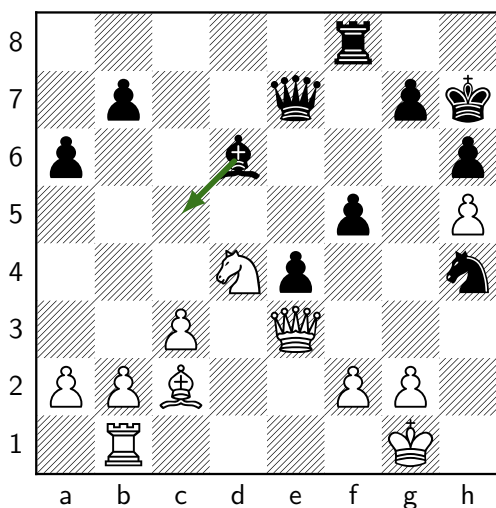
Figure 31: **Move 38, Black: d5**. (a) Our position; Melanie plays d5 (d7d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#327` (ply 88) [structural distance $d=103$, QAP obj=41]: the master played Nd6; under μ our *Pd2* plays the role of that game’s *Nc4* (a **cross-type** correspondence); cross-type roles: our *Qb6* as their *Ra1*, our *Nc4* as their *Rd1*; transferred move d7d5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#703` (ply 87) [structural distance $d=95$, QAP obj=76]: the master played Rc5; under μ our *Ra1* plays the role of that game’s *Rc2*; cross-type roles: our *Qb6* as their *Ra8*, our *ph4* as their *Bb7*; transferred move a8a6. (d) Retrieved from `ccrl_top_p0000#743` (ply 80) [structural distance $d=94$, QAP obj=66]: the master played Nd2; under μ our *Bg2* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *Nc4* as their *Qe2*, our *Qb6* as their *Bg3*; transferred move g7h8. Synthesis: 1 retrieved analog transferred to d5 (summed QAP objective 41, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



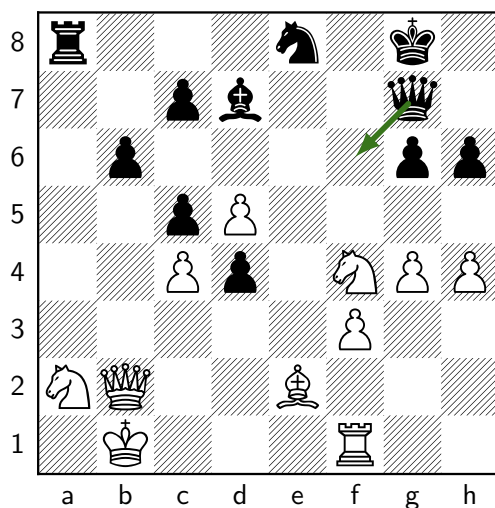
(a)



(b)

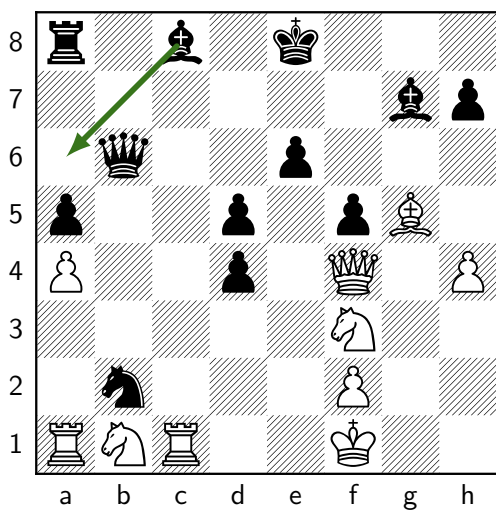


(c)

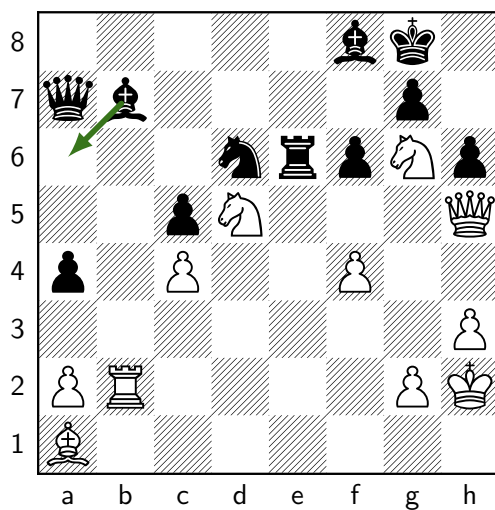


(d)

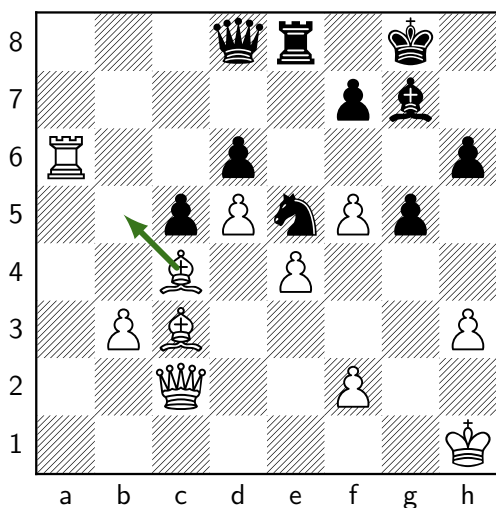
Figure 32: **Move 39, Black: Nxb2.** (a) Our position; Melanie plays Nxb2 (c4b2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#2274 (ply 93) [structural distance $d=108$, QAP obj=52]: the master played Qxh2; under μ our Nc5 plays the role of that game’s Qh5 (a **cross-type** correspondence); cross-type roles: our ph4 as their Rc8, our Bg7 as their Rh6; transferred move c4b2 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#1809 (ply 69) [structural distance $d=116$, QAP obj=75]: the master played Bc5; under μ our Ra1 plays the role of that game’s Bd3 (a **cross-type** correspondence); cross-type roles: our Bg7 as their Rf8, our bg5 as their pc3; transferred move a8a7. (d) Retrieved from ccr1_top_p0001#459 (ply 63) [structural distance $d=108$, QAP obj=65]: the master played Qf6; under μ our Qb3 plays the role of that game’s Qg2; cross-type roles: our Nc4 as their Ra8, our Pa5 as their Ne8; transferred move b6c5. Synthesis: 1 retrieved analog transferred to Nxb2 (summed QAP objective 52, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



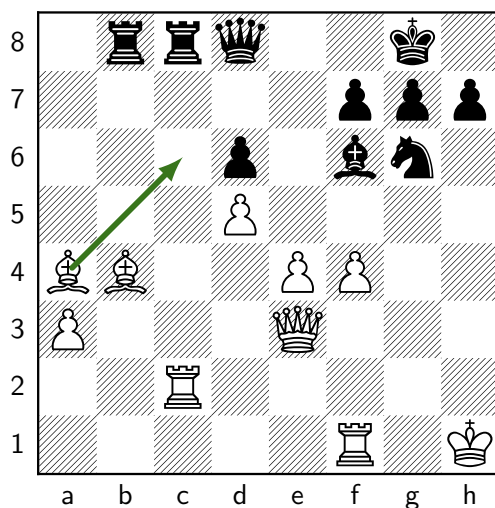
(a)



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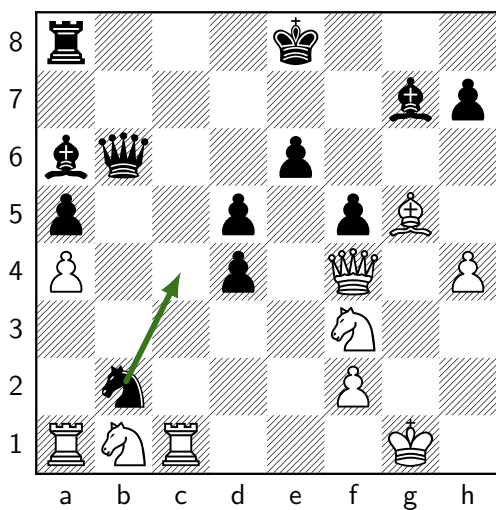


(c)

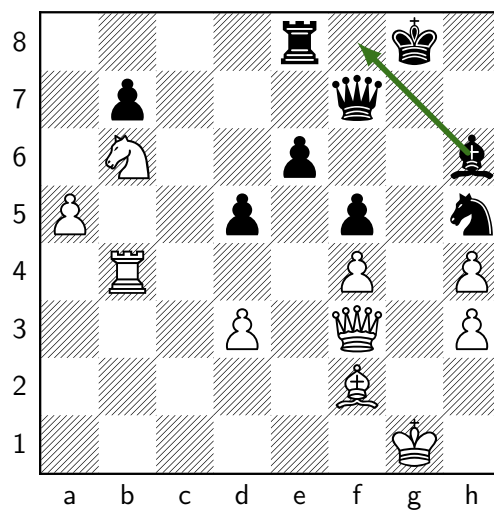


(d)

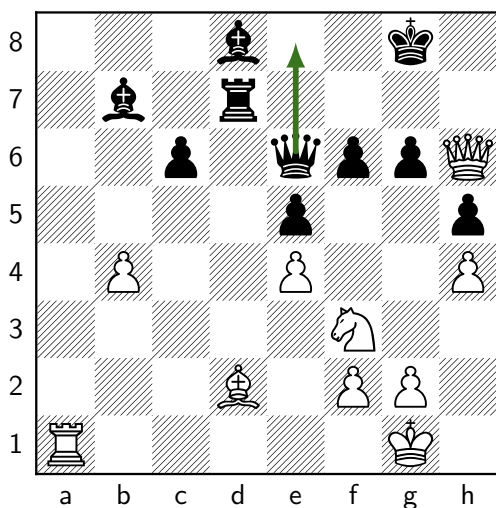
Figure 33: **Move 40, Black: Ba6+.** (a) Our position; Melanie plays Ba6+ (c8a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2825` (ply 67) [structural distance $d=108$, QAP obj=39]: the master played Ba6; under μ our Bc1 plays the role of that game’s Bb2; cross-type roles: our Pf5 as their Bf8, our Pd5 as their Re6; transferred move c8a6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2461` (ply 72) [structural distance $d=111$, QAP obj=20]: the master played Bb5; under μ our Bc1 plays the role of that game’s Bc4; cross-type roles: our pf2 as their Qc2, our Qb6 as their Pb3; transferred move c8a6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2498` (ply 86) [structural distance $d=103$, QAP obj=40]: the master played Bc6; under μ our Bg2 plays the role of that game’s Ba4; cross-type roles: our Pf5 as their Rc2, our nb1 as their Pa3; transferred move g7f6. Synthesis: 2 retrieved analogs transferred to Ba6+ (summed QAP objective 59, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



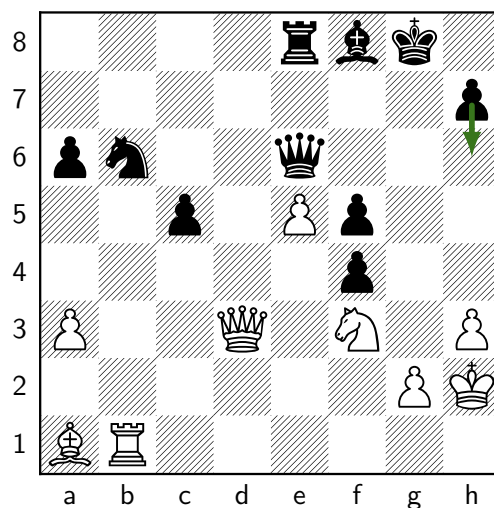
(a)



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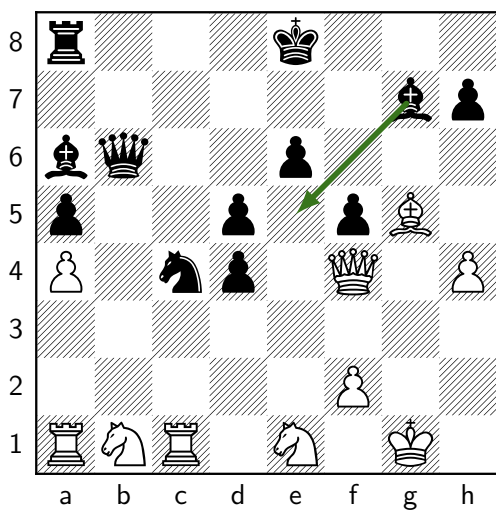


(c)

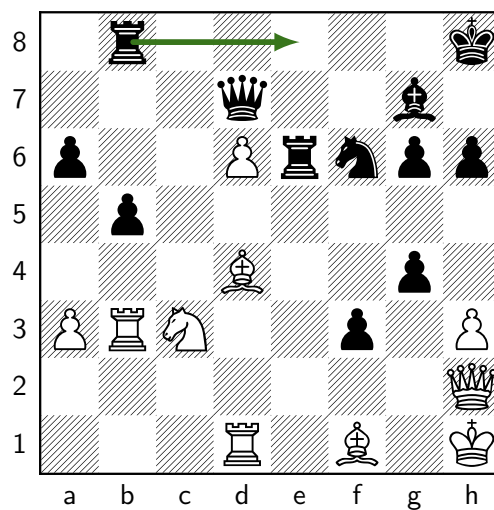


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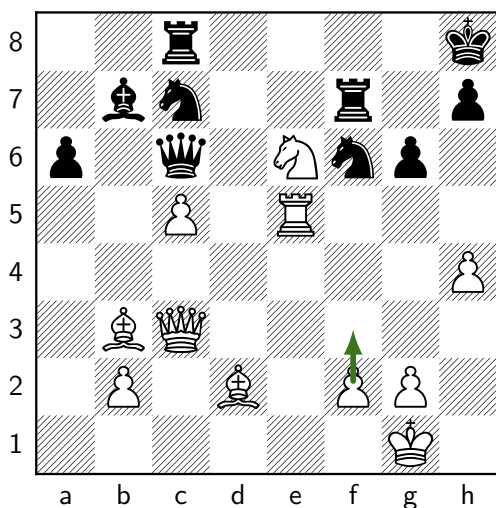
Figure 34: **Move 41, Black: Nc4.** (a) Our position; Melanie plays Nc4 (b2c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#711` (ply 93) [structural distance $d=107$, QAP obj=39]: the master played Bf8; under μ our Nb7 plays the role of that game’s Bh3 (a **cross-type** correspondence); cross-type roles: our Bg7 as their Pe6, our Pf5 as their Nh5; transferred move b2c4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3719` (ply 79) [structural distance $d=114$, QAP obj=58]: the master played Qe8; under μ our Qb3 plays the role of that game’s Qe3; cross-type roles: our ra1 as their Bd8, our Nb2 as their Rd7; transferred move b6b7. (d) Retrieved from `ccr1_top_p0000#3875` (ply 91) [structural distance $d=116$, QAP obj=43]: the master played h6; under μ our Ra1 plays the role of that game’s Ph2 (a **cross-type** correspondence); cross-type roles: our bg5 as their Re8, our Pe6 as their Bf8; transferred move a8b8. Synthesis: 1 retrieved analog transferred to Nc4 (summed QAP objective 39, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



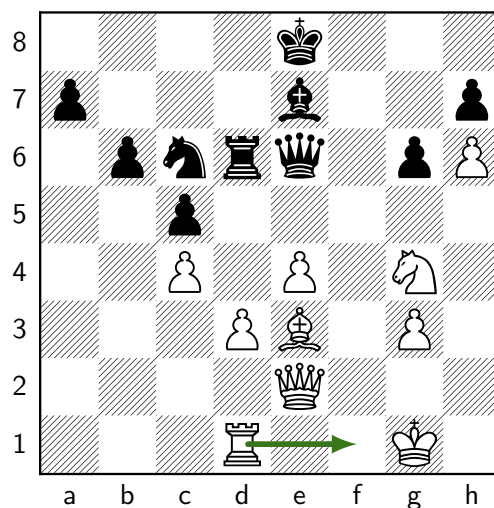
(a)



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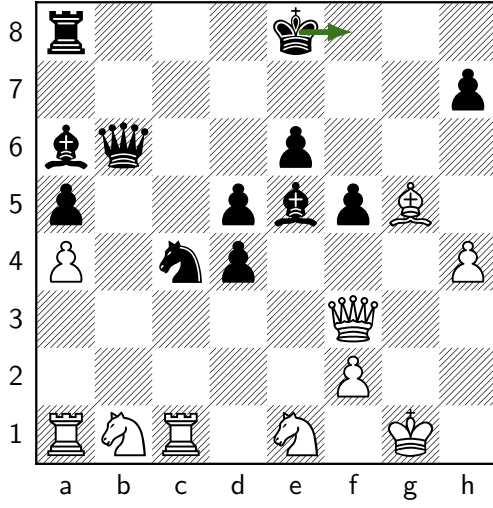


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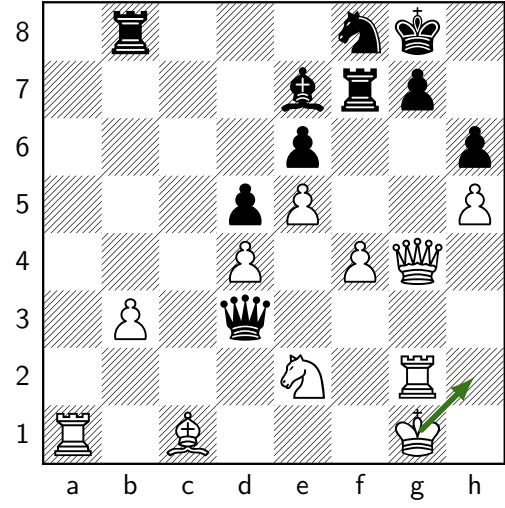


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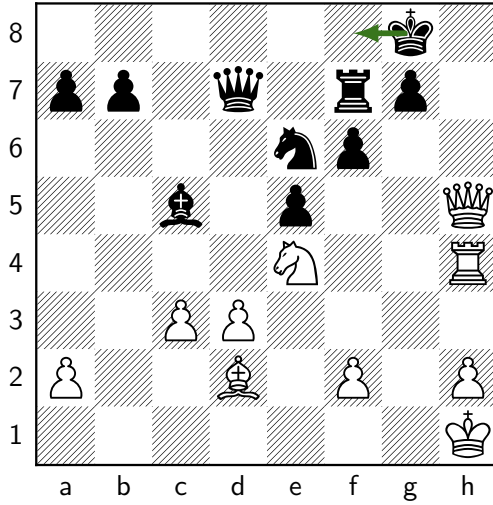
Figure 35: **Move 42, Black: Be5.** (a) Our position; Melanie plays Be5 (g7e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#271` (ply 77) [structural distance $d=100$, QAP obj=60]: the master played Rbe8; under μ our Bg2 plays the role of that game’s Rb1 (a **cross-type** correspondence); cross-type roles: our rc1 as their Qd7, our ra1 as their Bg7; transferred move g7e5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#6` (ply 76) [structural distance $d=85$, QAP obj=55]: the master played f3; under μ our Bg2 plays the role of that game’s Pf2 (a **cross-type** correspondence); cross-type roles: our Ra8 as their Pb2, our Nc4 as their Bd2; transferred move g7e5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#164` (ply 92) [structural distance $d=103$, QAP obj=81]: the master played Rf1; under μ our Ra1 plays the role of that game’s Rd1; cross-type roles: our Bg7 as their Qe2, our Qb6 as their Be3; transferred move a8d8. Synthesis: 2 retrieved analogs transferred to Be5 (summed QAP objective 115, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



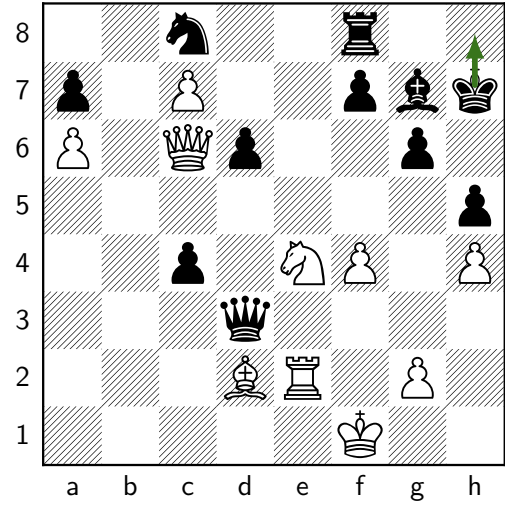
(a)



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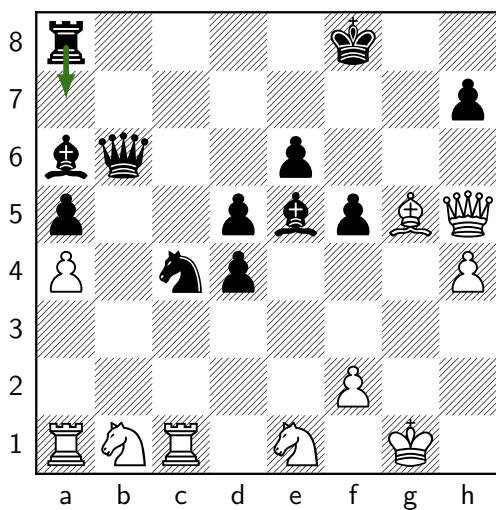


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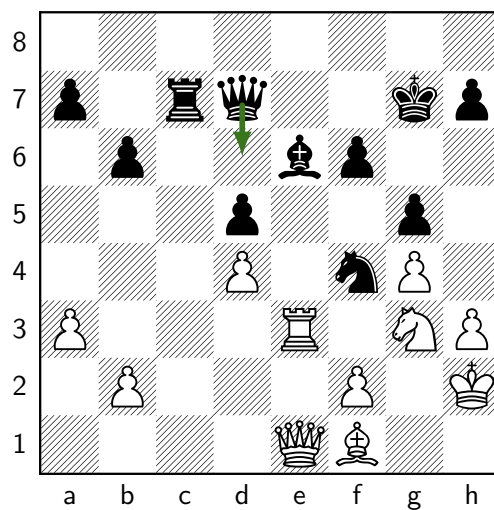


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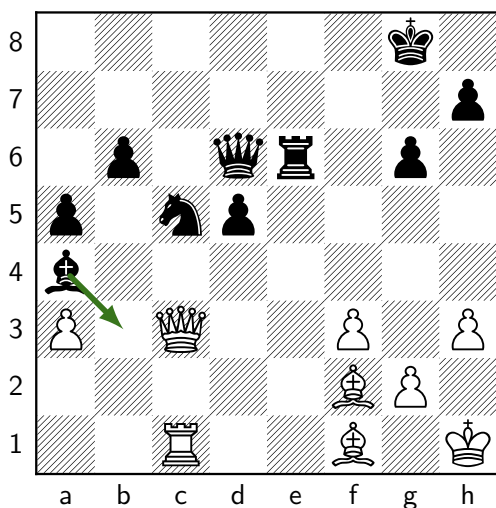
Figure 36: **Move 43, Black: Kf8.** (a) Our position; Melanie plays Kf8 (e8f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#591` (ply 84) [structural distance $d=126$, QAP obj=39]: the master played Kh2; under μ our *Ke1* plays the role of that game's *Kg1*; cross-type roles: our *pa4* as their *Bc1*, our *ne1* as their *Rg2*; transferred move e8f8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3725` (ply 69) [structural distance $d=113$, QAP obj=33]: the master played Kf8; under μ our *Ke1* plays the role of that game's *Kg1*; cross-type roles: our *Ra8* as their *Pb7*, our *Pf5* as their *Rf7*; transferred move e8f8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3731` (ply 97) [structural distance $d=127$, QAP obj=5]: the master played Kh8; under μ our *Ke1* plays the role of that game's *Kh2*; cross-type roles: our *pa4* as their *Nc8*, our *Ba6* as their *pc7*; transferred move e8f8 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kf8 (summed QAP objective 77, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



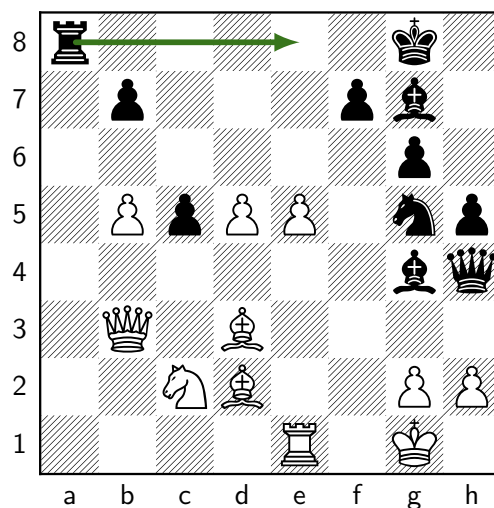
(a)



(b)

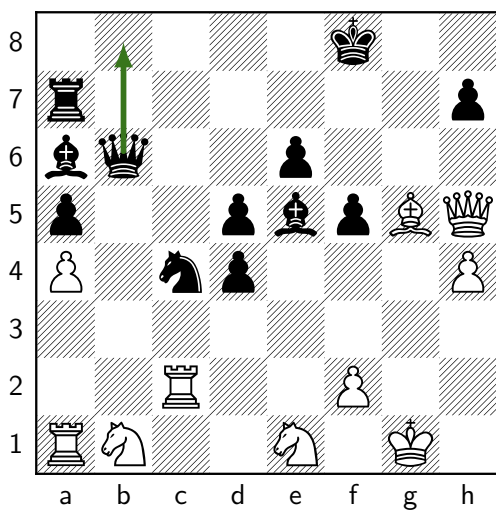


(c)

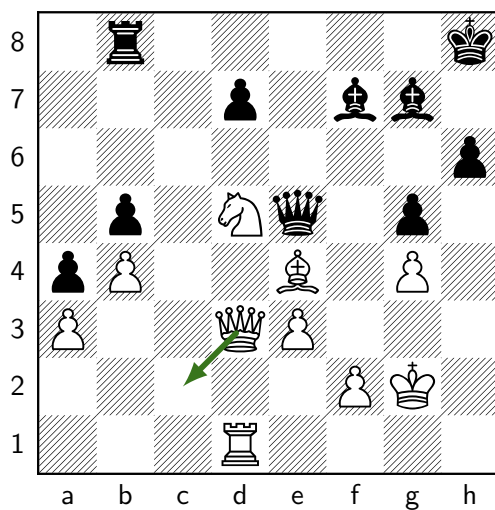


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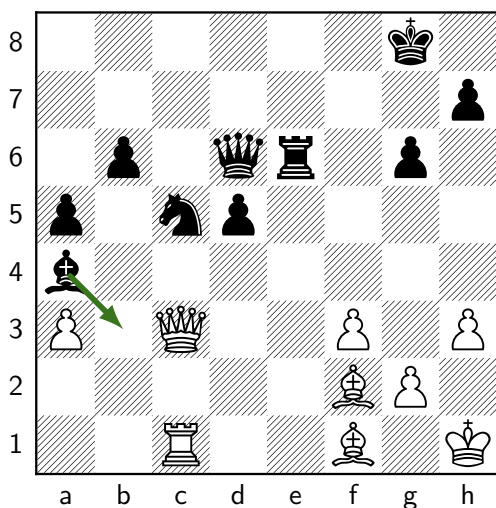
Figure 37: **Move 44, Black: Ra7.** (a) Our position; Melanie plays Ra7 (a8a7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#2963` (ply 81) [structural distance $d=100$, QAP obj=32]: the master played Qd6; under μ our Ra1 plays the role of that game’s Qd2 (a **cross-type** correspondence); cross-type roles: our qh5 as their Rc7, our bg5 as their Ph7; transferred move a8a7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#443` (ply 73) [structural distance $d=113$, QAP obj=33]: the master played Bb3; under μ our Be4 plays the role of that game’s Ba5; cross-type roles: our Ba6 as their Qd6, our Qb6 as their Re6; transferred move e5f6. (d) Retrieved from `ccrl_top_p0001#3988` (ply 71) [structural distance $d=120$, QAP obj=30]: the master played Re8; under μ our Ra1 plays the role of that game’s Ra1; cross-type roles: our qh5 as their Bg7, our ne1 as their Ph5; transferred move a8c8. Synthesis: 1 retrieved analog transferred to Ra7 (summed QAP objective 32, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



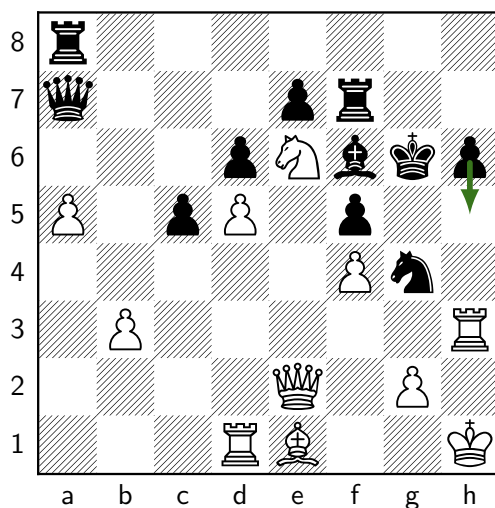
(a)



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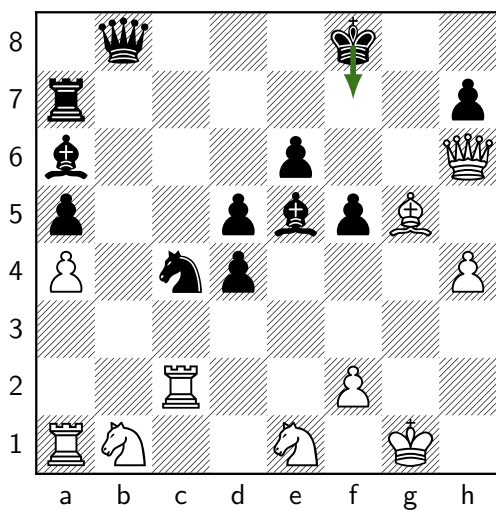


(c)

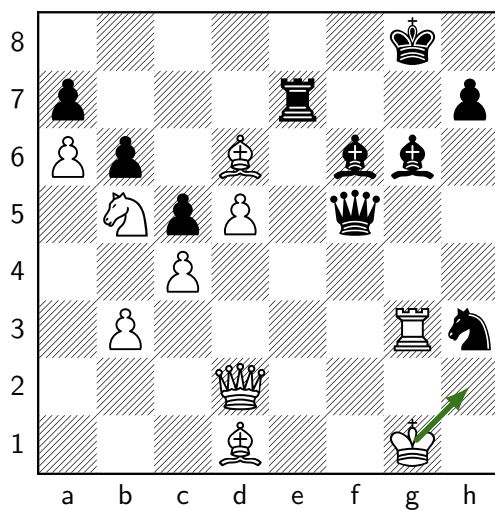


(d)

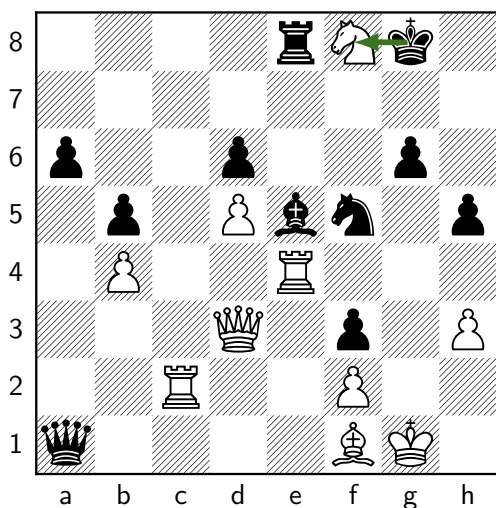
Figure 38: **Move 45, Black: Qb8.** (a) Our position; Melanie plays Qb8 (b6b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3580` (ply 88) [structural distance $d=108$, QAP obj=28]: the master played Qc2; under μ our Qb3 plays the role of that game's Qd3; cross-type roles: our Ba6 as their Rd1, our ra1 as their Pb4; transferred move b6b8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#443` (ply 73) [structural distance $d=94$, QAP obj=41]: the master played Bb3; under μ our Be4 plays the role of that game's Ba5; cross-type roles: our Ba6 as their Qd6, our Qb6 as their Re6; transferred move e5f6. (d) Retrieved from `ccr1_top_p0001#97` (ply 75) [structural distance $d=105$, QAP obj=34]: the master played h5; under μ our Pf4 plays the role of that game's Ph3; cross-type roles: our ph4 as their Ra8, our ne1 as their Rf7; transferred move f5f4. Synthesis: 1 retrieved analog transferred to Qb8 (summed QAP objective 28, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



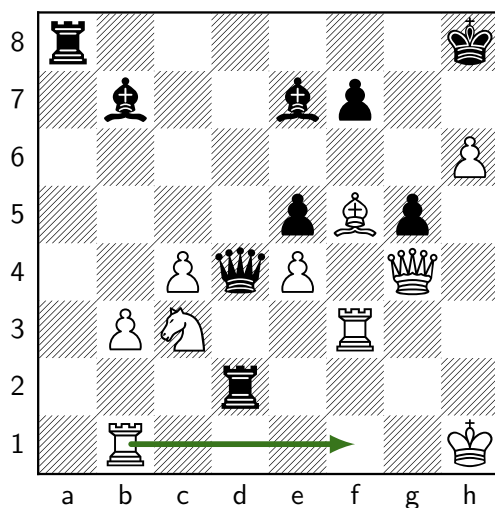
(a)



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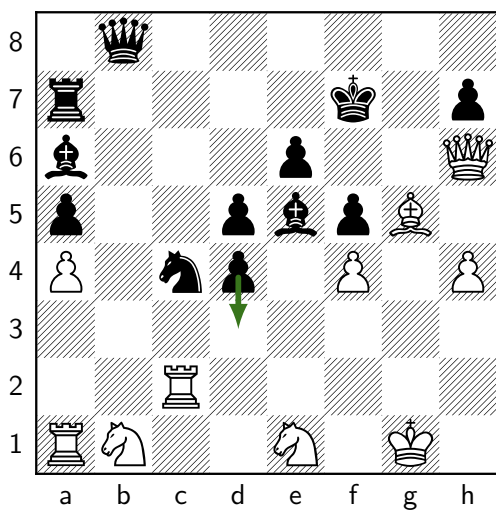


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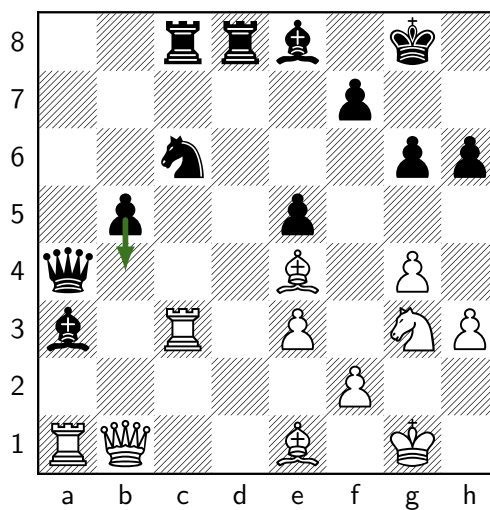


(d)

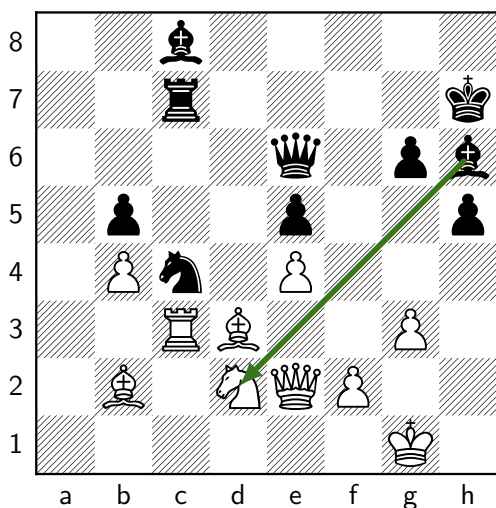
Figure 39: **Move 46, Black: Kf7**. (a) Our position; Melanie plays Kf7 (f8f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0002#2215` (ply 106) [structural distance $d=99$, QAP obj=72]: the master played Kh2; under μ our Kf1 plays the role of that game’s Kg1; cross-type roles: our Pf5 as their Bd1, our Nc4 as their Qd2; transferred move f8f7 (**endorses this move**). (c) Retrieved from `ccl1_top_p0000#1852` (ply 99) [structural distance $d=105$, QAP obj=5]: the master played Kxf8; under μ our Kf1 plays the role of that game’s Kg1; cross-type roles: our pa4 as their Re8, our qh6 as their nf8; transferred move f8f7 (**endorses this move**). (d) Retrieved from `ccl1_top_p0000#802` (ply 90) [structural distance $d=107$, QAP obj=19]: the master played Rbf1; under μ our Be4 plays the role of that game’s Rb1 (a **cross-type** correspondence); cross-type roles: our bg5 as their rd2, our Pd4 as their Nc3; transferred move e5g7. Synthesis: 2 retrieved analogs transferred to Kf7 (summed QAP objective 77, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



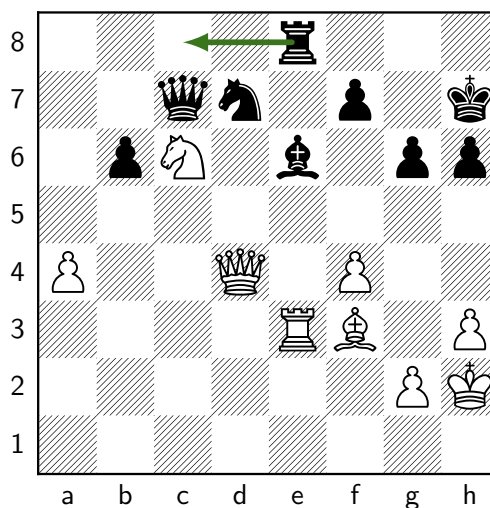
(a)



(b)

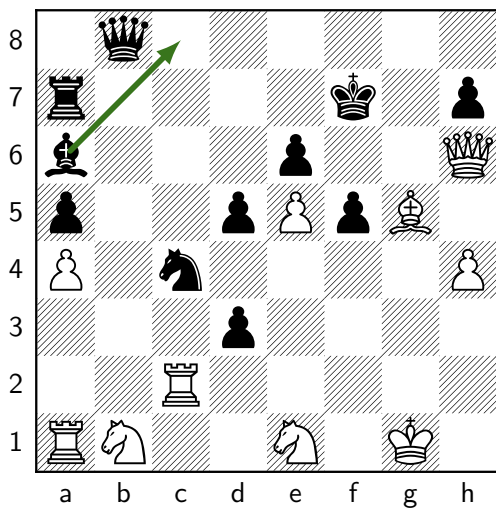


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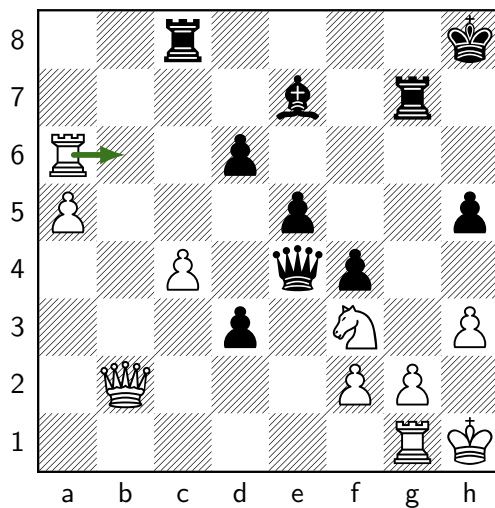


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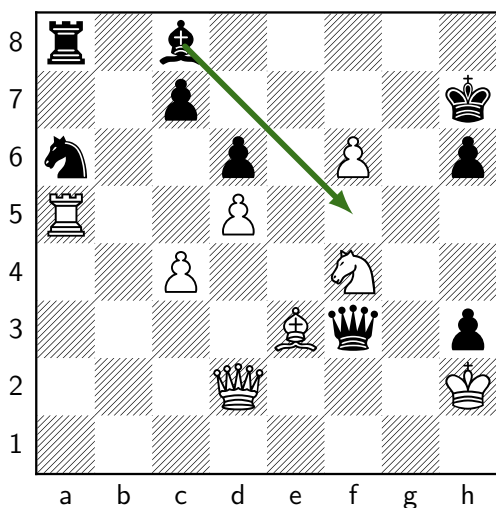
Figure 40: **Move 47, Black: d3.** (a) Our position; Melanie plays d3 (d4d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#2831` (ply 87) [structural distance $d=102$, QAP obj=28]: the master played b4; under μ our $Pd5$ plays the role of that game’s $Pb4$; cross-type roles: our $Qb8$ as their $Rc8$, our $Pd5$ as their $Be8$; transferred move d4d3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#943` (ply 95) [structural distance $d=105$, QAP obj=52]: the master played Bxd2; under μ our $Be4$ plays the role of that game’s $Bh3$; cross-type roles: our $Qb8$ as their $Bc8$, our $Nc4$ as their $Qe6$; transferred move e5f4. (d) Retrieved from `ccrl_top_p0000#740` (ply 79) [structural distance $d=107$, QAP obj=34]: the master played Rc8; under μ our $Ba3$ plays the role of that game’s $Re1$ (a **cross-type** correspondence); cross-type roles: our $pf4$ as their $Qc7$, our $bg5$ as their $Pb6$; transferred move a6c8. Synthesis: 1 retrieved analog transferred to d3 (summed QAP objective 28, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



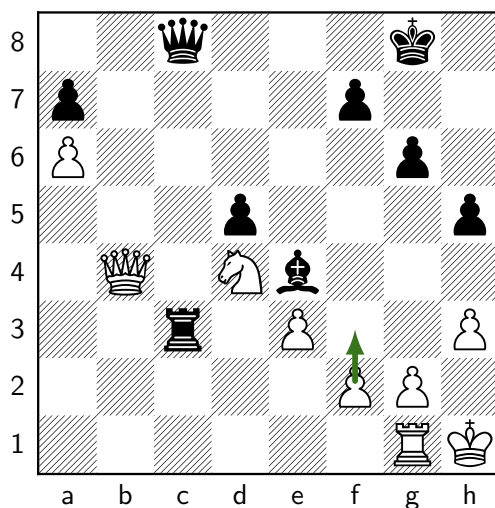
(a)



(b)

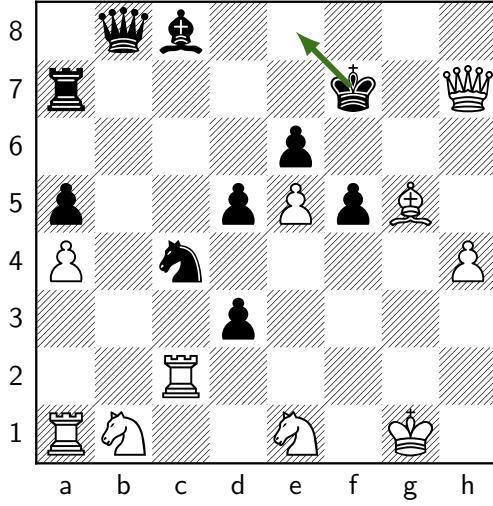


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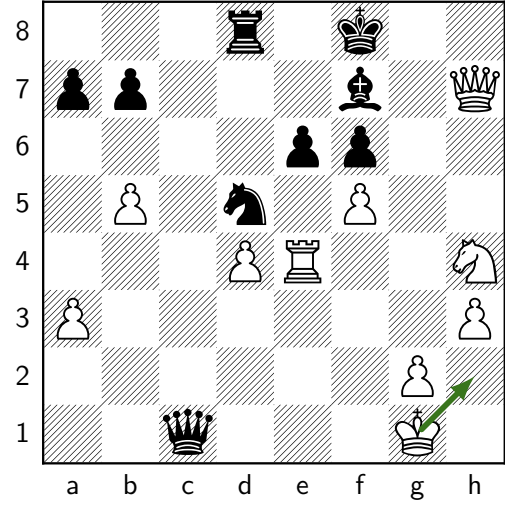


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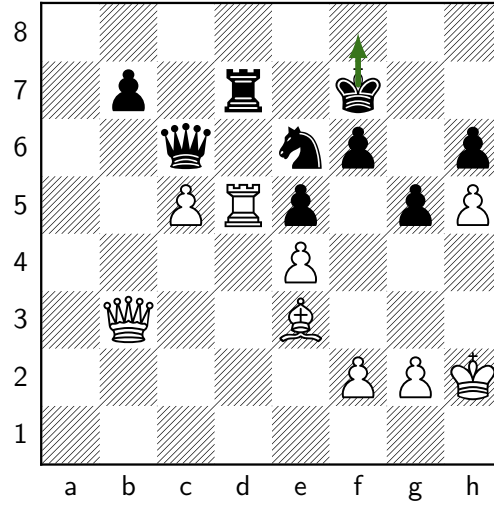
Figure 41: **Move 48, Black: Bc8.** (a) Our position; Melanie plays Bc8 (a6c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#925` (ply 80) [structural distance $d=95$, QAP obj=37]: the master played Rb6; under μ our Ba3 plays the role of that game’s Ra6 (a **cross-type** correspondence); cross-type roles: our bg5 as their Rg1, our Nc4 as their Pf2; transferred move a6c8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1319` (ply 79) [structural distance $d=95$, QAP obj=23]: the master played Bf5; under μ our Ba3 plays the role of that game’s Bc1; cross-type roles: our Qb8 as their Ra8, our Pe6 as their Na6; transferred move a6c8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2981` (ply 96) [structural distance $d=88$, QAP obj=38]: the master played f3; under μ our Ba3 plays the role of that game’s Pf2 (a **cross-type** correspondence); cross-type roles: our Ph7 as their Rg1, our Pd3 as their Qb4; transferred move a6b5. Synthesis: 2 retrieved analogs transferred to Bc8 (summed QAP objective 60, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



(a)

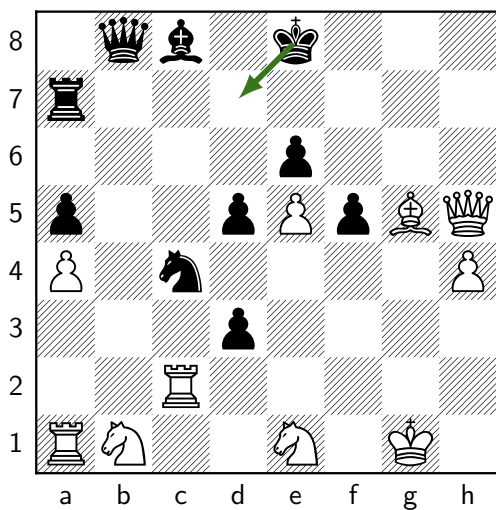


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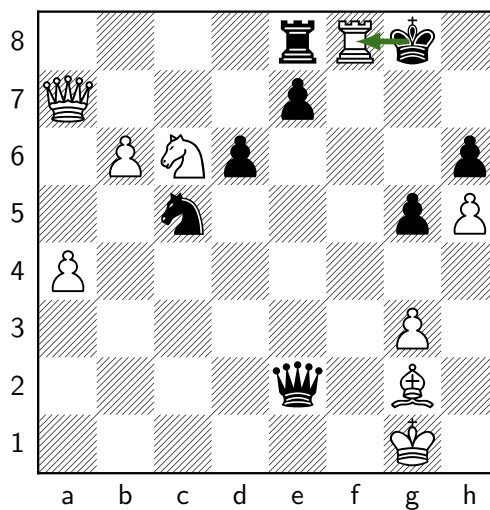


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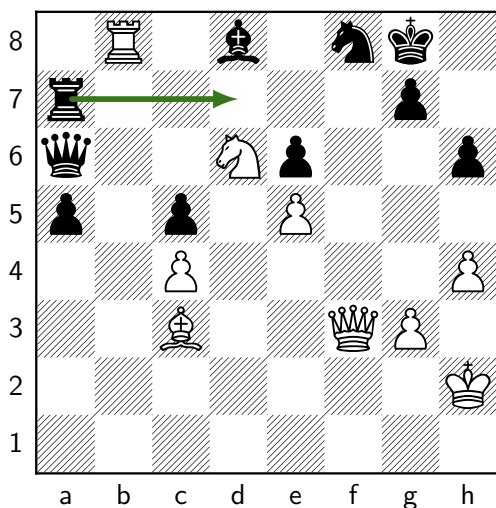
Figure 42: **Move 49, Black: Ke8.** (a) Our position; Melanie plays Ke8 (f7e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1.top_p0002#773` (ply 88) [structural distance $d=76$, QAP obj=42]: the master played Kh2; under μ our Kf2 plays the role of that game’s Kg1; cross-type roles: our Pd3 as their Re4, our pe5 as their Nh4; transferred move f7e8 (**endorses this move**). (c) Retrieved from `ccr1.top_p0000#1212` (ply 89) [structural distance $d=75$, QAP obj=12]: the master played Kf8; under μ our Kf2 plays the role of that game’s Kf2; cross-type roles: our Nc4 as their Rd7, our Pd3 as their Qc6; transferred move f7f8. Synthesis: 2 retrieved analogs transferred to Ke8 (summed QAP objective 85, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



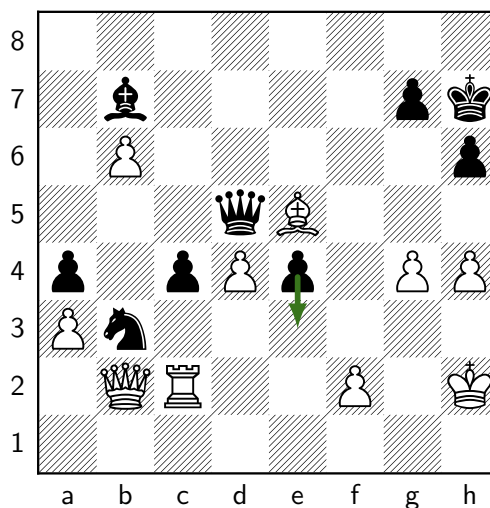
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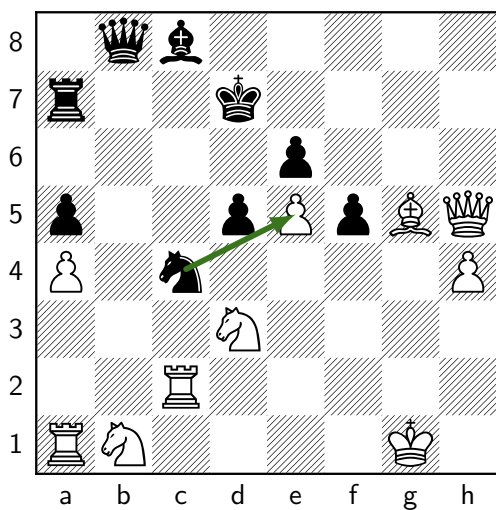


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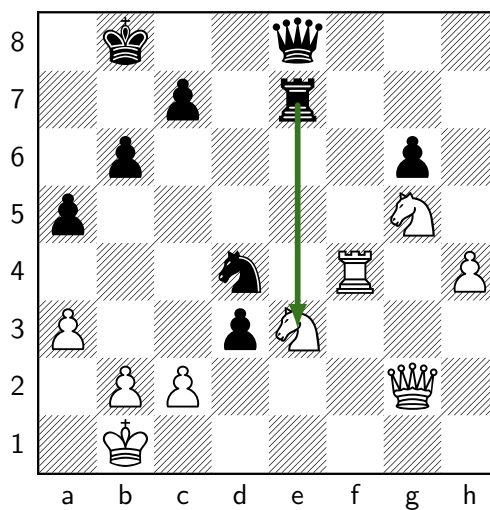


(d)

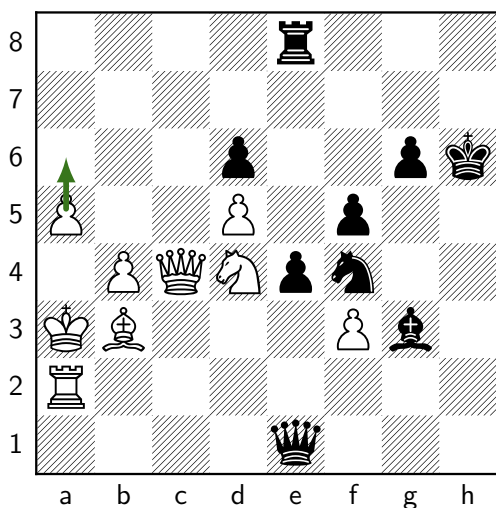
Figure 43: **Move 50, Black: Kd7.** (a) Our position; Melanie plays Kd7 (e8d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2694` (ply 83) [structural distance $d=77$, QAP obj=26]: the master played Kxf8; under μ our *Ke1* plays the role of that game's *Kg1*; cross-type roles: our *qh5* as their *rf8*, our *Ra7* as their *qa7*; transferred move e8f8. (c) Retrieved from `ccr1_top_p0000#291` (ply 111) [structural distance $d=72$, QAP obj=5]: the master played Rd7; under μ our *Ra2* plays the role of that game's *Ra2*; cross-type roles: our *qh5* as their *rb8*, our *ne1* as their *Bd8*; transferred move a7f7. (d) Retrieved from `ccr1_top_p0001#2706` (ply 97) [structural distance $d=72$, QAP obj=5]: the master played e3; under μ our *Ra2* plays the role of that game's *Pe5* (a **cross-type** correspondence); cross-type roles: our *ne1* as their *Ph6*, our *qh5* as their *be5*; transferred move a7f7. Synthesis: no retrieved analog transferred to Kd7 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



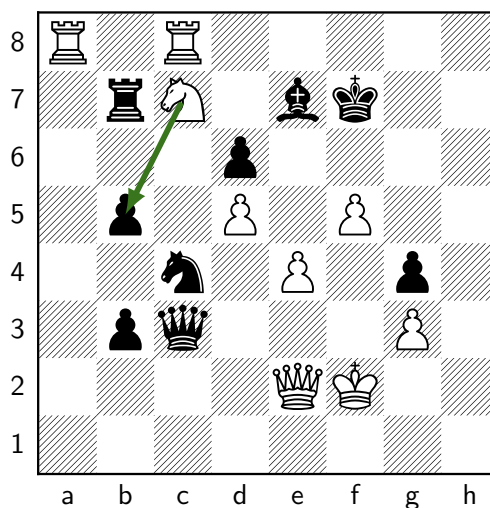
(a)



(b)

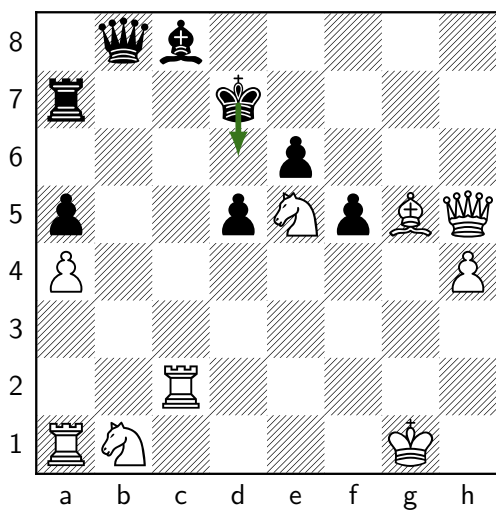


(c)

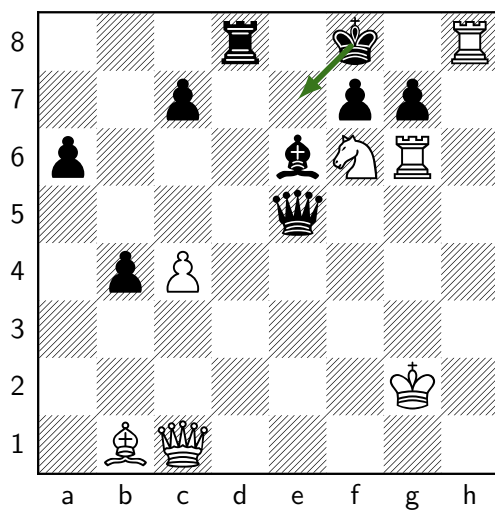


(d)

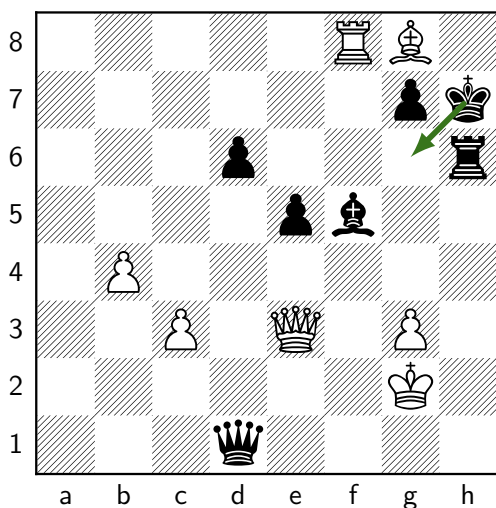
Figure 44: **Move 51, Black: Nxe5**. (a) Our position; Melanie plays Nxe5 (c4e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3605` (ply 117) [structural distance $d=76$, QAP obj=43]: the master played Rxe3; under μ our Qb1 plays the role of that game’s Re2 (a **cross-type** correspondence); cross-type roles: our Ra7 as their Qe8, our Bc8 as their Pc7; transferred move b8b3. (c) Retrieved from `ccr1_top_p0003#1070` (ply 110) [structural distance $d=78$, QAP obj=26]: the master played a6; under μ our Pf4 plays the role of that game’s Pa5; cross-type roles: our Bc8 as their Ra2, our pa4 as their Bb3; transferred move f5f4. (d) Retrieved from `ccr1_top_p0002#2779` (ply 102) [structural distance $d=68$, QAP obj=24]: the master played Nxb5; under μ our Nc5 plays the role of that game’s Nc7; cross-type roles: our Bc8 as their Qe2, our ra1 as their pb3; transferred move c4a3. Synthesis: no retrieved analog transferred to Nxe5 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



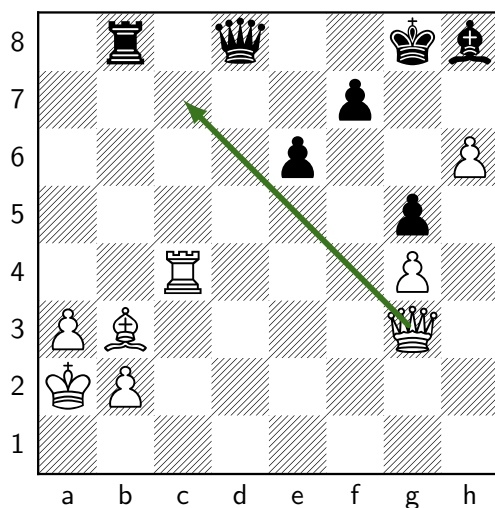
(a)



(b)

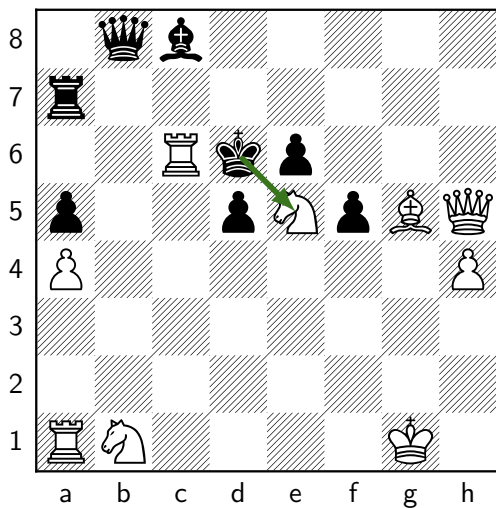


(c)

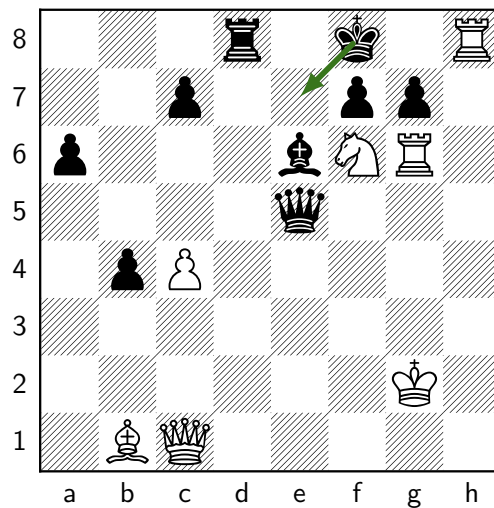


(d)

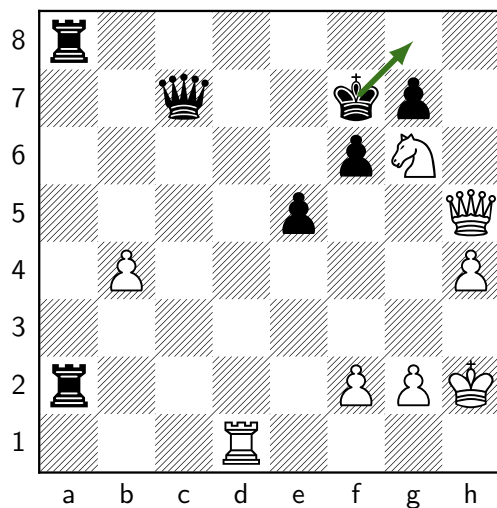
Figure 45: **Move 52, Black: Kd6**. (a) Our position; Melanie plays Kd6 (d7d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1335` (ply 97) [structural distance $d=56$, QAP obj=70]: the master played Ke7; under μ our Kd2 plays the role of that game’s Kf1; cross-type roles: our Bc8 as their Rd8, our ne5 as their rh8; transferred move d7d6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2530` (ply 87) [structural distance $d=70$, QAP obj=41]: the master played Kg6; under μ our Kd2 plays the role of that game’s Kh2; cross-type roles: our ne5 as their bg8, our qh5 as their Pg7; transferred move d7d6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2526` (ply 94) [structural distance $d=68$, QAP obj=6]: the master played Qc7; under μ our Qb1 plays the role of that game’s Qg3; cross-type roles: our bg5 as their pe6, our ra1 as their qd8; transferred move b8e5. Synthesis: 2 retrieved analogs transferred to Kd6 (summed QAP objective 112, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



(a)

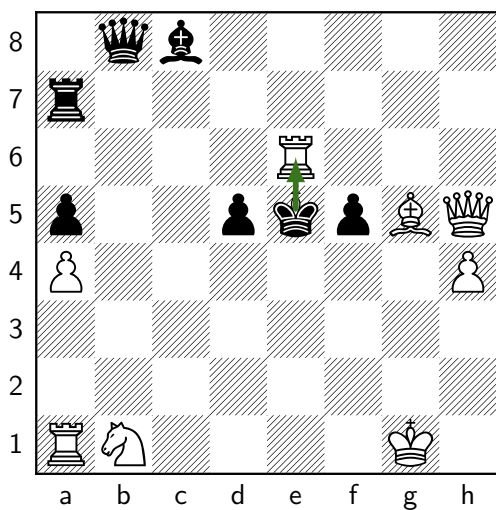


(b)

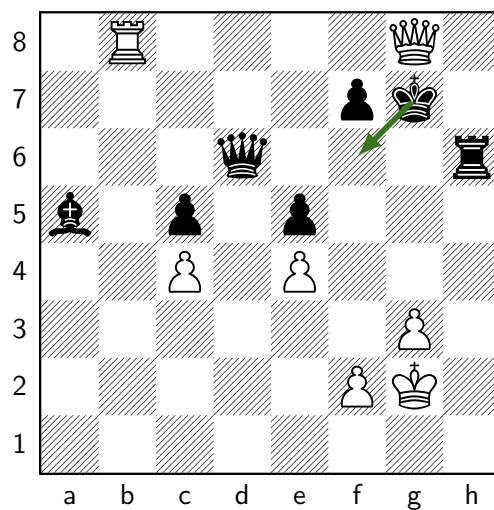


(c)

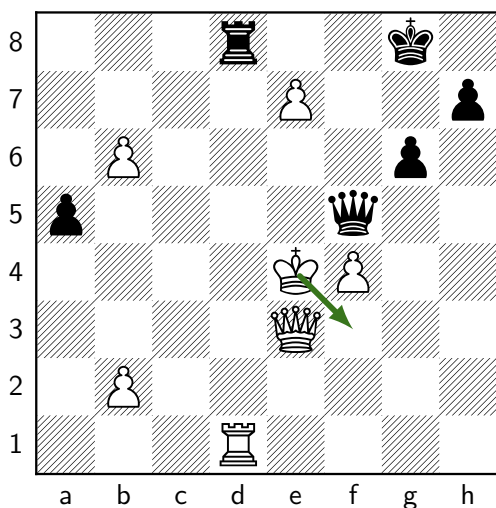
Figure 46: **Move 53, Black: Kxe5.** (a) Our position; Melanie plays Kxe5 (d6e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top-p0001#1335` (ply 97) [structural distance $d=55$, QAP obj=40]: the master played Ke7; under μ our Kd3 plays the role of that game’s Kf1; cross-type roles: our Qb8 as their Rd8, our Bc8 as their Pc7; transferred move d6e5 (**endorses this move**). (c) Retrieved from `ccr1_top-p0000#206` (ply 95) [structural distance $d=74$, QAP obj=4]: the master played Kg8; under μ our Kd3 plays the role of that game’s Kf2; cross-type roles: our qh5 as their Ra8, our rc6 as their ng6; transferred move d6e5 (**endorses this move**). Synthesis: 2 retrieved analogs transferred to Kxe5 (summed QAP objective 45, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



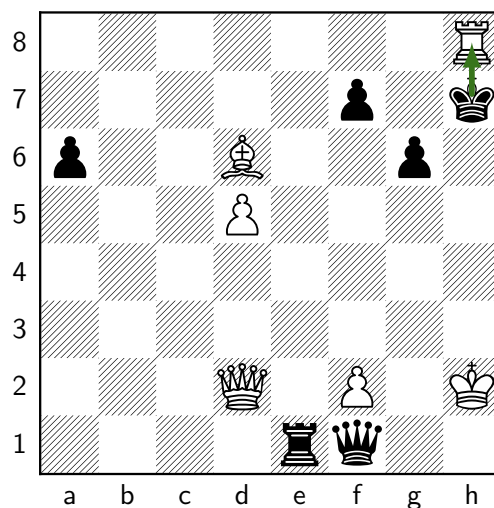
(a)



(b)

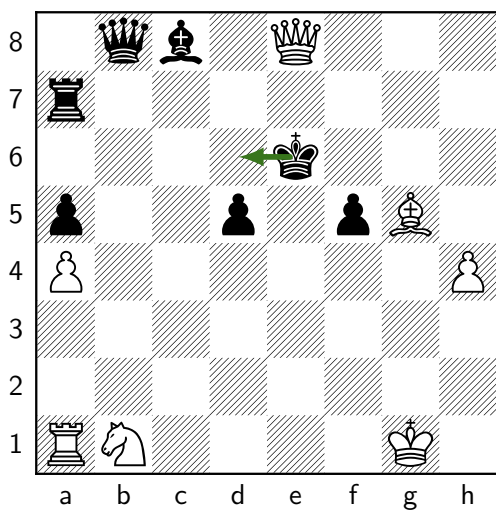


(c)

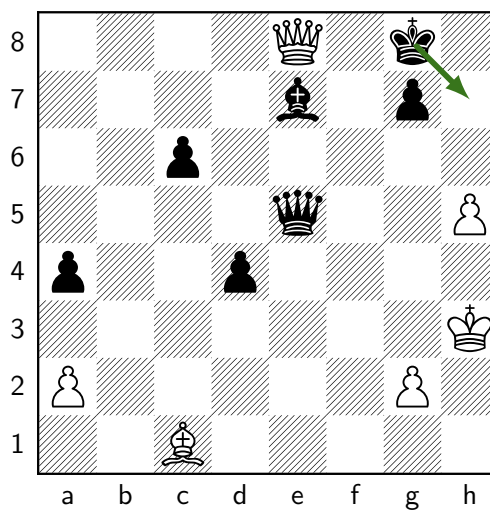


(d)

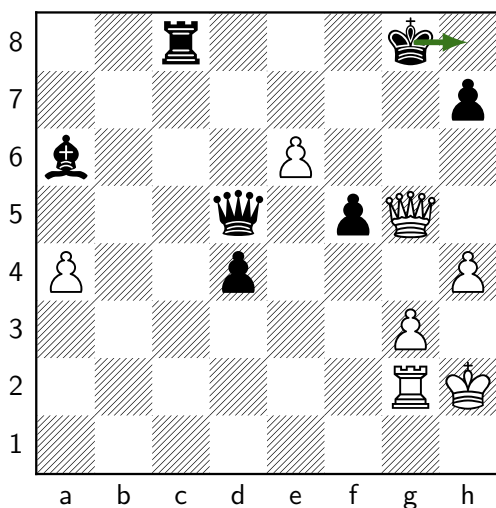
Figure 47: **Move 54, Black: Kxe6.** (a) Our position; Melanie plays Kxe6 (e5e6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1049` (ply 99) [structural distance $d=60$, QAP obj=71]: the master played Kf6; under μ our Ke4 plays the role of that game’s Kg2; cross-type roles: our bg5 as their rb8, our re6 as their qg8; transferred move e5e6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2731` (ply 110) [structural distance $d=65$, QAP obj=69]: the master played Kf3; under μ our Ke4 plays the role of that game’s Ke4; cross-type roles: our ph4 as their Rd1, our Bc8 as their Pb2; transferred move e5e6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1019` (ply 95) [structural distance $d=57$, QAP obj=68]: the master played Kxh8; under μ our Ke4 plays the role of that game’s Kh2; cross-type roles: our Qb8 as their bd6, our bg5 as their qd2; transferred move e5e6 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Kxe6 (summed QAP objective 357, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



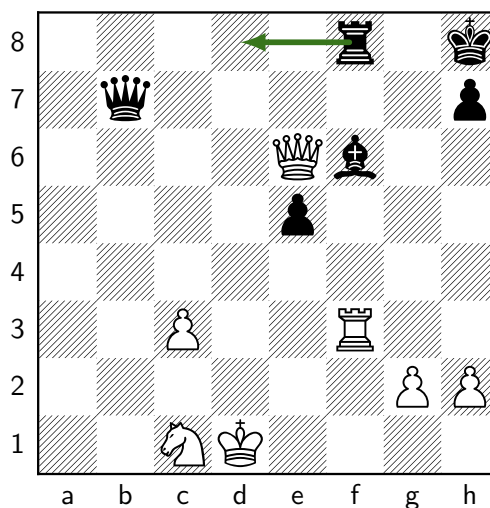
(a)



(b)

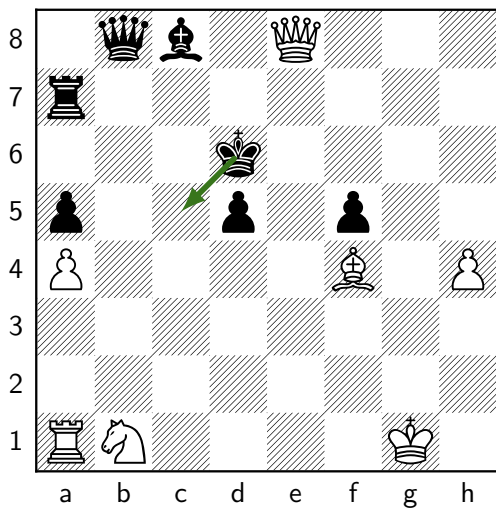


(c)

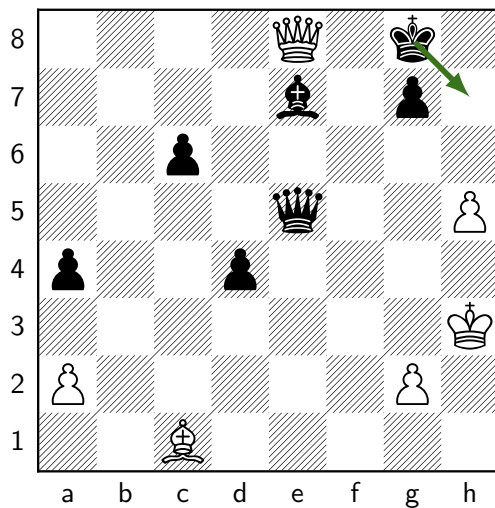


(d)

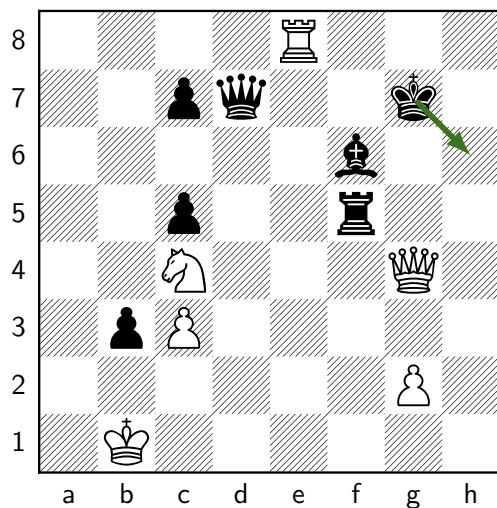
Figure 48: **Move 55, Black: Kd6.** (a) Our position; Melanie plays Kd6 (e6d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2240` (ply 107) [structural distance $d=48$, QAP obj=86]: the master played Kh7; under μ our Ke3 plays the role of that game’s Kg1; cross-type roles: our Ra7 as their Pg7, our bg5 as their pa2; transferred move e6d6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2772` (ply 101) [structural distance $d=49$, QAP obj=63]: the master played Kh8; under μ our Ke3 plays the role of that game’s Kg1; cross-type roles: our ph4 as their Rc8, our bg5 as their Ph7; transferred move e6d6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2303` (ply 101) [structural distance $d=48$, QAP obj=20]: the master played Rd8+; under μ our Ra2 plays the role of that game’s Rf1; cross-type roles: our bg5 as their Qb7, our ra1 as their qe6; transferred move a7e7. Synthesis: 3 retrieved analogs transferred to Kd6 (summed QAP objective 235, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



(a)

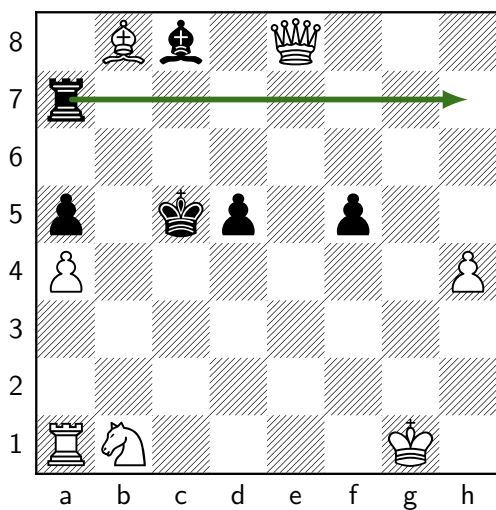


(b)

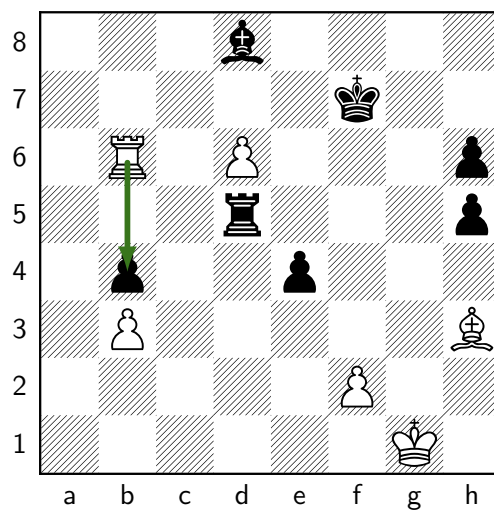


(c)

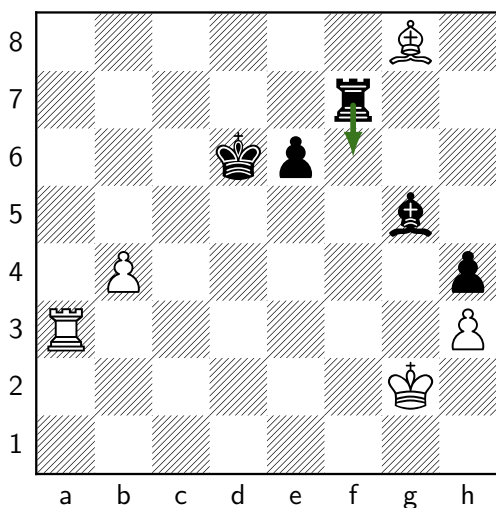
Figure 49: **Move 56, Black: Kc5.** (a) Our position; Melanie plays Kc5 (d6c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#2240` (ply 107) [structural distance $d=47$, QAP obj=71]: the master played Kh7; under μ our Kd3 plays the role of that game’s Kg1; cross-type roles: our bf4 as their qe8, our Ra7 as their Pc6; transferred move d6c5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#3494` (ply 109) [structural distance $d=50$, QAP obj=49]: the master played Kh6; under μ our Kd3 plays the role of that game’s Kg2; cross-type roles: our qe8 as their re8, our Ra7 as their Pc7; transferred move d6c5 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kc5 (summed QAP objective 335, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



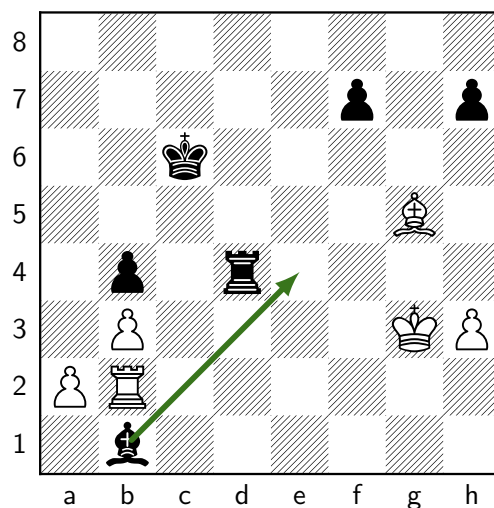
(a)



(b)

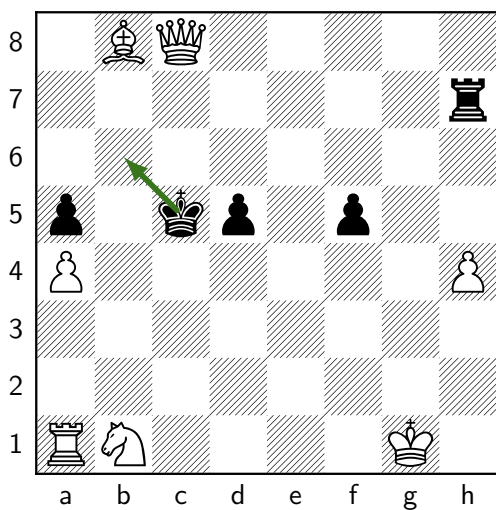


(c)

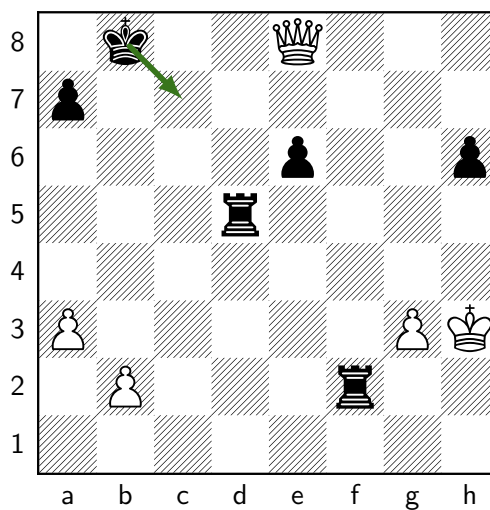


(d)

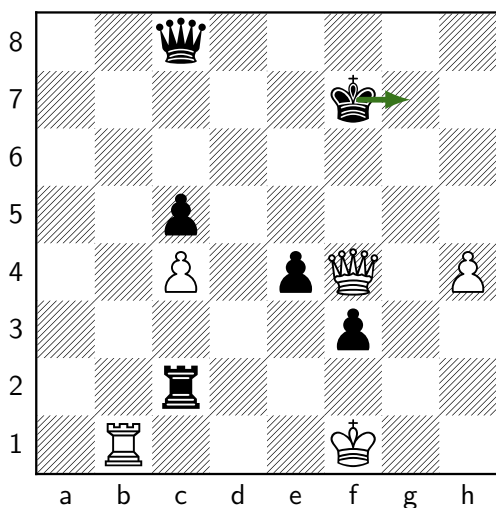
Figure 50: **Move 57, Black: Rh7.** (a) Our position; Melanie plays Rh7 (a7h7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2173` (ply 96) [structural distance $d=34$, QAP obj=11]: the master played Rxb4; under μ our Ra2 plays the role of that game's Rb6; cross-type roles: our qe8 as their Bh3, our Bc8 as their pb4; transferred move a7h7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#3105` (ply 107) [structural distance $d=35$, QAP obj=20]: the master played Rf6; under μ our Ra2 plays the role of that game's Rf2; cross-type roles: our ra1 as their Bg5, our nb1 as their Ph4; transferred move a7a6. (d) Retrieved from `ccrl_top_p0000#3457` (ply 101) [structural distance $d=37$, QAP obj=13]: the master played Be4; under μ our Ra2 plays the role of that game's Bb8 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Pf7, our qe8 as their bg5; transferred move a7a6. Synthesis: 1 retrieved analog transferred to Rh7 (summed QAP objective 11, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



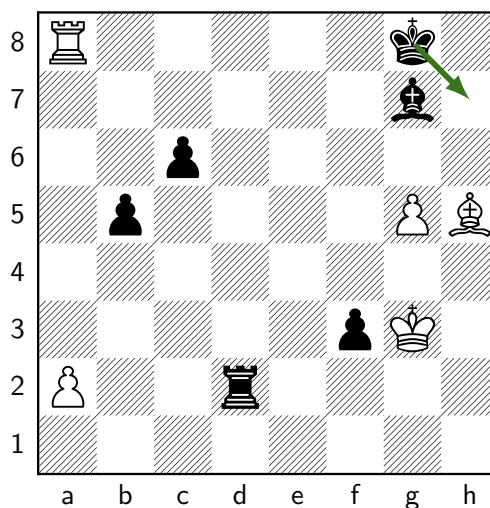
(a)



(b)



(c)



(d)

Figure 51: **Move 58, Black: Kb6**. (a) Our position; Melanie plays Kb6 (c5b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2895` (ply 119) [structural distance $d=30$, QAP obj=69]: the master played Kc7; under μ our Kc4 plays the role of that game’s Kb1; cross-type roles: our ra1 as their Pe6, our Rh7 as their Ph6; transferred move c5b6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1299` (ply 111) [structural distance $d=29$, QAP obj=66]: the master played Kg7; under μ our Kc4 plays the role of that game’s Kf2; cross-type roles: our Rh7 as their Qc8, our bb8 as their ph4; transferred move c5b6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3495` (ply 111) [structural distance $d=29$, QAP obj=37]: the master played Kh7; under μ our Kc4 plays the role of that game’s Kg1; cross-type roles: our qc8 as their ra8, our Pd5 as their Bg7; transferred move c5b6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kb6 (summed QAP objective 174, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

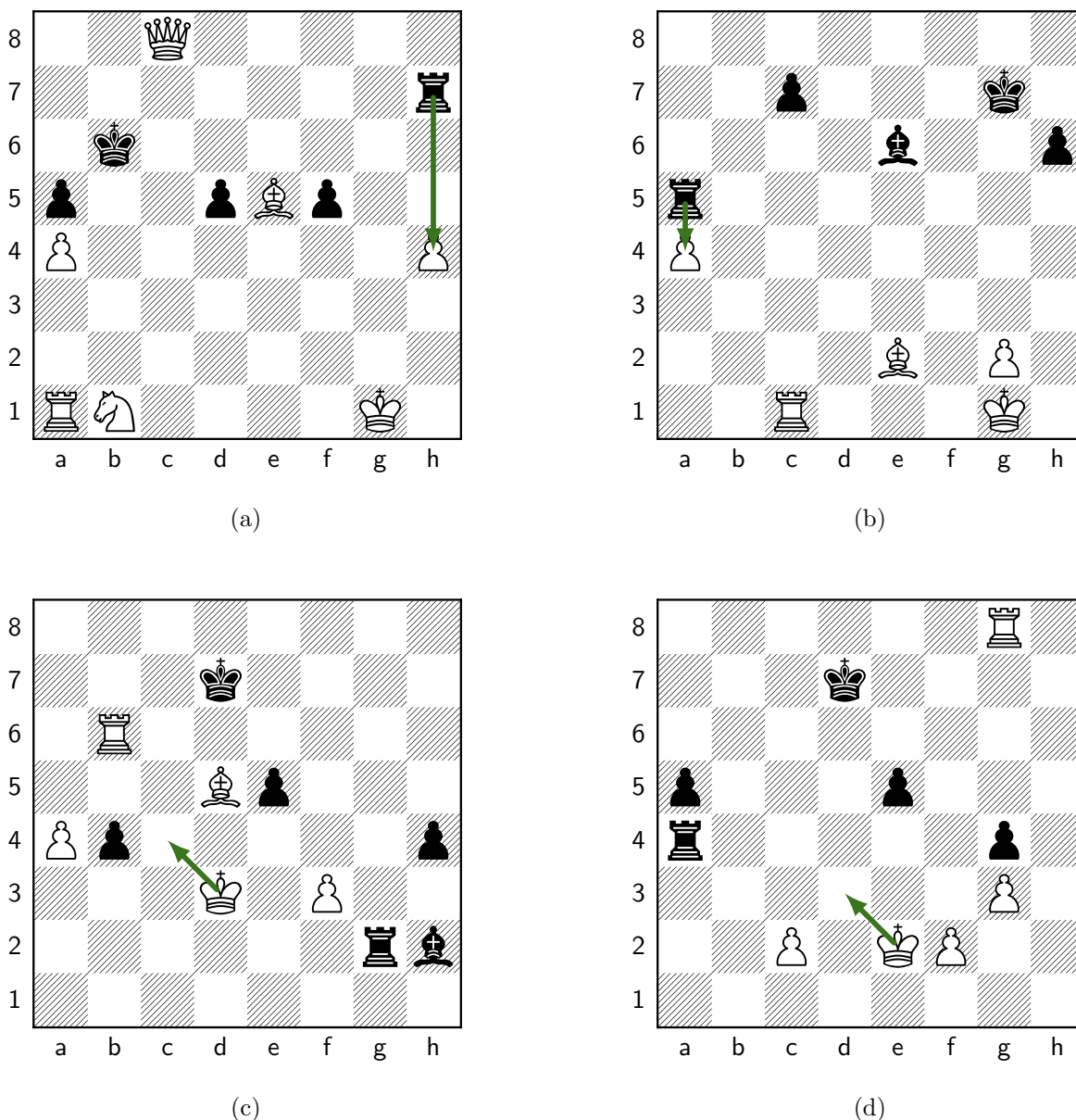
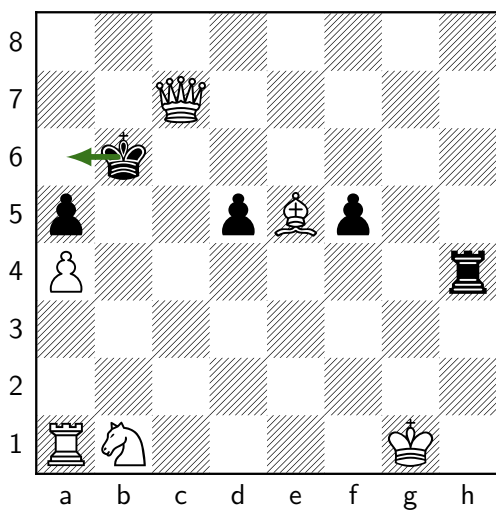
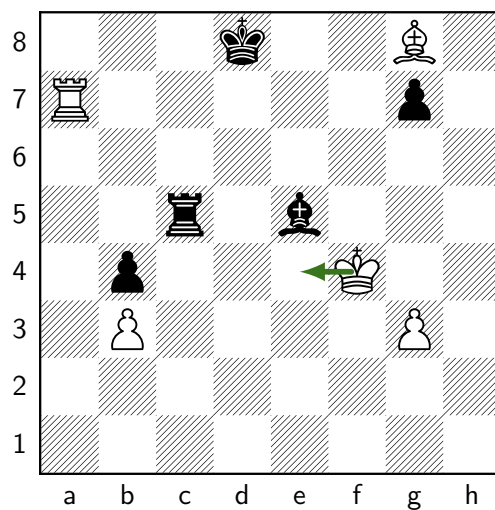


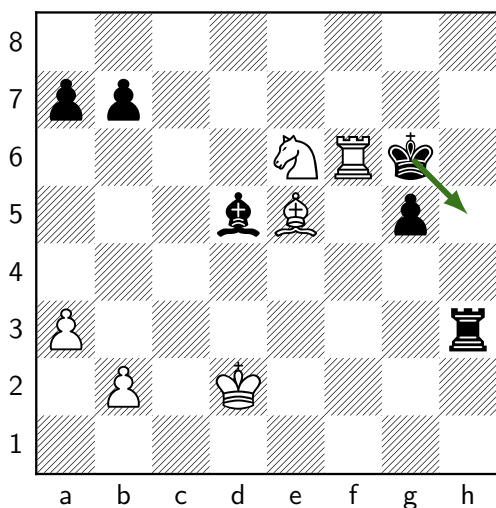
Figure 52: **Move 59, Black: Rxh4.** (a) Our position; Melanie plays Rxh4 (h7h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#17` (ply 101) [structural distance $d=30$, QAP obj=7]: the master played Rxa4; under μ our Rh2 plays the role of that game’s Ra4; cross-type roles: our Pa5 as their Be6, our qc8 as their rc1; transferred move h7h4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3128` (ply 130) [structural distance $d=27$, QAP obj=15]: the master played Kc4; under μ our Kb3 plays the role of that game’s Kd3; cross-type roles: our nb1 as their bh2, our qc8 as their Pa4; transferred move b6a7. (d) Retrieved from `ccr1_top_p0001#1792` (ply 126) [structural distance $d=28$, QAP obj=15]: the master played Kd3; under μ our Kb3 plays the role of that game’s Ke2; cross-type roles: our qc8 as their Pc2, our be5 as their Pg3; transferred move b6a7. Synthesis: 1 retrieved analog transferred to Rxh4 (summed QAP objective 7, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



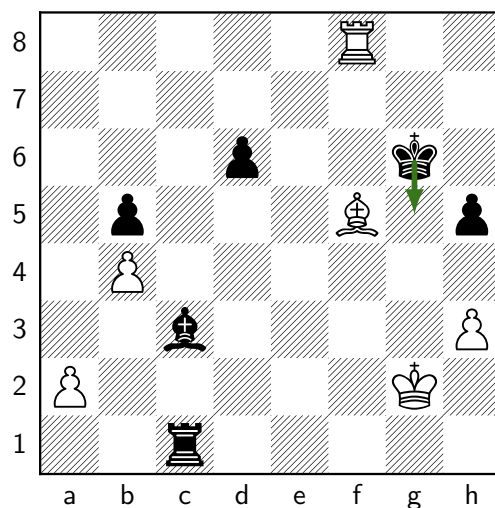
(a)



(b)



(c)



(d)

Figure 53: **Move 60, Black: Ka6**. (a) Our position; Melanie plays Ka6 (b6a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3649` (ply 126) [structural distance $d=32$, QAP obj=69]: the master played Kc4; under μ our Kb3 plays the role of that game's Kf4; cross-type roles: our Rh4 as their Pb3, our ra1 as their pb4; transferred move b6a6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3432` (ply 115) [structural distance $d=36$, QAP obj=69]: the master played Kh5; under μ our Kb3 plays the role of that game's Kg3; cross-type roles: our Rh4 as their Pb7, our ra1 as their ne6; transferred move b6a6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3424` (ply 115) [structural distance $d=39$, QAP obj=69]: the master played Kg5; under μ our Kb3 plays the role of that game's Kg3; cross-type roles: our be5 as their Pd6, our qc7 as their bf5; transferred move b6a6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Ka6 (summed QAP objective 274, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

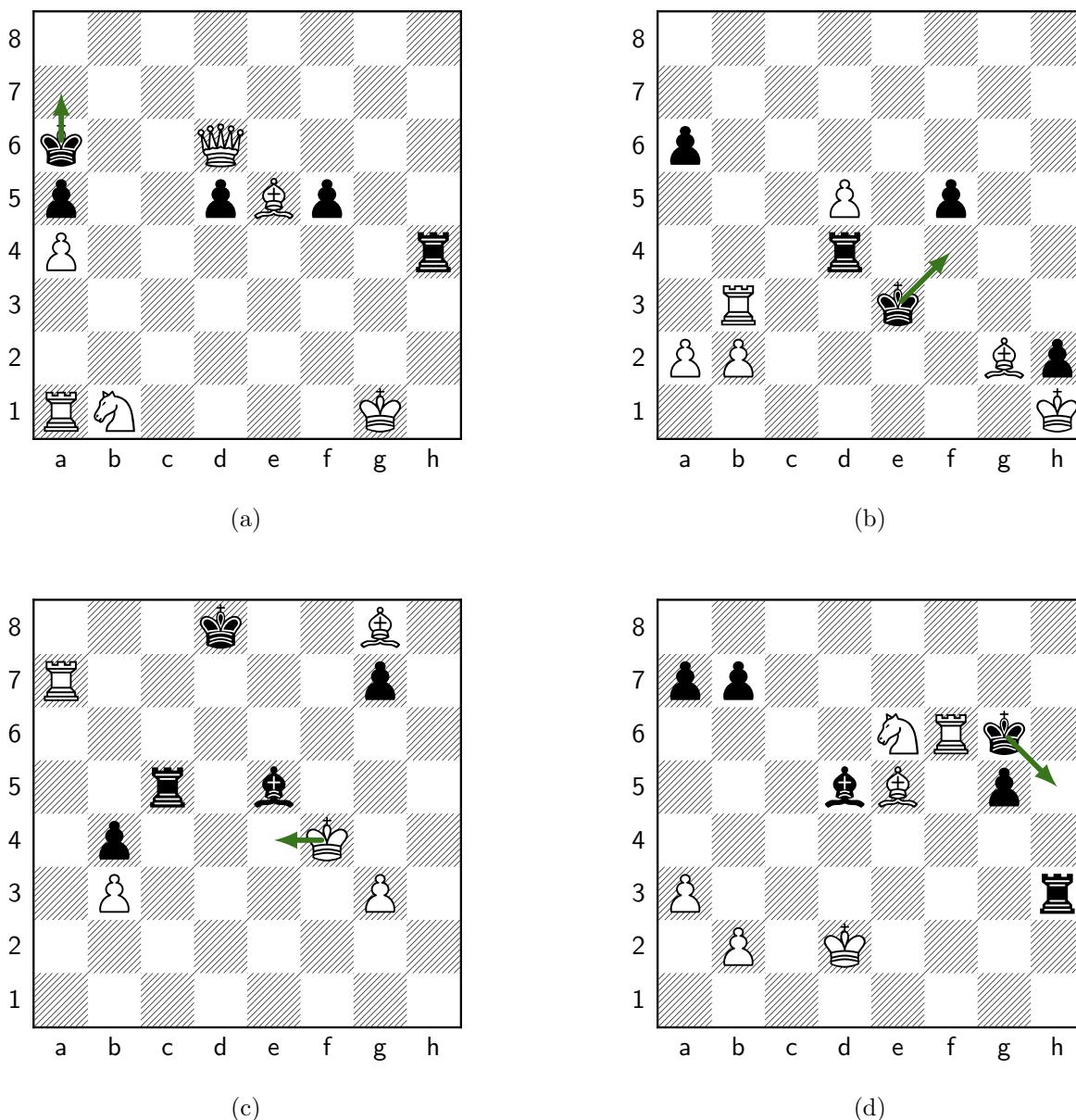
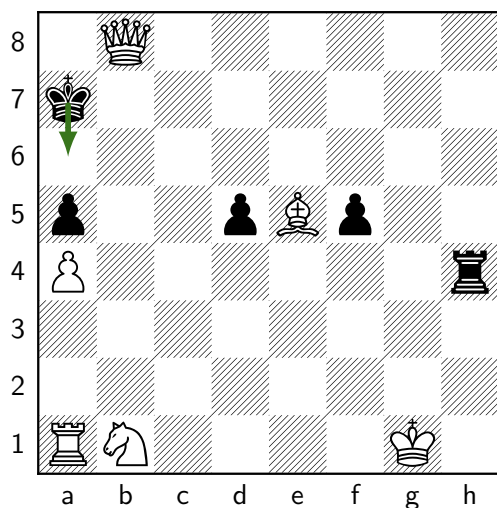
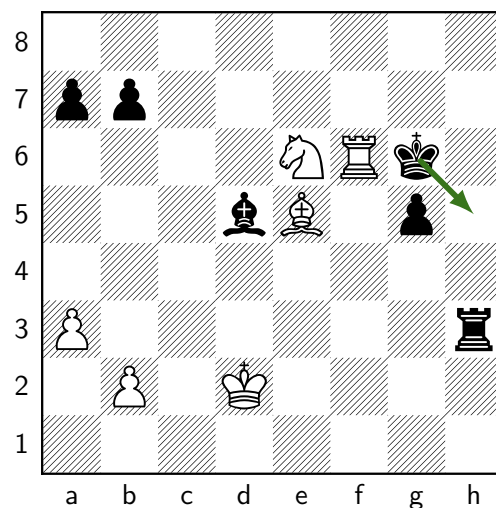


Figure 54: **Move 61, Black: Ka7.** (a) Our position; Melanie plays **Ka7** (**a6a7**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3775` (ply 119) [structural distance $d=38$, QAP obj=67]: the master played **Kf4**; under μ our *Ka3* plays the role of that game’s *Ke6*; cross-type roles: our *qd6* as their *rb3*, our *be5* as their *pa2*; transferred move **a6a7** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3649` (ply 126) [structural distance $d=35$, QAP obj=25]: the master played **Ke4**; under μ our *Ka3* plays the role of that game’s *Kf4*; cross-type roles: our *be5* as their *Pb3*, our *pa4* as their *rc5*; transferred move **a6a7** (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3432` (ply 115) [structural distance $d=37$, QAP obj=25]: the master played **Kh5**; under μ our *Ka3* plays the role of that game’s *Kg3*; cross-type roles: our *pa4* as their *ne6*, our *qd6* as their *rf6*; transferred move **a6a7** (**endorses this move**). Synthesis: 3 retrieved analogs transferred to **Ka7** (summed QAP objective 116, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

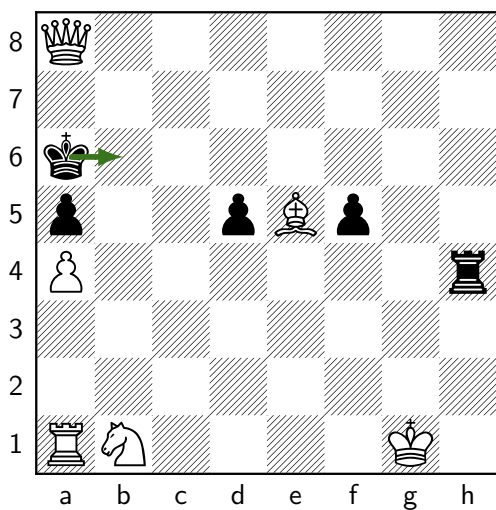


(a)

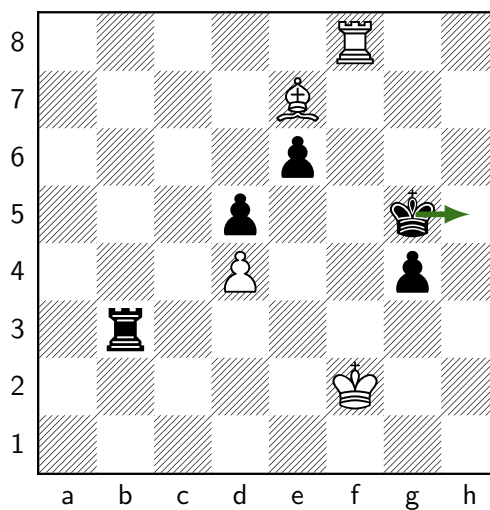


(b)

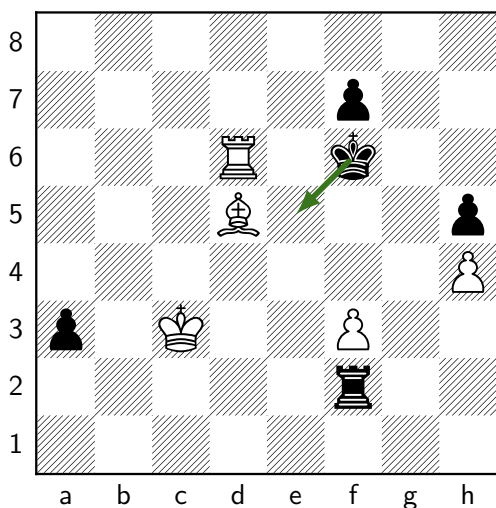
Figure 55: **Move 62, Black: Ka6.** (a) Our position; Melanie plays Ka6 (a7a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3432` (ply 115) [structural distance $d=50$, QAP obj=68]: the master played Kh5; under μ our Ka2 plays the role of that game’s Kg3; cross-type roles: our Rh4 as their Pb7, our ra1 as their ne6; transferred move a7a6 (**endorses this move**). Synthesis: 1 retrieved analog transferred to Ka6 (summed QAP objective 68, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



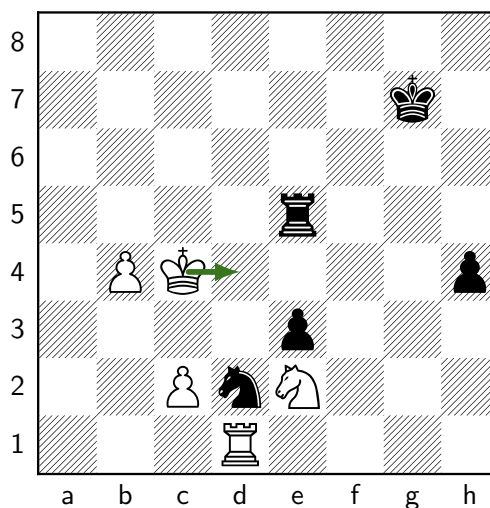
(a)



(b)

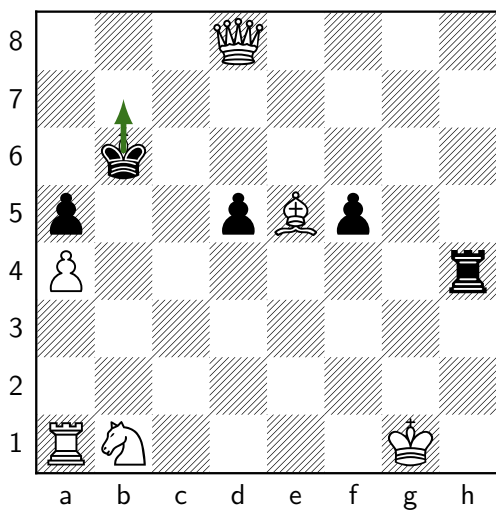


(c)

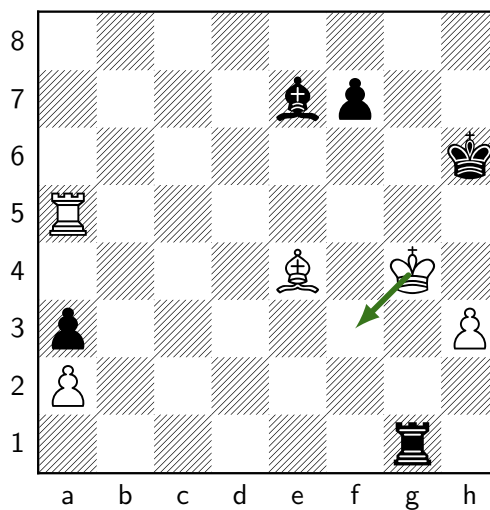


(d)

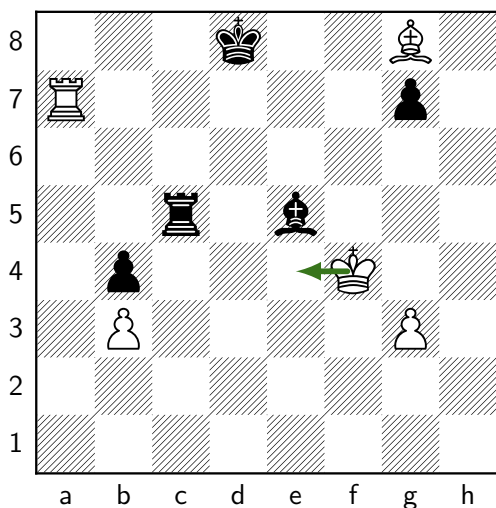
Figure 56: **Move 63, Black: Kb6**. (a) Our position; Melanie plays Kb6 (a6b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1986` (ply 113) [structural distance $d=32$, QAP obj=69]: the master played Kh5; under μ our Ka3 plays the role of that game’s Kg4; cross-type roles: our pa4 as their rf8, our qa8 as their be7; transferred move a6b6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1233` (ply 121) [structural distance $d=27$, QAP obj=68]: the master played Ke5; under μ our Ka3 plays the role of that game’s Kf3; cross-type roles: our qa8 as their rd6, our ra1 as their bd5; transferred move a6b6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#773` (ply 124) [structural distance $d=32$, QAP obj=68]: the master played Kd4; under μ our Ka3 plays the role of that game’s Kc4; cross-type roles: our qa8 as their nd2, our be5 as their Ne2; transferred move a6b6 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Kb6 (summed QAP objective 458, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



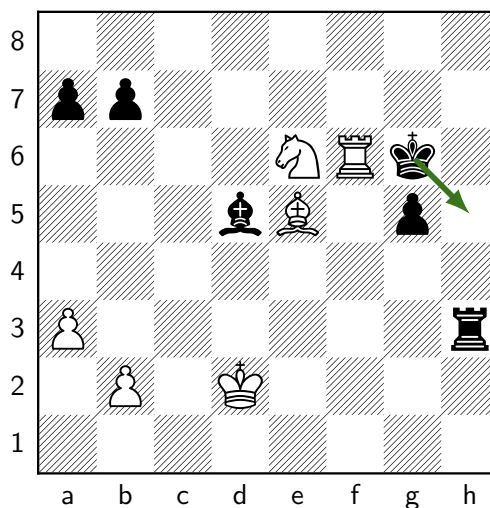
(a)



(b)

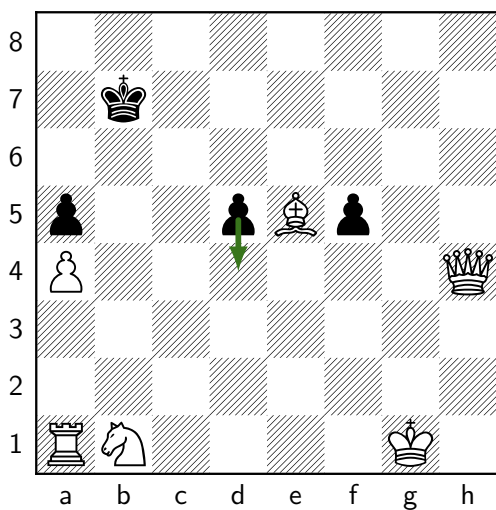


(c)

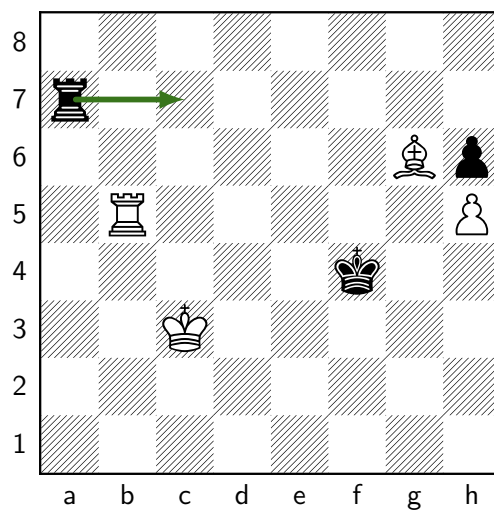


(d)

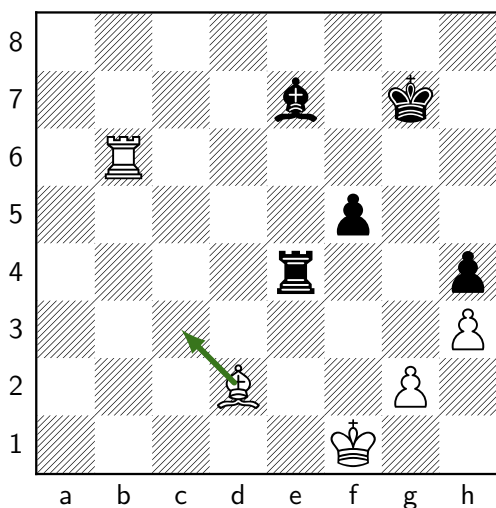
Figure 57: **Move 64, Black: Kb7.** (a) Our position; Melanie plays Kb7 (b6b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2031` (ply 110) [structural distance $d=40$, QAP obj=48]: the master played Kf3; under μ our *Kb3* plays the role of that game’s *Kg4*; cross-type roles: our *qd8* as their *rg1*, our *be5* as their *pa3*; transferred move b6c6. (c) Retrieved from `ccr1_top_p0000#3649` (ply 126) [structural distance $d=37$, QAP obj=36]: the master played Ke4; under μ our *Kb3* plays the role of that game’s *Kf4*; cross-type roles: our *be5* as their *pb4*, our *pa4* as their *rc5*; transferred move b6c5. (d) Retrieved from `ccr1_top_p0001#3432` (ply 115) [structural distance $d=35$, QAP obj=35]: the master played Kh5; under μ our *Kb3* plays the role of that game’s *Kg3*; cross-type roles: our *be5* as their *Pb7*, our *pa4* as their *ne6*; transferred move b6a6. Synthesis: no retrieved analog transferred to Kb7 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



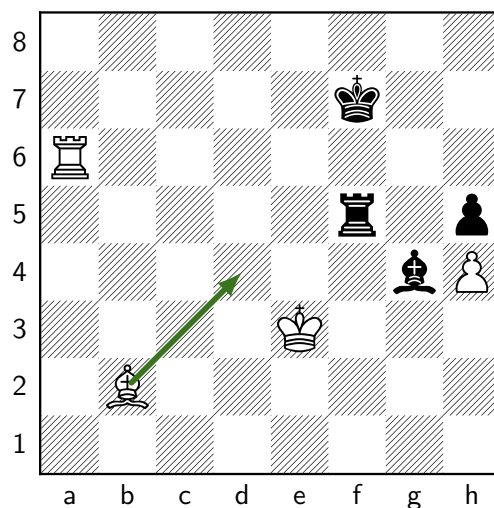
(a)



(b)

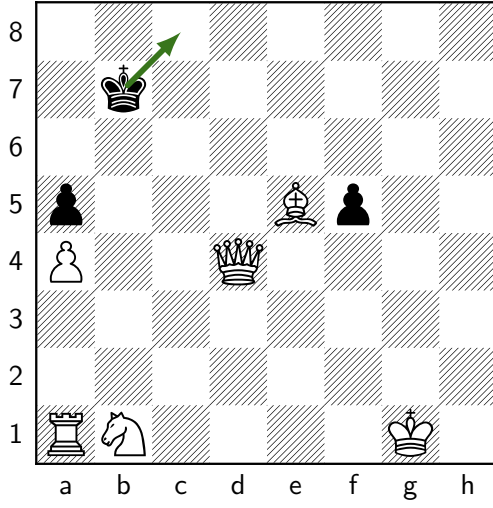


(c)

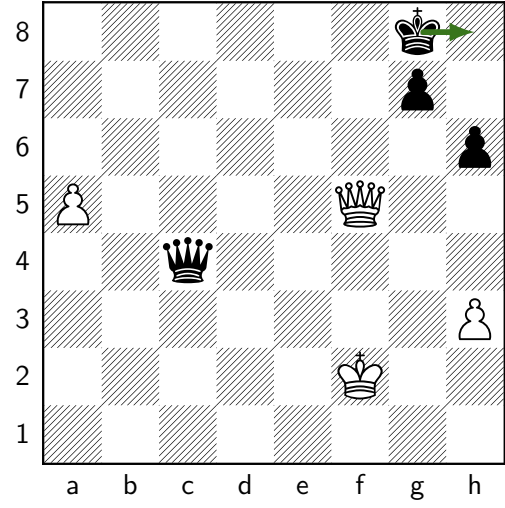


(d)

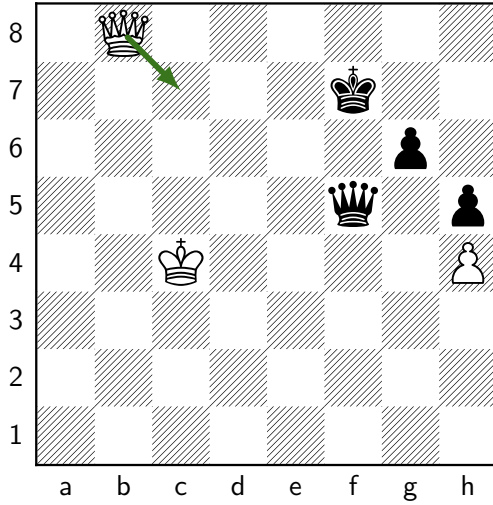
Figure 58: **Move 65, Black: d4**. (a) Our position; Melanie plays d4 (d5d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#810` (ply 135) [structural distance $d=44$, QAP obj=2]: the master played Rc7+; under μ our *Pd4* plays the role of that game's *Ra2* (a **cross-type** correspondence); cross-type roles: our *nb1* as their *bg6*, our *qh4* as their *rb5*; transferred move d5d4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2603` (ply 136) [structural distance $d=44$, QAP obj=2]: the master played Bc3+; under μ our *Pd4* plays the role of that game's *Bd2* (a **cross-type** correspondence); cross-type roles: our *qh4* as their *Pg2*, our *be5* as their *re4*; transferred move d5d4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#729` (ply 128) [structural distance $d=42$, QAP obj=1]: the master played Bd4; under μ our *Pd4* plays the role of that game's *Bb2* (a **cross-type** correspondence); cross-type roles: our *ra1* as their *bg4*, our *qh4* as their *rf5*; transferred move d5d4 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to d4 (summed QAP objective 7, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



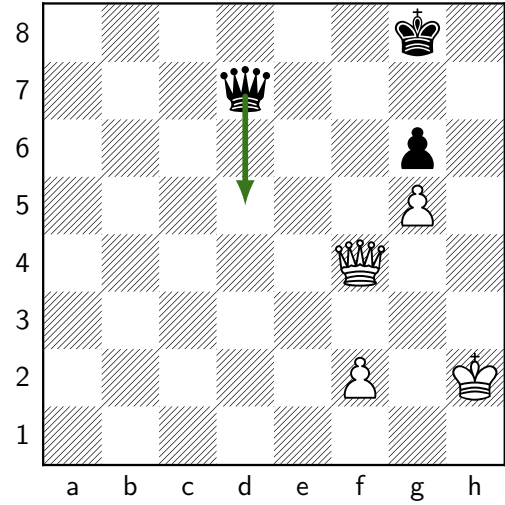
(a)



(b)

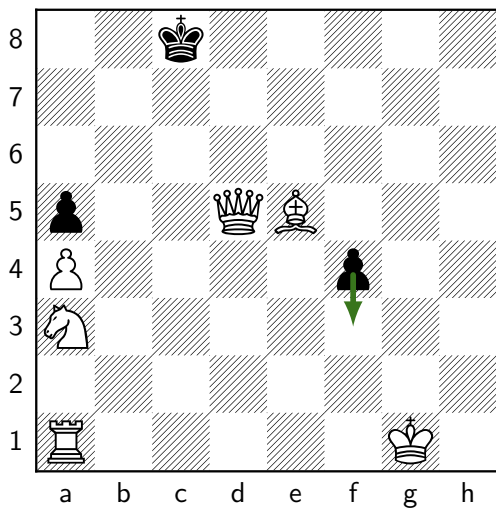


(c)

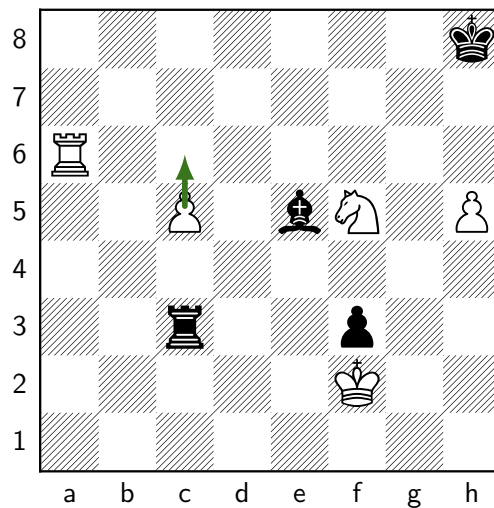


(d)

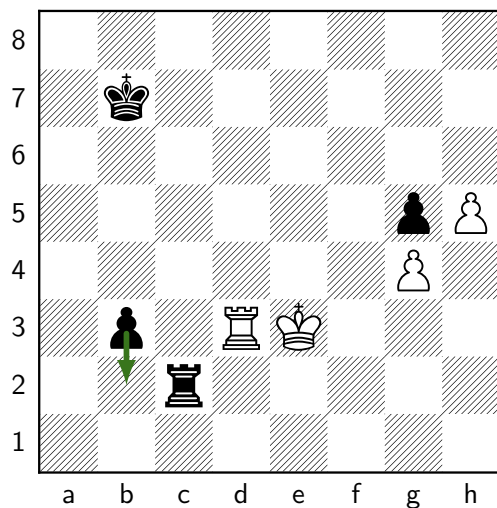
Figure 59: **Move 66, Black: Kc8**. (a) Our position; Melanie plays Kc8 (b7c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3481` (ply 119) [structural distance $d=48$, QAP obj=3]: the master played Kh8; under μ our Kb2 plays the role of that game’s Kg1; cross-type roles: our ra1 as their Pg7, our Pa5 as their Qc4; transferred move b7a8. (c) Retrieved from `ccr1_top_p0002#3942` (ply 120) [structural distance $d=46$, QAP obj=3]: the master played Qc7+; under μ our Pf4 plays the role of that game’s Qb8 (a **cross-type** correspondence); cross-type roles: our ra1 as their pg6; transferred move f5f4. (d) Retrieved from `ccr1_top_p0000#1226` (ply 117) [structural distance $d=38$, QAP obj=2]: the master played Qd5; under μ our Pf4 plays the role of that game’s Qd2 (a **cross-type** correspondence); cross-type roles: our be5 as their pf2; transferred move f5f4. Synthesis: no retrieved analog transferred to Kc8 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



(a)



(b)



(c)

Figure 60: **Move 68, Black: f3**. (a) Our position; Melanie plays f3 (f4f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top.p0003#1586` (ply 128) [structural distance $d=27$, QAP obj=12]: the master played c6; under μ our $Pf5$ plays the role of that game’s $Pc5$; cross-type roles: our $be5$ as their $rc3$, our $ra1$ as their $pf3$; transferred move f4f3 (**endorses this move**). (c) Retrieved from `ccr1_top.p0000#3901` (ply 127) [structural distance $d=31$, QAP obj=10]: the master played b2; under μ our $Pf5$ plays the role of that game’s $Pb6$; cross-type roles: our $na3$ as their $ph5$, our $be5$ as their $rd3$; transferred move f4f3 (**endorses this move**). Synthesis: 2 retrieved analogs transferred to f3 (summed QAP objective 22, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.